FRAGILITY ASSESSMENT AND RELIABILITY ANALYSIS OF TRANSMISSION LINES SUBJECTED TO CLIMATIC HAZARDS

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ABSTRACT

Overhead transmission lines are the backbone of electric power networks. As reported after several climatic hazards including the ice storm of January 1998 in Canada, the wind storms of December 1999 in France and the ice storm of January 2008 in China, the economic impact of the failure of transmission lines can be significant which highlights the importance of understanding and predicting the behaviour of these structural systems under weather-related loads. The randomness in environmental loads applied to transmission lines and the variability of the strength of line components or elements call for the application of probabilistic methods in investigating their behaviour.

The main objective of this study is focused on developing a probabilistic framework for reliability analysis and fragility assessment of line components and systems subjected to wind and ice storms. The principal step to achieve this objective is to address the issue of unknown implicit limit state functions. Two main approaches are proposed for this purpose: the component-based approach and the system-based approach. The component-based method focuses on the novel application of the concepts of Statistical Learning Theory (SLT) and Vapnik-Chervonenkis (VC) generalization bounds for model complexity control in the field of structural engineering and for reliability analysis purposes. SLT and VC-generalization bounds have been previously used for model selection in solving practical problems such as motion estimation and tracking as well as traffic flow time series prediction. Our goal is to select a surrogate demand model of the finite element solver for each tower element or line component from several possible models that has good generalization (prediction) properties. Using this approach, component and system fragility curves are developed and further integrated with climatic hazards to estimate component and system failure probabilities. The relative importance

of different line components is further quantified and components are ranked based on the improvement potential importance measure and the Bayesian reliability importance measure. The proposed rankings are respectively useful for the purpose of upgrading lines with limited resources and fault diagnosis after an extreme climatic event. The system-based approach focuses on the novel application of a probabilistic support vector machine (PSVM) classification technique called the Venn Predictor (VP) in system-based fragility assessment and reliability analysis of transmission lines systems. SVM is a state-of-the-art machine learning technique which is used to discover patterns in highly dimensional and complex data sets. In this application, PSVM is used to determine the limit state function of the line and to provide estimates of the probability that the line fails or survives for given climatic conditions. This method allows developing system fragility curves without requiring information about component fragilities and is particularly useful in fragility analysis of complex structural systems with hundreds of members such as power networks. Both approaches are applied to a nine span segment of a typical 120 kV single circuit transmission line located in North America.

This study also shows how the proposed methodologies can be used by researchers and engineers to solve some of the current challenges. For example, component and system fragility curves are developed for different scenarios of unbalanced ice loading that can result from non-uniform ice formation and/or ice shedding. In this application, the effect of wind direction on non-uniform ice formation and the resulting fragility curves is taken into account. Another application of the proposed methodologies is to investigate the impact of climate change on the performance and reliability of transmission lines by performing a sensitivity analysis of the system reliability index over a range of assumed changes in the mean and standard deviation of the climatic variables of interest.

RÉSUMÉ

Les lignes de transport sont des éléments critiques du réseau électrique. La défaillance de ces lignes peut avoir un impact majeur sur la sécurité du public et sur les activités économiques tel que démontré par les événements reliés à la tempête de verglas de janvier 1998 au Canada, les tempêtes de vent de décembre 1999 en France et la tempête de verglas de 2008 en Chine. Ces exemples démontrent la besoin de mieux comprendre et prédire le comportement de ces systèmes face à l'aléa associé aux charges associées au vent et au verglas. Les méthodes probabilistes sont les mieux adaptées pour modéliser les incertitudes et la variabilité associées aux charges appliquées, aux caractéristiques des matériaux et au comportement structural.

Un élément important de l'analyse probabiliste est la définition des fonctions d'état limite. Deux approches sont proposées: une approche par composante et une approche systémique. L'approche par composante utilise une approche novatrice basée sur les concepts de la théorie d'apprentissage statistique (SLT) et la généralisation des intervalles Vapnik-Chervonenkis (VC) pour contrôler le niveau de complexité du modèle. La procédure est utilisée pour identifier un modèle représentatif équivalent à un modèle plus complexe basé sur les éléments finis. Le modèle représentatif équivalent est utilisé afin de dériver les courbes de fragilité des composants et du système et leur couplage avec les fonctions de récurrence du vent et du verglas pour obtenir les probabilités de défaillance des composants et du système. L'importance relative des composants est évaluée et utilisée pour ordonner ces derniers en fonction de critères basés sur l'impact d'une réfection sur la performance globale de la ligne et selon un critère Bayesien. Cette information est utile afin d'effectuer les travaux de réfection plus efficacement et également afin de guider les diagnostics de défaillance suite à des événements climatiques extrêmes. L'approche systémique utilise une application novatrice des techniques de

classification par machines à vecteur de support probabiliste (PSVM), en particulier par la technique Venn Predictor (VP).

PSVM est une technique d'apprentissage par machine qui identifie les interrelations parmi un grand ensemble multidimensionnel de données. Dans la présente application, PSVM est utilisé pour identifier la fonction d'état limite d'une ligne et pour estimer la probabilité conditionnelle de défaillance en fonction de l'intensité du vent et du verglas. Cette technique peut également être utilisée afin dériver directement la fragilité d'une ligne directement sans avoir à évaluer la fiabilité des composants.

Les deux méthodes sont appliquées à un segment typique pour un circuit simple 120kV avec 9 portées. Le modèle est également utilisé afin d'évaluer le comportement de la ligne pour différents scénarios de charges non balancées, de direction du vent, et d'accumulation non-uniforme du verglas. Une autre application du modèle est également démontrée pour l'analyse de la résilience de la ligne pour différents scénarios de changement climatique.

PREFACE

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DEDICATION

To my best friend and life companion, Nima, with whom I have truly experienced love and joy ,and

In memory of my father

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SYMBOLS AND ABBREVIATIONS

A_c	Conductor or ground wire load (N) due to wind pressure
A_i	Insulator string load (N) due to wind pressure
A_t	Wind load (N) in the direction of the wind, applied at the centre of gravity of each tower panel
ASCE	American society of civil engineers
ATC	Applied technology council
b	Bias
C_k	Index set for the *k*-th minimum cut set
C_{xc}	Drag coefficient of conductors
C_{xi}	Drag coefficient of insulator strings
C($\mathbf{x}_c$)	Load carrying capacity function
CIGRE	Conseil international des grands reseaux electriques
COV	coefficient of variation
CV	Cross validation
d	Conductor / ground wire diameter
D($\mathbf{x}_d$)	Structural demand function
$E(\omega)$	Prediction error
$E_{emp}(\omega)$	Empirical error
EPRI	Electric power research institute
ERM	Empirical risk minimization
F_i	Failure event i
$f_I(i)$	Marginal probability density function of ice thickness
$f_R(r)$	Marginal probability density function of capacity

$f_{W_a}(w_a)$	Marginal probability density function of wind angle
$f_{W_s}(w_s)$	Marginal probability density function of wind speed
$f_{W_s W_a IR}$	Joint probability density function of wind speed, wind angle, ice thickness and capacity
$f(x)$	Probability density function
$f(x, \omega)$	Approximating meta-model
fn	False negatives, examples that are incorrectly assigned to the "survival" class
FORM	First order reliability method
fp	False positives, examples that are incorrectly assigned to the "failure" class
F_β	F-measure which takes into account both sensitivity and precision
g	Ice weight per unit length of conductor (N/m)
g_R	Reference ice load on a 25 mm conductor 10 m above ground
G(x)	Limit state function
G_c	Combined wind factor for conductors
G_i	Combined wind factor for insulator strings
G_L	Span factor
G_t	Combined wind factor for supports
GCM	Global climate model
GDP	Generalized pareto distribution
GEV	Generalized extreme value distribution
h	VC dimension
HS	Heavy suspension type tower
$I_{Bay}(i)$	Bayesian reliability importance of component *i*
$I_{ip}(i)$	Improvement potential importance of component *i*

IEEE	Institute of electrical and electronics engineers
$i.i.d$	Independent and identically distributed
IPCC	Intergovernmental panel on climate change
J_{best}	The best column in the probability matrix with the highest quality
K_d	Diameter factor
K_h	Height factor
K_R	Roughness factor
K(x,x')	Kernel function
L	Wind span (m)
LHS	Latin hypercube sampling
LS	Light suspension type tower
LSSVR	Least square support vector machine for regression
MC	Monte Carlo
MCS	Monte Carlo simulation
MIS	Method of independent storms
MLP	Multi-layer perceptrons
MPP	Most probable point/ Design point
MSE	Mean square error
MST	Total mean square of variation in observations
n	Number of training samples
N	Total number of simulations
N_f	Number of events for which structural demand exceeds structural capacity
n_{te}	Number of observed demands in the test subset
p	Number of parameters in the regression model
p_f	Failure probability

p_j	Precipitation rate (mm in the jth hour)
PDF	Probability density function
POT	Peak-over-threshold
PSVM	Probabilistic support vector machine
q_0	Dynamic reference wind pressure (N/m^2)
R^2	Coefficient of multiple determination
$\bar{R}^2$	Adjusted coefficient of multiple determination
R_p^2	Prediction coefficient of multiple determination
R_{eq}	Equivalent radial accreted ice thickness (mm) using the Jones simple model
RBF	Radial basis function
RSM	Response surface method
S_i	Area of the insulator string projected horizontally on a vertical plane parallel to the axis of the string (m^2)
S_{t1}	Total surface area projected normally on face 1 of the panel (m^2) which is parallel to the line direction
S_{t2}	Total surface area projected normally on face 2 of the panel (m^2) which is perpendicular to the line direction
SFE	Stochastic finite elements
SLT	Statistical learning theory
SORM	Second order reliability method
SRM	Structural risk minimization
STR	Dead end/angle tower
SVM	Support vector machine
SVR	Support vector regression
t	uniform radial ice thickness (mm)

T	return period
tn	True negatives, correctly classified "survival" examples
tp	True positives, correctly classified " failure" examples
u^*	Unit normal at the most probable point directed towards the failure domain
V_R	Reference wind speed (m/s)
V_{RB}	Reference wind speed (m/s) for terrain category B corresponding to return period *T*
VC	Vapnik-Chervonenkis
VP	Venn predictor
W	Weight vector
WMO	World meteorological organization
X	Vector of basic random variables
$\mathbf{X}_c$	Vector of random variables related to the capacity
$\mathbf{X}_d$	Vector of random variables related to the demand
y_{ite}	Observed demands in the test subset
$\hat{y}_{itr}$	Predictions of the demands in the test subset using the fitted model to the training subset
$\bar{y}_{te}$	Mean of the observed demands in the test subset
β	Reliability index
γ	Kernel parameter
δ	Ice density (kg/m^3)
θ	Angle of incidence of the wind direction with the perpendicular to line direction in a horizontal plane
μ	Mean
μ_a	Air mass per unit volume

ξ_i	Slack variable
ρ_0	Density of water (1 gr/cm^3)
ρ_i	Density of glaze ice (0.9 gr/cm^3)
σ	Standard deviation
τ	Air density correction factor
φ_1	Wire direction
φ_2	Wind direction
χ	Solidity ratio of tower panels
ω_j	Liquid water content (gr/m^3) of the rain-filled air in the jth hour
$\Phi(x)$	Feature function that maps data from input space into a higher dimensional feature space
$\Omega(x)$	Failure domain

CHAPTER 1 INTRODUCTION

1.1. Overview

The backbone of modern societies is formed by different infrastructure systems (lifelines) such as electric power networks, telecommunications, municipal water systems and gas distribution networks which provide essential services that are vital to the functioning of societies and economies. Proper functionality of infrastructure systems are threatened by natural hazards among other types of disruptions. Due to the interdependency of different infrastructure systems, high economical losses can be imposed on societies once one system is damaged (Adachi and Ellingwood 2008, Leelardcharoen 2011). Therefore, it is essential to study the structural behaviour and vulnerability of each infrastructure system. In an evaluation of lifeline behaviour for the 1995 Kobe earthquake, Nojima and Sugito (2002) reported that electric power delivery systems are usually restored much more quickly than other lifelines. Because so many other lifelines depend on power delivery to function, and the rapid restoration of electric power following a hazardous event is critical to the rapid recovery of the affected communities.

Overhead transmission and distribution lines are the cornerstone of electric power networks. According to a recent technical brochure by CIGRE (Brennan 2004), most developed countries with major electrical infrastructure will face three coinciding critical issues within the next decade. 1) the age of most of the electrical infrastructure is over 50 years. 2) the design life of much of the electrical infrastructure is in many cases about 50 years and has matured beyond the engineering serviceability or economic life and requires some form of life extension. 3) the need to increase the capacity of the existing electrical infrastructure places extraordinary demands on

utilities to establish strategies to uprate existing electrical infrastructure. In addition, our climate is changing, which can result in higher climatic loads in certain regions around the world (Cheng et al. 2007, 2011, 2012, 2014, Eichelberger 2008, Zamuda et al. 2013), and can consequently affect the performance of overhead transmission and distribution lines (Boyle et al. 2013, Kezunovic et al. 2008, Rezaei et al. 2016, Schaeffer et al. 2012).

The economic impact of transmission line failure can be significant as shown by several recent extreme climatic events. During the ice storm of January 1998 which hit Ontario, Quebec, the Maritimes, New York and New England, 31% of the distribution lines of Ontario hydro were damaged in addition to 130 transmission towers and 2100 transformers which were destroyed. Hydro-Québec also lost 600 transmission towers, 4000 transformers, over 3000 km of high voltage transmission lines, over 16000 wooden transmission poles and over 2000 additional supporting structures. The estimated costs of damages were 140$CDN million for Ontario Hydro, and 800$CDN million for Hydro-Québec (Mirza 2008, Peters et al. 2006). In December 1999, France was hit by two wind storms. Gusts reached 200 km/h causing failure in 35 extra high voltage lines, 180 high voltage lines and more than 100 high voltage/ medium voltage substations. The cost of the failure to Electricitie de France (EDF) was 1.1 billion Euros (Le Du et al. 2002, Sundell et al. 2006). In addition, in November 2004, Poland was affected by a heavy snow storm that caused serious damages to both national and international transmission lines. The costs resulting from these events were estimated at around €20 M (Sundell et al. 2006). In January 2008, power networks of 19 provinces in China suffered from substantial damages due to an extreme ice storm. The number of collapsed towers was respectively, 295 and 586 for 500 kV and 220 kV transmission lines. Lower power networks were also damaged resulting in a total economic loss of 53.7 billion RMB (Jiang et al. 2009, Lu et al. 2008). These are only a few

examples of extreme climatic events that caused electric transmission and distribution line failures, which highlights the importance of understanding and predicting the structural behaviour of these complex structural systems.

1.2. Problem description and motivation

Until recently, the deterministic approach largely dominated structural engineering design. This was reflected by a generation of deterministic design codes, aiming to comply, as a minimum, with the safety or overload factors required for each loading case. Experience and technical studies have shown that deterministic methods have inherent deficiencies that can only be resolved by the reliability based design methods (Ghannoum et. al. 2006). Deterministic methods cannot efficiently cope with the variations in environmental loads applied to structures and result in difficulties to adjust design to local environmental conditions. Besides, it is very difficult to attribute a reliability level to any design based on deterministic principles, except by a general inference from very long experience. Also, deterministic designs often result in inconsistencies and unbalance in strength of components because there are no requirements for a preferred sequence of failure. Another limitation of deterministic methods is that it is difficult to adjust the overall reliability of complex and heterogeneous structural systems such as overhead transmission lines using these methods. And finally, deterministic methods are unable to integrate data on the quality of material in the design process because they express strength by only one value (namely nominal, characteristic or guaranteed strength). All these shortcomings call for the application of probabilistic methods in the design and analysis of many structures including transmission line systems. The probabilistic methods address the limitations of deterministic methods by recognizing loads and strengths as random variables represented by

appropriate probability distribution functions instead of single nominal values, by specifying limit states for each structural component instead of incorporating safety and/or overload factors, and by selecting the design loads based on a required level of reliability. In other words, the application of probabilistic methods and of reliability theory offers a methodology to systematically study the probability of non-performance of various structures considering the uncertainties associated with all the variables involved in the performance criteria or definition of limit states. During the last decades, many organizations such as the International Electrical Commission (IEC), the Institute of Electrical and Electronics Engineers (IEEE) (Ghannoum 1983.a, 1983.b, 1984), the American Society of Civil Engineers (ASCE) and the Conseil International des Grands Reseaux Electriques (CIGRE) study committees (Ghannoum et. al. 2006) introduced and applied the reliability/probability based design concepts to the transmission lines design. As a result, many reliability based guidelines such as the North-American Standard ASCE 74 (2010) , the European Standard EN 50341 (2001), CEI/IEC 60826 (2003), the Canadian standard CAN/CSA-C22.3 (2006), and the reports for the reliability based design of transmission line foundations provided by the Electric Power Research Institute (EPRI) (Kulhawy et. al. 2006) were developed. The development of probabilistic codes is not the only application of structural reliability analysis. A probabilistic framework for determining the failure probability of transmission lines, not only helps engineers achieve an optimized design through the application of reliability-based optimal design procedures (Enevoldsen and Sørensen 1994, Lee et al. 2002, Liu and Moses 1992, Royset et al. 2001a, Royset et al. 2001b, Royset 2002), but also has the potential to serve as a practical decision tool for utility owners to assess the risk, reliability or damage state of their lines (Dagher et al. 1991, Dagher and Lu 1993, Foschi 2004, Li and Bhuyan 2006, Van Der Wal et al. 2004), optimize their inspection and

maintenance regimes (Hathout 2006, Goodfellow 2000, Sherif and Smith 1981, Sommer 1993, Straub & Faber 2005, Onoufriou and Frangopol 2002) and estimate the probabilistic life-cycle performance of existing transmission lines (Frangopol and Maute 2003, Hathout 2006). In addition, a structural reliability analysis methodology is helpful for uprating, upgrading, refurbishment or life extension of existing transmission lines to adapt to higher electricity demand due to population increase or to higher climatic loads as a result of climate change (Bjarnadottir et al. 2014, Brennan 2004, Haldar 2006, Peters et al. 2006, Madsen 2014, Rezaei et al. 2016). Finally, reliability models are also helpful for evaluating icing hazards mitigation strategies (Hou et al. 2013, Hou et al. 2015, Liu and Peters 1999). Although the idea of applying probabilistic methods and reliability theory to design, condition assessment and maintenance scheduling of structures is already well developed and applied to single type structures such as bridges (Choi et al. 2004, Kong and Frangopol 2003, Liu and Frangopol 2006, Mackie and Stojadinović 2005, Markow et al. 1993, Robelin and Madanat 2008, Stewart 1998), dams (Tekie and Ellingwood 2003), industrial buildings (Lee et al. 2013) and telecommunication towers (EIFashny, 2005), the complex nature of transmission line systems has made these structures less attractive to probability researchers and experts. Very few studies are available on the reliability analysis of transmission or distribution lines which are mainly focused on specific components of the line (towers or poles) neglecting the nonlinearities involved, the interaction between line components and their failure correlation. Foschi (2004) applied the reliability concepts to calculate the reliability of lattice transmission towers from a linear analysis of the structural response and of wood poles under wind and ice loads. He also suggested that the probabilistic concepts can be used to assess the existing structures for continuing use under a change in load regime. Li et al (2006) calculated the reliability of wood poles at 15 locations across Canada

using the first order and second order reliability methods and actual climatic data of the region. Shafieezadeh et al. (2014) developed age-dependent fragility curves of utility wood poles under wind induced loads. Dagher et al (1991) and Dagher and Lu (1993) proposed an approximate relationship between the structural system reliability of a transmission/distribution line and that of a fully utilized tower/pole within the line neglecting the contribution of other line components. Dagher and Lu (1993) also investigated the effect of neglecting tower redundancies in determining the system failure probability of lattice suspension towers and reported that the failure probability of towers when assumed as series systems are only slightly higher than when redundancies are taken into account which verifies the widely used assumption that many latticed towers can be approximately modelled as series systems. Haldar (2006) proposed upgrading strategies for a transmission line based on the reliability of the most critical members of suspension and dead-end towers and the reliability of conductors under vertical ice loads. Xie et al (2012) developed seismic fragility curves of transmission towers using nonlinear buckling analysis and nonlinear dynamic analysis and Monte Carlo simulation.

Considering that transmission line systems consist of hundreds of members with implicit limit state functions, this makes it unfeasible or even impossible to apply many of the current reliability based methods to transmission lines (further discussed in chapter 2). Consequently, a rationally structured probabilistic framework must be developed that allows for the quantification of different uncertainty types in order to assist transmission line owners in making optimal decisions regarding their assets.

1.3. Research objectives and scope

The main objective of this study is to develop component-based and system-based probabilistic frameworks that can be used for fragility assessment and reliability analysis of electric transmission line systems under multiple climatic hazards, using the concepts of reliability theory and machine learning methods. Although the methodology is applied to a nine span segment of a typical 120 kV single circuit transmission line located in north America and subjected to extreme wind and ice storms, it can easily be modified and used for other natural and/or man-made external loads such as earthquakes, impact, etc. In this study, various concepts and probabilistic tools are used to provide a better understanding of the structural behaviour of transmission lines and their components considering the significant uncertainties involved.

- Component and system fragility curves are developed to determine the conditional probability of failure for given intensity measures of wind speed and ice thickness. These fragility curves are particularly useful as they are independent of the location of the line.
- The multi-hazard fragility surface and the wind and ice hazards at a specific site are combined to estimate component and system annual failure probabilities and reliability indices. The estimated system failure probability is then compared with the target reliability proposed by current reliability-based transmission line design guidelines.
- Various importance measures are used to determine the relative importance of different components of the system and their contribution to the overall system failure.
- The design point which is defined as the combination of climatic variables that results in the failure of system and has the highest likelihood is determined for the studied transmission line.

This study also aims to show how the proposed methodology can be used by researchers and engineers to solve some of the current challenges.

- To improve current design approaches that assume extreme winds are applied perpendicularly to the conductors, this study considers the direction of the wind as a probabilistic variable in studying the performance of structural components and fragility curves are presented as a function of wind direction as well as wind speed and ice thickness.
- Component and system fragility curves are also developed for different scenarios of unbalanced ice loading that can result from non-uniform ice formation and/or ice shedding. In this application, the effect of wind direction on non-uniform ice formation and the resulting fragility curves is taken into account.
- The impact of climate change on the performance and reliability of transmission lines is investigated by performing a sensitivity analysis of the system reliability index under a range of assumed changes in the mean and standard deviation of climatic variables such as annual extreme wind speed and ice thickness.

1.4. Assumptions and limitations

It is noted that this study is focused on the fragility assessment and reliability analysis of transmission lines as a system including components such as suspension towers, angle/dead-end towers, ground wires, conductors and insulators. However, the failure probability of foundations is not addressed in this study and may be considered in future studies when experimental data becomes available.

As mentioned in section 1.3, the climatic hazards of interest are extreme wind and ice storms. Although during extreme ice storms trees falling or leaning on wires cause problems in electric power distribution systems, failure due to falling trees is less common in the electric power transmission system because the wires are usually on taller poles in right-of-way clear-cuts. As a result, electric transmission line failures are more likely due to structure overload caused by severe ice and wind-on-ice loads on the wires, conductors and towers. Therefore, this study does not address falling trees as a failure scenario during extreme ice storms.

Galloping is another cause of line outages due to phase to phase contacts and various types of mechanical damage to the conductors, hardware and supporting structures during winter storms (Havard 2002). As first explained by Den Hartog (1932), galloping of conductors causes large dynamic loads at low ice loads and moderate, steady winds. Negative lift on the airfoil shape that is formed by a thin layer of ice accreting on one side of a conductor or guy leads to a positive-feedback, low-frequency, high-amplitude oscillation that can tear the attached structure apart. At high ice loads, sustained galloping is less likely because of the weight of the ice-covered conductor and the additional damping from the accreted ice (Farzaneh 2008). Ice shedding may have similar consequences to galloping. It may provoke flashovers and might be responsible for tower arm failures or even a cascade failure of several towers in a transmission line (Jamaleddine et al. 1993, Mirshafiei 2010, Roberge 2006, Roshan Fekr et al. 1998). It is noted that this study does not address failure scenarios due to dynamic loads imposed by galloping or ice shedding.

The three dimensional model of the studied line developed in the computer software SAP2000 accounts for the nonlinear behaviour of the wire system and large deformations using nonlinear static analyses. Since the wire system is not modeled separately from the towers, the interaction between transmission line components are also taken into account. As proposed by current

design guidelines (CAN/CSA-C22.3 2006, CEI/IEC 60826 2003, ASCE74 2010), the transmission line model is loaded with equivalent static wind loads and equivalent radial ice thickness resulting from freezing rain. Investigating the reliability of transmission lines under other types of atmospheric icing such as incloud icing and wet snow is an ongoing challenge for researchers due to insufficient field data and is beyond the scope of this study.

In this study, wind speed, ice thickness, wind direction and structural component capacities are considered as basic random variables. The type and parameters of the probability distribution functions for each random variable is selected according to suggestions from current reliability-based design guidelines (CAN/CSA-C22.3, 2006, CEI/IEC 60826, 2003). Due to the lack of data, other sources of uncertainty such as statistical and model uncertainties are not addressed in this study (further discussed in chapter 2). However, these uncertainties can be easily considered with the proposed methodology and could be included as more information becomes available for future studies.

From a probabilistic point of view, the lattice overhead towers and the entire line are represented as a series system in this study. This assumption implies that the system fails once anyone of its members fails. It is noted that, this assumption is conservative for some overhead towers based on their configuration and design, because the contributions of tower redundancy and robustness to structural safety is neglected. However, for most suspension towers in this study, modeling the tower as a series system seems a reasonable assumption due to their weak redundancy. If the system is not deemed to be a series system, then failure of one component may not trigger the failure of the system. In such a case, probabilistic progressive collapse analysis can be applied which substantially increases the required computational effort and is beyond the scope of this

research (Biondini et al. 2008, Ellingwood 2006, Ellingwood 2009, Frangopol and Curley 1987, Frangopol and Nakib 1991, Miao 2014).

1.5. Dissertation outline

The eight chapters and two appendices of this Dissertation describe the work done to achieve the objectives of this research study. Each of the chapters deals with separate but inherently integrated tasks. Chapters 4-7 of this thesis are four separate journal papers which are either published or under the review process. The outline of this dissertation is as follows:

Chapter 1 presents the overview, the problem statement, the research objectives and scope and the assumptions and limitations of this study.

Chapter 2 presents a detailed review of the current state of the art on the reliability analysis and fragility assessment methods of structural systems. A brief review of some of the concepts and mathematical tools which are used in this dissertation and can help researchers and engineers better understand the behaviour of structural components and systems in a probabilistic context are also presented in this chapter.

Chapter 3 explains how weather related loads such as wind loads, ice loads and combined wind and ice loads are applied on different transmission line components. In addition, it provides a detailed review on available probabilistic climatic load models for overhead electric transmission lines.

Chapter 4 describes the novel application of Statistical Learning Theory (SLT) and VC generalization bounds to component-based structural reliability analysis of transmission line systems considering the uncertainties of climatic variables such as wind speed, ice thickness and

wind angle, and of the resistance of structural elements. It summarizes how the proposed STL-based structural reliability analysis approach can overcome the limitations of traditional methods for reliability assessment of complex structural systems with implicit limit state functions. The outline of a typical 9-span transmission line segment located in North America which is used as a case study in this dissertation, and the description of overhead lattice towers within the line are also presented in this chapter. It is shown that the prediction ability of the component-based approach is investigated by calculating the adjusted coefficient of multiple determination, $\bar{R}^2$, and the prediction coefficient of multiple determination, $\bar{R}_p^2$, using a 10-fold cross-validation technique. Component and system failure probabilities and reliability indices are estimated for the studied line. In order to show the effect of failure correlation between different elements of towers or different components of the line, the calculated failure probabilities are compared with first and second order Ditlevson bounds. In order to quantify the relative importance of components and their contribution to system failure, and provide the engineers and utility owners with practical decision tools, this chapter also presents two checklists based on the improvement potential importance and the Bayesian reliability importance measures. These checklists are useful for upgrading the line with limited resources and for fault diagnosis purposed, respectively. Finally, the design point, which is the most likely climatic scenario that results in the failure of the line, is determined and compared to suggested design combinations from current design guidelines.

Chapter 5 describes how the component-based approach presented in chapter 4 can be used to develop component and system fragility curves for the studied line. Fragility curves are practical probabilistic tools that improve our understandings of the behaviour of transmission line systems and their components under uncertainties. This chapter also presents the novel application of

probabilistic support vector machine (PSVM) classification techniques to system-based fragility assessment and reliability analysis of transmission lines systems. Using the system based-approach, the limit state surface of the whole transmission line is determined and further used to predict the survival or damage state of the line for new climatic conditions with a probability interval that this prediction is correct. The prediction ability of the system-based approach is thoroughly investigated by estimating the prediction error, sensitivity, specificity, precision and F-measure. The advantages and disadvantages of both component-based and system-based methods are compared and suggestions are made on the appropriate application of each method.

Chapters 6 and 7 present two practical applications of the proposed probabilistic framework. Chapter 6 presents component and system fragility curves of the studied transmission line subjected to unbalanced ice loads resulting from non-uniform ice formation or ice shedding. The results are presented in the form of fragility curves for 3 different unbalanced loading scenarios of longitudinal, transverse and torsional loadings. This chapter also studies the effect that line exposure to wind has on non-uniform ice accumulation. And, the results are compared to fragility curves developed assuming that ice accretes uniformly on all conductors regardless of their exposure to wind. Finally, the effect of various design parameters such as wind speed and direction, and ice loading proportion, on the developed fragility curves, are also investigated.

Chapter 7 quantifies the impact of climate change on the reliability of transmission lines by investigating the sensitivity of the reliability of the studied transmission line to different climate change scenarios using the developed component-based approach. The effects of wind direction and exposure on ice accumulation are included in the model. The results are presented in the form of design specifications as a function of a range of plausible climate change scenarios. The information presented in this chapter is useful to develop new policies, climate adaptation

strategies and risk mitigation measures that are cost-effective for a wide range of likely climate change scenarios.

Chapter 8 summarizes the conclusions of this dissertation, presents its original contributions and offers recommendations for future research work on this subject.

Appendix A provides additional technical information about equations used to determine climatic loads on transmission line components.

Appendix B provides technical information about the statistical distribution functions and their application to climatic loads and strength of overhead lines.

Appendix C provides the flowcharts for both component-based and system-based approaches.

CHAPTER 2 REVIEW OF TECHNICAL LITERATURE

2.1 Overview

Structural reliability analysis of electric transmission lines is concerned with the rational treatment of uncertainties in structural engineering design and the associated problems of rational decision making. The randomness in the occurrence of climatic events in time and space, the vast uncertainty in predicting the response of structural elements and our inability to accurately assess capacities of structural components, all compel us to make use of probabilistic methods in order to consistently account for the underlying uncertainties and make quantitative assessment of structural reliability.

The general formulation for time invariant structural reliability problems is the probability integral presented in equation (2-1).

$$p = \int_{\Omega(\mathbf{X})} f_{\boldsymbol{X}}(\mathbf{X}) d\mathbf{X} \tag{2-1}$$

where p is the failure probability, $f_{\mathbf{X}}(\mathbf{X})$ is the joint probability density function (PDF) of the vector of basic random variables $\mathbf{X} = [X_1, X_2, \cdots, X_n]^T$ which represent uncertain quantities such as loads, material property constants and geometric dimensions, and $\Omega(\mathbf{X})$ is the failure domain of the structure which is generally defined in the form

$$\Omega(\boldsymbol{X}) = \left\{ x \middle| \cup_k \cap_{i \in C_k} \{g_i(\mathbf{x}) \leq 0\} \right\} \tag{2-2}$$

where $g_i(\mathbf{x})$, i=1,…,m, are a set of limit-state functions formulated so that $\{g_i(\mathbf{x}) \leq 0\}$ defines the failure of component i and $\{g_i(\mathbf{x}) > 0\}$ defines its survival, m denotes the number of

components, and C_k is the index set for the k-th minimum cut set, where each minimum cut set represents a minimal set of components whose joint failure constitutes failure of the structural system. This description of the failure domain defines a system reliability problem. Series and parallel systems are special cases. Series systems refer to cases where failure of any component constitutes failure of the system and each component is a cut set. On the other hand, when all components collectively form a single minimum cut set, the structure constitutes a parallel system. It is noted that, when $\Omega(\boldsymbol{X})$ is described through a single function, a component reliability problem is at hand. Figures 2-1 and 2-2 present graphical illustrations of component and system reliability problems, respectively.

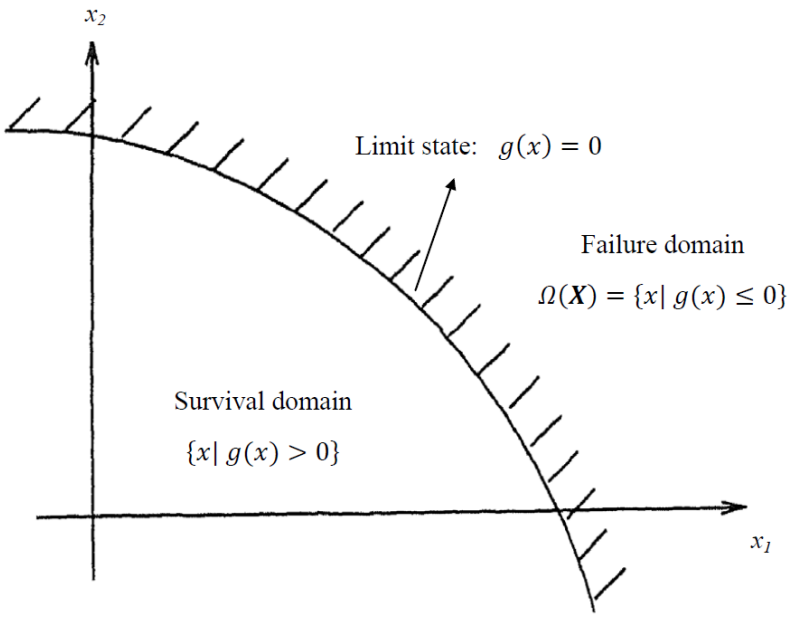


Figure 2-1. Component reliability problem

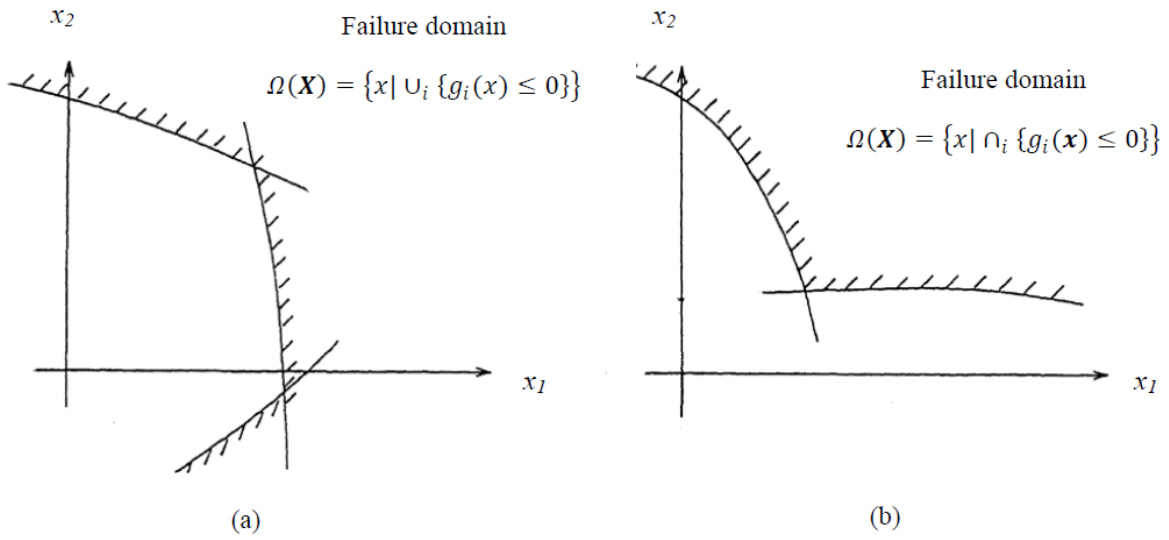


Figure 2-2. System reliability problem (a) Series system (b) Parallel system

A related measure of interest is the generalized reliability index defined by

$$\beta = \Phi^{-1}(1 - p_f) \tag{2-3}$$

where Φ^{-1} denotes the inverse of the standard normal cumulative probability. Although the concept and formulation presented in this section seems quite straight forward and easy, in reality especially for complex structural systems such as transmission lines, the calculation of equation (2-1) is far from being simple, because neither the joint probability density function of random variables $f_{\mathbf{X}}(\mathbf{X})$ nor the component or system limit-state functions $g_i(\mathbf{x})$ are precisely known. Instead, idealized models of these functions must be developed, typically through statistical inference using real-world observations or by using the suggestions of current guidelines. A great deal of effort has been devoted to develop efficient algorithms for computing the probability integral of equation (2-1). These methods differ from each other in their efficiency and degree of approximation. A review of the most widely used methods is presented in this chapter. First, the different sources of uncertainty involved in the reliability assessment of structures are reviewed.

2.2 Types of uncertainty

For the purpose of structural reliability analysis it is required to distinguish between at least three types of uncertainty. 1) physical uncertainty, 2) statistical uncertainty and 3) model uncertainty (Der Kiureghian 1996, Thoft-Christensen 2012, Thoft-Cristensen and Baker 2012).

Physical uncertainty is the inherent randomness that arises from intrinsic variability in environmental effects such as wind and ice loads and in the relevant material properties that govern structural capacities. This variability can be described in terms of probability distributions. Physical variability can be quantified only by examining sample data. However, sample sizes are usually limited by practical and economic considerations which gives rise to so-called statistical uncertainty.

Statistical uncertainty arises when estimating parameters of probability distributions from limited sample data, and unlike physical uncertainty, is only due to lack of information. Therefore, the distribution parameters may themselves be considered as random variables with uncertainty that is dependent on the amount of data available and any prior knowledge.

Structural analysis and design apply mathematical models to relate desired output quantities to the values of a set of input quantities or basic variables. Model uncertainty arises from the imperfection of the mathematical models used to describe complex physical phenomena, such as models describing loads and capacities of structures. In other words, the response of typical structures and structural elements contains a component of uncertainty in addition to those components arising from uncertainties in the values of the basic loading and strength variable.

The uncertainty due to inherent randomness is irreducible and generally characterized as aleatory uncertainty (Der Kiureghian and Ditlevsen 2009). On the other hand, statistical uncertainties can be reduced by gathering more information. Model uncertainties can be quantified either by comparisons with other models that exhibit a closer representation of nature or by comparisons with collected data from the field or the laboratory and can be reduced by using more refined models. Therefore, statistical and model uncertainties are generally categorized as epistemic uncertainties (Der Kiureghian and Ditlevsen 2009, Thoft-Christensen 2012). As mentioned in section 1.4, due to lack of data, this study does not address statistical and model uncertainties and the interested reader is referred to available literature on the subject including but not limited to research done by Der Kiureghian (1989, 2008), Der Kiureghian and Liu (1986), Ditlevsen (1981, 1983) and Kaminski et al. (2008).

2.3 Structural reliability methods

2.3.1 Simulation based methods

Monte Carlo simulation (MCS) is the basic and universal approach to solving the reliability problem (Rubinstein and Kroese, 2008). For this approach, N independent samples of the vector of random variables **X** are generated from the joint probability density function $f_{\mathbf{X}}(\mathbf{X})$. For each sample x_1, the limit state function $g_x(x_1)$, is computed and if $g(x_1) < 0$, then it is counted as a failure event. The ratio of the number of failure events to the total number of samples N is then used as the estimate of failure probability (p_f). As the number of samples is increased, the estimate of the probability of failure approaches the true value. The advantage of the MCS approach comes from the fact that it allows to realistically take into account many phenomena that can occur without additional complications in the modelling and in the solution procedure.

However, according to Sudret (2012), the slow convergence rate of this method makes its application inefficient for rare events since a typical evaluation of a probability of failure of the order of magnitude 10^{-r} with less than 10% coefficient of variation for failure probability requires about 10^{r+2} simulations, and the computational cost of each evaluation of the limit state function is not negligible when finite element analysis is involved. In order to improve the convergence of MCS, other simulation based methods are also proposed by researchers, including the directional simulation (Bjerager 1988, Ditlevsen et al. 1987), importance sampling (Au and Beck 1999, Hohenbichler and Rackwitz 1988, Maes et al. 1993, Melchers 1990), subset simulation (Au and Beck 2001, Hsu and Ching 2010, Katafygiotis and Cheung 2005) and line sampling (Pradlwarter et al. 2007). A few examples of the application of MCS and its derivative simulation methods to structural and system reliability analyses are presented by Biondini (2008), Laumakis and Harlow (2002), Lu et al. (1994), Zio (2013) and Zio and Pedroni (2010).

As already mentioned in this section, simulation based methods rely upon repeated calls to the underlying computational model of the structure or system. The large number of calls that is required for precise predictions is usually not compatible with costly computational models such as finite element models, even when high-performance computing platforms are at hand. This has promoted the use of approximate methods to reduce the computational cost by introducing some approximation in the reliability estimation through first or second order reliability methods, or by using the meta modelling techniques that aim to find a surrogate model that can mimic the behaviour of the computational model while being inexpensive to evaluate.

2.3.2 First and second order reliability methods

First order reliability methods (FORM) and second order reliability methods (SORM) are based on linear and quadratic approximations of the limit state functions at the most probable point (MPP) or design point, *u**, in the transformed random variable space. In the standard normal space, the optimal point on the limit-state surface for fitting the approximating surface is the point of minimum distance to the origin, because the probability density decays exponentially with distance from the origin (Liu and Der Kiureghian, 1991). The two main steps in FORM and SORM include: 1) the transformation of basic random variables to the standard normal space and 2) searching for the MPP. The transformation to the standard normal space can be performed using various methods including the Rackwitz and Fiessler method (Rackwitz and Fiessler 1976), the Chen and Lind Method (Chen and Lind 1982) or the Rosenblatt transformation (Rosenblatt, 1952). In addition, Liu and Der Kiureghian (1991) present various algorithms to perform the MPP searches. Figure 2-3 is a graphical presentation of the concept behind FORM and SORM.

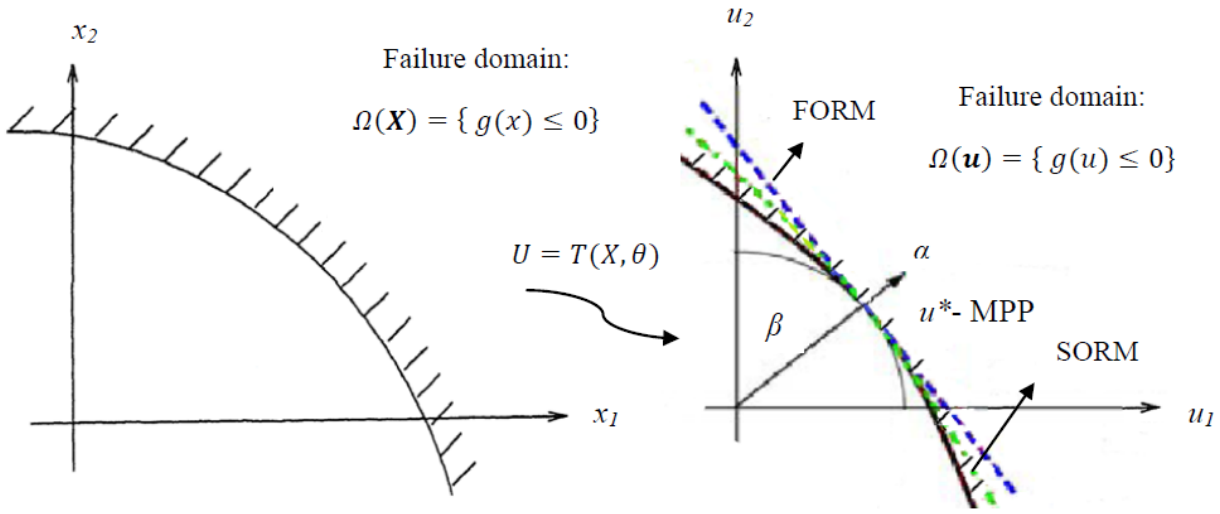


Figure 2-3. Illustration of original to standard space transformation and MPP, FORM reliability index and FORM and SORM approximations in standard space

The first order approximation of the failure probability is given by

$$p_f = \Phi(-\beta) \quad , \ \beta = \alpha u^* \tag{2-4}$$

The first order reliability index, β, is the distance from the origin in the standard normal space and u^* is the unit normal at the MMP directed towards the failure domain. The second order approximation of the failure probability is given in terms of β and the principal curvatures of the paraboloid (Breitung 1984). Various methods for selecting the approximating failure surface and calculating the principal curvatures in the second order reliability method are available (Der Kiureghian et al. 1987, Der Kiureghian and Stefano 1991, Fiessler et al. 1979, Geyskens et al. 1991). Examples of the application of FORM/SORM to reliability analysis of simple structural components or systems can be found in Du (2009), Madsen (1985) and Maymon (1993) among others.

The main advantage of FORM/ SORM compared to simulation methods is that the probability integral (equation 2-1) is estimated with a relatively small number of calculations of the limit state function and its gradient which is independent of the magnitude of the failure probability. The additional advantage of these methods is that they readily provide the coordinates of the most probable failure point, the shape of the failure surface in the neighbourhood of the MMP and the sensitivities of the failure probability (Der Kiureghian 1996). The need for evaluating the limit state function and its gradients in FORM/ SORM makes its application challenging when a closed-form explicit limit state function is not available which is the case in reliability analysis of most complex structural systems. In addition, when the limit state surface in the standard normal space is strongly nonlinear, multiple design points on the surface may occur with significant

neighbourhood densities. This situation further complicates the search for local design points because it is difficult to prevent the convergence of the optimization algorithm to the global design points (Der Kiureghian 1996). These challenges have motivated the researchers to develop more efficient methods such as the response surface method (RSM) or advanced meta modelling techniques for reliability assessment of structural systems. These methods are further elaborated in sections 2.3.3 and 2.3.4.

2.3.3 Response surface method

The principle of the response surface method (RSM), as a meta modelling technique, consists in the substitution of the real implicit limit state function by approximate simple explicit functions only at the neighbourhood of the design points, where their contribution to the total failure probability is more important. The use of quadratic polynomial response surfaces as a surrogate of the limit state function has been pioneered by Faravelli (1989). In the original conceptual form of RSM, quadratic polynomials with or without the cross terms are used to approximate real limit state functions (Bucher and Bourgund 1990, Khuri and Cornell 1996, Wong 1985) (equation 2-5).

$$y = \beta_0 + \sum_{i=1}^{k} \beta_i x_i + \sum_{i=1}^{k} \beta_{ii} x_i^2 + \left(\sum \sum_{i<j} \beta_{ij} x_i x_j\right) + \epsilon \qquad (2\text{-}5)$$

where $\mathbf{x}=[x_1, x_2, ..., x_k]$ is the vector of basic random variables and y represents the response of interest. The key is to adjust correctly the polynomials to the limit state function using the sample points, particularly in the neighbourhood of design points. Several sampling techniques have been proposed for the fitting procedure including the full and partial factorial designs, the central composite designs and the Bucher and Bourgund method among others (Bucher and Bourgund 1990, Hibbert 2012, Myers et al. 2016).

Many scientists have successfully applied the RSM along with simulation methods or FORM/SORM to study the reliability of different structural systems. Bucher and Bourgund (1990) applied the RSM in conjunction with importance sampling for reliability assessment of various structural systems. Rajashekhar et al. (1996) compared the reliabilities of reinforced concrete slabs for various limit states using the response surface approach. Roussouly et al. (2013) applied the response surface method and Latin Hypercube Sampling (LHS) for reliability analysis of truss and frame structures. Towashiraporn (2004) used RSM and LHS technique for developing seismic fragility curves for unreinforced masonry (URM) buildings. Wong et al. (2005) proposed an adaptive experimental design to apply the RSM and FORM/SORM for reliability analysis of reinforced concrete bridges subjected to multiple loading sequences.

Since the response surface method applies the Empirical risk minimization (ERM) inductive principle to fit a quadratic polynomial model to some non-random samples fixed by an experimental plan, it can be regarded as a rigid, non-adaptive regression technique (Hurtado 2004). The rigidity and non-adaptivity of the model can explain the application problems of the RSM reported by Guan and Melchers (2001) which include the high sensitivity of the value of the reliability index and the failure probability to the experimental design plan. In addition, the strict assumptions over the class of approximating functions imposed by the ERM inductive principle can lead to overfitting and poor prediction abilities (large model bias) for the meta model, even when the number of training samples is fairly large (Cherkassky and Mulier 2007). This is due to the fact that under the ERM inductive principle, the optimal surrogate model is determined by minimizing the empirical error, which is the average of squared error for the training data, without checking the model for its prediction abilities.

On the other hand, flexible and adaptive methods based on structural risk minimization (SRM) inductive principle can reduce the model bias by adapting the model complexity to the training samples at hand (Cherkassky and Mulier 2007) and are regarded as a better alternative to RSM (Hurtado 2004). Statistical Learning Theory (SLT), also known as Vapnik-Chervonenkis (VC) theory, (Cherkassky and Mulier 2007, Cherkassky et al. 1999) and Support Vector Machines (SVM) (Vapnik 1995, 1998) are among these methods which are further elaborated in section 2.3.4. In addition, a more detailed description of the concept of machine learning and ERM and SRM inductive principles are presented in section 4.2.

2.3.4 Advanced meta modelling techniques

2.3.4.1 Issue of model complexity control

As discussed in section 2.3.3, in conventional RSM and other parametric methods based on the ERM inductive principle, the model is specified first and then the parameters are estimated from the data. However, in many situations, there is a mismatch between the assumed parametric model and the true physical model which introduces large modelling bias even when the number of samples is fairly large. Flexible methods, however, reduce the bias by adapting the model complexity to the training samples at hand. Hence, the problem consists of both determining the model flexibility or complexity and its parameters. In general, from a large class of approximating functions, we should seek simpler models over complex ones and optimize the trade-off between model complexity and the accuracy of model's description of the training data. In other words, our task is to choose a model of optimal complexity for the given finite data that has the minimum prediction risk. There are several general approaches for estimating the prediction risk from finite data including: 1) the analytical model selection criteria (Akaike 1970, Craven and Wahba 1979, Schwartz 1978, Shibata 1981), 2) the upper bound prediction risk

provided by the statistical learning theory (Cherkassky & Mulier 2007, Cherkassky et al. 1999, Vapnik 1998), and 3) the resampling techniques (Efron 1979, Efron 1983, Efron and Tibshirani 1997, Stone 1974, Xu and Liang 2001).

Cherkassky and Ma (2003), Cherkassky and Mulier (2007), Cherkassky et al. (1999), and Shao et al. (2000) presented empirical comparisons for several analytical model selection methods and concluded that model selection based on SLT is superior to other methods. The main advantage of using the upper bound prediction risk provided by SLT or resampling techniques over the analytical methods for model selection is that they do not depend on assumptions about the statistics of the data or specific properties of approximating functions. Model selection via resampling techniques is based on the estimation of the model using a proportion of the available samples (learning or training set) and the estimation of the prediction risk for this model using the remaining samples (validation or test set). Molinaro et. al (2005) compared different resampling methods and concluded that 10 fold cross-validation generally performs well and is preferred especially for computationally burdensome analyses. Moreover, Rodrıguez et. al (2010) studied the statistical properties such as bias and variance of the k-fold cross-validation and recommended to use a 10-fold cross validation technique for estimating the error because of its small bias. In this study both SLT and 10-fold cross-validation technique are used for selecting the optimal demand function or limit state function in the proposed component-based methodology (chapter 4) and system-based methodology (chapter 5), respectively. In addition to model selection, in this study, resampling techniques are also used to provide an unbiased estimate of the prediction error for the purpose of model validation. It is noted that, when resampling technique is applied for both model selection and model validation, a double resampling technique (Friedman 1994) should be used which is further elaborated in chapter 5.

2.3.4.2 Statistical learning theory

The Statistical Learning Theory (SLT), also known as Vapnik– Chervonenkis (VC) theory (Vapnik and Chervonenkis 1968) is the best currently available theory for flexible statistical estimation with finite samples (Cherkassky and Mulier 2007). Vapnik and Chervonenkis (1968) introduced a new measure of model complexity, called the VC dimension (h), which is the maximum number of samples for which all possible binary labelings can be induced without error by a set of indicator functions. In the case of linear real-valued functions, the VC dimension equals the number of free parameters.

In addition, as mentioned in section 2.3.4.1, VC theory provides analytical generalization bounds for model selection, which relate unknown prediction risk and known quantities such as the number of training samples, empirical error and the VC-dimension. These generalization bounds are particularly important under the Structural Risk Minimization (SRM) inductive principle as they provide quantitative characterizations of the trade-off between the complexity of the approximating function and the quality of fitting the training data. The suggested upper bound on prediction risk for the regression problem and further explanations are presented in section 4.2 of chapter 4 and repetition is avoided in this section.

The inductive principle called SRM, which has been originally proposed and applied for classification problems (Vapnik and Chervonenkis 1979), provides a formal mechanism for choosing an optimal model complexity for finite samples. There are two implementations of SRM inductive principle. Under the first implementation of SRM, the model complexity (VC dimension) is kept fixed and the empirical error is minimized. For this reason, the set S of approximating functions $f(\mathbf{x},\omega)$ includes nested elements S_k ordered according to their VC dimension. For a given training data, SRM selects the function $f(\mathbf{x},\omega^*)$ which minimizes the

empirical risk for the functions from element S_k,. Then, the prediction error is calculated for each element S_k using the VC generalization bound. The optimal structure element S_{opt} is the one providing minimum prediction error. This structure element has the optimal complexity (VC dimension). A graphical representation of the first implementation of SRM inductive principle is presented in section 4.2. Under the second implementation of SRM, the empirical error is kept constant (small) and the VC dimension is minimized. The later implementation of SRM forms the basis of Support Vector Machines and margin-based loss functions.

2.3.4.3 Linear binary SVM classification for separable and non-separable data

Support vector machine, proposed by Vapnik (1995, 1998), provides a meaningful characterization of the models complexity independent of the problems dimensionality via the number of support vectors. The SVM approach combines several fundamental ideas including: margin-based loss function, kernel representation, and duality. The concept of margin-based loss function and duality are represented in this section, while the kernel representation is discussed in section 2.3.4.4. Although SVMs can be used for both classification and regression problems, this study is only focused on SVM classification and the reader is referred to Cherkassky and Mulier (2007) for more details on Support Vector Regression (SVR).

Given a set of training data $(x_1,y_1),(x_2,y_2),...,(x_n,y_n)$ where $x_i \in R^n$ is the training data and $y_i \in \{+1,-1\}$ is the class label for x_i, SVM aims to find the optimal hyper-plane that separates one class (+1) from the other (-1) correctly and can be represented as

$$f(x,w) = w.x + b = 0 \quad (2\text{-}6)$$

Where (.) is the inner product of two vectors, w is the weight vector, and b is the bias. For this reason, a margin-based loss function, as shown in figure 2-4 is implemented by SVM such that

most of the training data can be separated perfectly (with zero empirical loss) and the rest of the training data falls within the margin and can only be explained with some uncertainty (non-zero loss). In other words, the introduced margin-based loss function can achieve a balance between two conflicting goals of a good predictive model (Cherkassky and Mulier, 2007):

1. Minimizing the total empirical loss for the samples that lie within the margin.

2. Achieving maximum separation (margin) between training samples that are perfectly classified by the model.

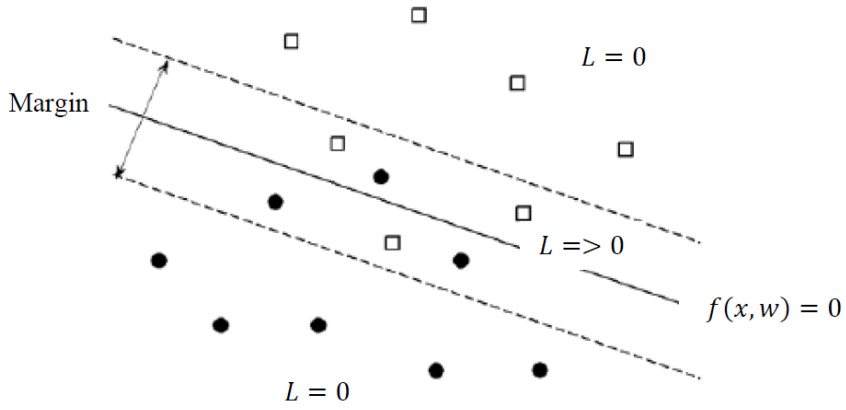


Figure 2-4. Margin based loss function for SVM classification

The training set is linearly separable when a linear function matches the sign for the class of all training samples. In this case, the first goal of the predictive model is already achieved and the optimal separating hyper-plane is the one which maximizes the distance between the hyper-plane and the nearest sample points of each class. It is noted that, the size of the margin plays an important role in controlling the model complexity, in that models with smaller margins have more flexibility (higher VC dimension) than models with larger margins. The data points that are on the edge of the margin and therefore define the margin are called the support vectors. The optimization problem for linearly separable samples is presented in equation 2-7.

$$Min(\frac{1}{2}\mathrm{w}^T.\mathrm{w}) \qquad \text{subjected to} \quad y_i(\mathrm{w}^T.\mathrm{x}_i + b) \geq 1 \tag{2-7}$$

where w follows the general form of equation 2-8 in which α_i are the parameters of the hyper-plane.

$$w = \sum_{i=1}^{n} \alpha_i y_i \mathrm{x}_i \tag{2-8}$$

Finding the optimal hyper-plane for separable data points is a quadratic optimization problem with linear constraints. It is noted that for very high dimensional spaces, the problem should be translated into its dual form which is finding parameters α_i that maximize the functional

$$L(\alpha) = \sum_{i=1}^{n} \alpha_i - 1/2 \sum_{i,j=1}^{n} \alpha_i \alpha_j y_i y_j (\mathrm{x}_i . \mathrm{x}_j) \qquad \text{subjected to} \quad \sum_{i=1}^{n} y_i \alpha_i = 0, \quad \alpha_i \geq 0 \tag{2-9}$$

The optimal hyper-plane is then obtained from

$$f(x) = \sum_{i=1}^{n} \alpha_i y_i (\mathrm{x}_i . \mathrm{x}_j) + b \tag{2-10}$$

where *b* is defined by equation

$$b = \mathrm{y}_s - \sum_{i=1}^{n} \alpha_i y_i (\mathrm{x}_i . \mathrm{x}_j) \tag{2-11}$$

Figure 2-5 presents an example of the optimal hyper-plane of two linearly separable data sets.

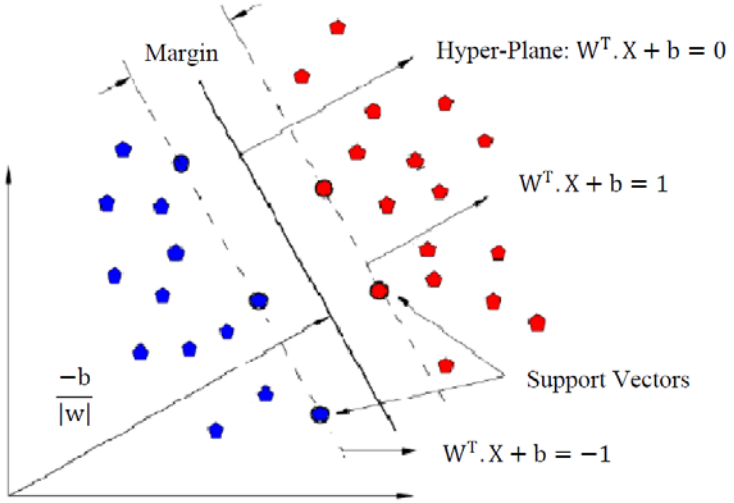


Figure 2-5. Linearly separable SVM classification

For classifications with non-separable training data, the empirical risk cannot be minimized to zero and some training samples are allowed to fall within the margin, the empirical risk of which are quantified as the deviation from the margin borders (ξi). Figure 2-6 shows the optimal hyper-plane for two non-separable data sets.

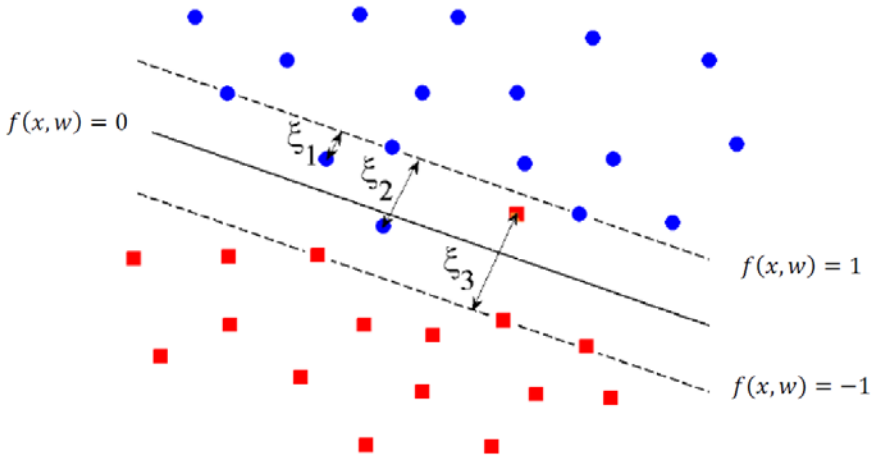


Figure 2-6. Non-separable SVM classification

In this case, the optimization problem can be formulated as

$$Min(\frac{1}{2}\mathrm{w}^T.\mathrm{w} + C\sum_{i=1}^{n}\xi_i) \qquad \text{subjected to } \; y_i(\mathrm{w}^T.\mathrm{x_i} + b) + \xi_i - 1 \geq 0 \;\; , \quad \xi_i \geq 0 \tag{2-12}$$

where $\xi_i = \max(1 - y_i f(x_i, w), 0)$ and *C* is a user selected parameter that controls the trade-off between complexity and the proportion of non-separable samples. The dual formulation of equation 2-12 is to find the parameters α_i that maximize the following functional

$$L(\alpha) = \sum_{i=1}^{n}\alpha_i - 1/2\sum_{i,j=1}^{n}\alpha_i\alpha_j y_i y_j(\mathrm{x_i}.\mathrm{x_j}) \qquad \text{subjected to} \sum_{i=1}^{n} y_i\alpha_i = 0, \quad 0 \leq \alpha_i \leq \mathrm{C} \tag{2-13}$$

2.3.4.4 Nonlinear SVM classification

Section 2.3.4.3 deals with linear decision functions (hyper-planes) in the original input space. However, if the decision function in the input space is nonlinear, the input space should be mapped into a high dimensional feature space using the nonlinear transform $\boldsymbol{x} \rightarrow \Phi(\boldsymbol{x})$ where $\Phi(\boldsymbol{x})$ is the feature function. The optimal hyper-planes in the feature space will then result in nonlinear decision boundaries in the input space. As indicated in equation 2-10, building the optimal hyper-plane in the input space only involves calculating the inner product of two vectors in this space. The same concept applies when building the optimal hyper-plane in the feature space for which Kernel functions can be used as shown in equation 2-14.

$$f(x) = \sum_{i=1}^{n}\alpha_i y_i \phi(\mathrm{x_i})^T.\phi(x) + b = \sum_{i=1}^{n}\alpha_i y_i \, \mathrm{K}(x, x_i) + b \tag{2-14}$$

The selection of the type of kernel function corresponds to the selection of the class of functions used for feature construction. The kernel functions for polynomial functions and Radial Basis Functions (RBF) which are used in this study are presented in section 5.3 of chapter 5. In

addition, chapter 5 provides more details on how to properly select the parameters affecting the quality of the SVM model including the type of kernel function, the kernel parameter (γ) and the penalty parameter.

2.3.4.5 Overview on the development and applications of the Support vector machines

The main concept of margin was first developed in the early 1960s for the classification problem with separable data (Vapnik and Chervonenkis 1964, Vapnik and Lerner 1963). It took another 30 years until two additional improvements, the kernel representation and the ability to handle non-separable data, were incorporated into the SVM method (Boser et al. 1992; Cortes and Vapnik 1995). Since then, SVM methodology has been adapted to solve regression problems and has been successfully used for numerous applications.

SVMs have been widely used in various areas such as text categorization (Joachims 2002), bioinformatics (Ding and Dubchak 2001, Furey et al. 2000, Pavlidis et al. 2001) image recognition (Moghaddam and Yang 2002, Osuna et al. 1997) and finance (Trafalis and Ince 2000, Yang et al. 2002). Many researchers have applied the method for structural health monitoring and damage detection purposes (Noori et al. 2010, Sharma 2008, Shimada and Mita 2005, Yang et al. 2005). The application of SVM in the reliability analysis was first introduced by Rocco and Moreno (2002). Li et al. (2006) proposed SVM-based MCS and SVM-based FORM analyses for reliability assessment of simple structures. Hurtado and Alvarez (2003) adopted SVM along with stochastic finite element for performing structural reliability analysis. Jack and Nandi (2002) and Widodo and Yang (2007) suggested the use of SVM classification for machine condition monitoring and fault diagnosis. Guo and Bai (2009) introduced the application of least squares support vector machine for regression (LSSVR) for reliability analysis of engineering structures.

2.3.4.6 Probabilistic Support Vector Machines (PSVM)s

As the outputs from the standard classification SVM analysis is the value of $sign(f(x))$, where f is the decision function presented in equation 2-14, SVM is in effect regarded as a non-probabilistic classifier. The attempt to present probabilistic extensions of the SVM classifier can be seen in the research done by Platt (1996), Drish (2001), Zadrozny and Elkan (2002), and Vovk et al. (2003, 2005) among others. Using Probabilistic Support Vector Machines (PSVMs), the researcher can determine not only the limit state surface, but also an estimate of the probability that a new observation belongs to each of the failure or safe domains. The application of PSVMs in reliability analysis and fragility assessment of structural components and systems is fairly new. Mahmoudi and Chouinard (2016) developed seismic fragility curves for an existing bridge in Montreal using the PSVM approach proposed by Platt. Zhou et. al. (2011) and Lambrou et al. (2012) compared various existing PSVMs and demonstrated the superiority of Venn machines since the resulting probability estimates are well calibrated, reliable and unbiased under the independent and identically distributed (i.i.d) assumption. Venn predictors (VPs) proposed by Vovk et al. (2003, 2005) follow three simple steps: 1) divide examples (observations) into categories based on a taxonomy, 2) classify the new example into one of the categories, and then 3) use the frequencies of labels in the chosen category as probabilities for the label of the new observation. It is noted that, when determining the frequencies of labels in the category containing the new example, the new example should be included along with the old examples already in that category. However, since at the time of prediction the label of the new example is unknown, the frequencies should be calculated several times, once for each label the new example might have. Each set of frequencies can be regarded as a probability distribution for the unknown label of the new example. In general, each taxonomy defines a different VP

such as the *k*-Nearest Neighbours-based VP (Dashevskiy and Luo 2009), the Nearest Centroid-based VP (Dashevskiy and Luo 2008), the Neural Network-based VP (Papadopoulos 2011) and the SVM-based VP (Zhou et. al. 2011, Lambrou et al. 2012). This study applies SVM-based Venn-machines for probabilistic structural analysis of transmission lines, and presents more details about the mathematical formulation of SVM-based VPs and their application in fragility assessment and reliability analysis of electric transmission lines in sections 5.4, 5.6.2 and 5.7.2.

2.4 The mathematical tools of structural reliability theory

This section briefly presents some of the concepts and tools which are used in the following chapters of this dissertation and can help researchers and engineers better understand the behaviour of structural components and systems in a probabilistic context.

2.4.1 Fragility curves

Fragility curves are statistical tools representing the conditional probability of a structure exceeding a given damage state (or performance) as a function of an engineering demand parameter.

$$P_F = P[G(\boldsymbol{x}) < 0 | D = d] \tag{2-15}$$

Where $G(\boldsymbol{x}) < 0$ represents the non-performance of the structural component or system and d is the engineering demand parameter of interest. Fragility curves are practical decision tools for updating existing structures and optimizing the design of new ones. Component and system level fragility curves can also be used to optimize and prioritize inspections and structural assessments in the aftermath of an extreme event. Finally, by incorporating fragilities with hazard functions at a specific location, component and system reliabilities can be estimated and used for risk

mitigation and decision analysis. Generally, fragility curves are classified into three major categories: 1) Empirical fragility curves, 2) Expert based fragility curves, and 3) Analytical fragility curves.

Empirical fragility curves are based on actual damage data obtained from field investigations after hazards or from experiments. Fragility curves derived from actual damage data provide valuable perception of potential impacts from future hazards. However, damage data obtained from the field needs to constitute a statistically large sample size since the limited number of sampled structures may lead to unreliable predictions of failure probabilities. Few examples of empirical seismic fragility curves developed for bridges can be found in Basöz and Kiremidjian (1999), Shinozuka et al. (2000c), and Yamazaki et al. (1999). When the required actual damage data are not available, fragility curves can be developed from experience and judgments of engineering experts (Jaiswal et al. 2013). For instance, Applied Technology Council (ATC) used some questionnaires filled by experts in order to develop expert based fragility curves in ATC-13 (ATC 1985) and ATC- 25 (ATC 1991). However, the resulted fragility curves could be highly subjective and biased because of the limited number of experts and limited types of structures considered in the questionnaires. Another approach would be to develop analytical fragility curves from response analysis of structures. Analytical fragility curves are widely used for vulnerability assessment of structural components and systems under different hazards. Nielson and DesRoches (2007), Padgett and DesRoches (2008) and Ramanathan et al. (2010) developed analytical seismic fragility curves for bridges. Hwang and Huo (1994a, 1994b) proposed an analytical method for establishing seismic fragility curves for a five-story reinforced concrete frame building. Li and Ellingwood (2009) and Ham et al. (2009) respectively generated fragility curves for residential home components and industrial buildings subjected to wind hazards.

Shafieezadeh et al. (2014) developed a framework for time-dependent fragility analysis of wood utility poles against hurricanes and strong winds.

In this study analytical component and system fragility curves are developed for transmission lines for given intensity measures of wind speed considering its angle of incidence and uniform ice thickness in chapter 5. Chapter 6 presents component and system fragility curves for transmission lines subjected to unbalanced ice loads resulting from non-uniform ice formation or ice shedding.

2.4.2 Component Failure probability

As mentioned in section 2.1 and shown in figure 2-1, when the failure domain is described through a single function, a component reliability problem is at hand. It is noted that the meaning of the term "component" from this perspective is different from the meaning of a structural component such as an angle member in a lattice transmission tower. In other words, if a structural component fails in more than one mode, the failure domain is no longer described through a single function, and the failure probability of this structural component is estimated by solving a system reliability problem rather than a component reliability problem. Component failure probability can be determined by incorporating component fragilities with hazard functions at a specific location.

$$(\mathrm{P_f})_{\mathrm{Component}} = P[G(\boldsymbol{x}) < 0] = P[G(\boldsymbol{x}) < 0 | D = d].P[D = d] \quad (2\text{-}16)$$

Where $P[G(\boldsymbol{x}) < 0 | D = d]$ is the fragility function and $P[D = d]$ is the hazard.

2.4.3 System failure probability

As mentioned in section 2.1, a failure is a system failure if it involves more than one failure equation (domain). In general, a structure may fail in various modes, each of them requiring several component failures. Therefore, a structure failure is usually a system failure. In addition, if a physical component fails in various modes such as yielding and buckling, each defined by a particular failure equation, this component is a system in a reliability analysis sense. There are two approaches for determining the failure probability of a system. The first approach is to express the failure of the system as a combination of the single component failure events. In other words, the probability of failure of the system is approximated from the failure probabilities of individual components of the system. This approach is referred to as component-based approach. As mentioned in section 2.1, a general system can be probabilistically represented as a series system, a parallel system or a combination of both. Here, the attention is focused on series systems which are used in subsequent chapters of this dissertation. It is important to note that, the exact failure probability of a system is difficult to calculate in general since common material sources, similar fabrication procedures and the same external loading, results in some correlation between different component failure events which should be considered in estimating the failure probability of the system (equation 2-17).

$$(P_f)_{\text{System}} = P(\cup F_i) = \sum_{i=1}^{m} p(F_i) - \sum\sum_{i<j}^{m} P(F_i \cap F_j) + \sum\sum\sum_{i<j<k}^{m} P(F_i \cap F_j \cap F_k) - \cdots \quad (2\text{-}17)$$

As a results, many researchers have suggested different probability bounds for determining system failure probabilities, a review of which can be found in Ramachandran (1984). In this study the first and second order Ditlevsen's bounds (Ditlevson,1979) have been applied to compute the system failure probability. The first order bounds does not include the probability of

intersection of events (equation 2-18). It is noted that, the lower bound corresponds to perfectly correlated failure events, while the upper bound corresponds to independent failure events.

$$max_{i=1}^{m} P(F_i) \leq P(\cup F_i) \leq \sum_{i=1}^{m} P(F_i) \quad (2\text{-}18)$$

The second order Ditlevsen's bounds takes into account the pair wise intersection of events resulting in narrower and more practical bounds (equations 2-19 and 2-20).

$$P(F) \geq P(F_1) + \sum_{i=2}^{m} \max \{P(F_i) - \sum_{j=1}^{i-1} P(F_i \cap F_j), 0\} \quad (2\text{-}19)$$

$$P(F) \leq \sum_{i}^{m} P(F_i) - \sum_{i=2}^{m} \max_{j<i} P(F_i \cap F_j) \quad (2\text{-}20)$$

where failure probability of *m* components are arranged in the order of increasing reliability to produce narrower bounds.

On the other hand, if the failure of a structure can be described by a single equation, for example through a single plastic mechanism equation or through a classification approach, the structure is a component in a reliability analysis sense and the approach used for determining its failure probability is referred to as system-based approach. Chapter 5 presents more details about the component-based approach and the system-based approach for structural reliability analysis and fragility assessment.

2.4.4 Importance measures and critical components

Generally, structural systems should be designed, enhanced, or maintained efficiently using limited resources. Besides, in the crisis management, it is most economic to quickly identify the improvement for the system performance, instead of searching for a global solution. Therefore, it is desirable to allocate resources on the small subset of components that are most important to

the system. To evaluate the relative importance of components in a structural system, importance measures are used. Importance measures are useful decision tools in various stages of reliability-based design (Kuo et al. 2006, Kuo and Zhu 2012a), including component assignment problems (CAP), redundancy allocation, system upgrading, and fault diagnosis and maintenance. It is noted that, no single type of importance measure is universal and the design of importance measures for a given situation highly depends on the type of the problem under consideration. A review of different types of importance measures can be found in (Kuo and Zhu 2012b, Kuo and Zhu 2012c). In this study, two different reliability importance measures are determined for each component of the line section: the improvement potential importance and the Bayesian reliability importance. More details about the definition of these two reliability importance measures and how to apply them for ranking critical components in a structural system are provided in section 4.5.1.

2.4.5 Most probable failure event (design point)

As mentioned in section 2.3., the most probable point (MPP) also known as the design point is the closest point on the limit-state surface to the origin in the standard normal space. The distance between the MPP and the origin in the standard normal space is also referred to as the first order reliability index. Determination of MPP involves standard non-linear constrained optimization and is an important step in most MPP-based reliability analysis methods including FORM and SORM. In this study, the design point which is the combination of climatic variables (wind speed, wind angle and ice thickness) that results in the failure of the system and has the highest likelihood is determined using MCS method. Section 4.6 provides more details about how system design point is determined for the studied transmission line and how it compares to

the design climatic combinations proposed by current design guidelines (CAN/CSA-C22.3, 2006, CEI/IEC 60826, 2003).

CHAPTER 3 A PROBABILISTIC EXTREME LOAD MODEL FOR ELECTRIC TRANSMISSION LINES

3.1 Climatic loads applied on transmission lines

Generally, for the design of transmission lines, three weather-related loading conditions are recognized: wind, ice, and wind and ice combined. Current available loading guidelines for the reliability-based design of transmission lines include ASCE 74 (2010), CAN/CSA-C22.3 (2006), CEI/IEC 60826 (2003), and CENELEC EN 50341 (2001). Detailed comparison of loading criteria based on these standards is presented in Ghannoum (2005), Ghannoum et al. (2006), Peyrot et al. (1991), and Rezaei et al. (2011). This section mainly presents the formulas and assumptions used for the calculation of wind and ice loads on transmission line components following CAN/CSA-C22.3 (2006) and CEI/IEC 60826 (2003). It is noted that, other natural or man-made hazards including earthquakes, localized weather events such as tornados, maintenance and construction loads, and accidental loads for example due to cable rupture can contribute to the failure of transmission lines. However, studying the performance of overhead transmission lines subjected to these loading scenarios is beyond the scope of this study and further information can be found in ASCE 74 (2010), CAN/CSA-C22.3 (2006), Hamada and El Damatty (2011), Ishac and White (1995), Mirshafiei et al. (2013), and Tian et al. (2010).

3.1.1 Wind loads on conductors

Wind effects on conductors consist of loads due to wind pressure as well as the effect of the increase in the mechanical tension. Equation 3-1 presents the conductor or ground wire load (*Ac*)

in N due to the effect of the wind pressure upon a wind span, when wind is blowing at an angle Ω with the conductors.

$$A_c = q_0 C_{xc} G_c G_L dL sin^2 \Omega \quad (3\text{-}1)$$

where C_{xc} is the drag coefficient of the conductor which depends on the shape and surface properties of the element being considered and is assumed equal to 1 as recommended by CEI/IEC 60826 (2003), CAN/CSA-C22.3 (2006) and ASCE 74 (2010). The combined wind factor for the conductors, G_c, as presented in appendix A depends on height z and terrain categories and takes into account the load effects due to wind turbulence and dynamic response of cables. The span factor, G_L, which is also presented in appendix A accounts for the fact that gusts generally do not envelop the entire span between transmission structures and some reduction reflecting the spatial extent of gusts should be included in the calculation of wind loads. Also in equation 3-1, d is the diameter of the conductor (m), L is the wind span of the support which is equal to half the sum of the length of adjacent spans of the support, and q_0 is the dynamic reference wind pressure (N/m2). This dynamic pressure is given in terms of the reference wind speed, V_R, modified by roughness factor K_R (presented in appendix A) corresponding to the terrain category at the location of the line (equation 3-2).

$$q_0 = \frac{1}{2} \tau \mu_a (K_R V_{RB})^2 \quad (3\text{-}2)$$

where μ_a is the air mass per unit volume equal to 1,225 kg/m^3 at a temperature of 15 °C and an atmospheric pressure of 101.3 kPa at sea level, τ is the air density correction factor and V_{RB} is the reference wind speed (m/s) for terrain category B corresponding to a return period T. The reference wind speed can be determined from statistical analyses of relevant wind speed data at

10 m above ground and with an averaging period of 10 min. In addition, wind acting on conductors will cause an increase in their mechanical tension that can be considered with a nonlinear analysis of the structure using the computer software SAP2000.

3.1.2 Wind loads on insulators

Wind loads acting on insulator strings originate from the load A_c transferred by the conductors (section 3.1.1) and from the wind pressure acting directly on the insulator strings (equation 3-3).

$$A_i = q_0 C_{xi} G_t S_i \tag{3-3}$$

where q_0 is the dynamic reference wind pressure (N/m^2), C_{xi}, is the drag coefficient of the insulators, considered equal to 1.20 as recommended by CAN/CSA-C22.3 (2006) and CEI/IEC 60826 (2003), G_t, is the combined wind factor for insulators (presented in appendix A), and S_i is the area of the insulator string projected horizontally on a vertical plane parallel to the axis of the string (m^2).

3.1.3 Wind loads on Lattice towers of rectangular cross-section

In order to determine the wind load on latticed towers, they are divided into different panels at the intersections of the legs and bracings. The wind loading A_t (N) in the direction of the wind, applied at the centre of gravity of each panel is calculated using equation 3-4.

$$A_t = q_0(1 + 0.2sin^2 2\theta)(S_{t1} C_{xt1} cos^2\theta + S_{t2} C_{xt2} sin^2\theta) G_t \tag{3-4}$$

where θ is the angle of incidence of the wind direction with the perpendicular to line direction in a horizontal plane, S_{t1} is the total surface area projected normally on face 1 of the panel (m^2) which is parallel to the line direction, S_{t2} is the total surface area projected normally on face 2 of

the same panel (m^2) which is perpendicular to the line direction, C_{xt1} and C_{xt2} are the drag coefficients of parallel and perpendicular faces of the panel to the line direction (presented in appendix A as a function of χ). The solidity ratio of a panel, χ, is equal to the ratio between the total surface of the support members and the circumscribed area of the face of the considered panel. Finally, G_t is the combined wind factor for the supports (presented in appendix A).

3.1.4 Ice loads on components/elements

Ice accretion on a transmission line is often a governing loading criterion in structure design. In addition to imposing substantial vertical loads on the structural system, the ice build-up on the conductors presents a greater projected area exposed to the wind.

In general, the atmospheric icing accretion can be classified in the following types: 1) precipitation icing 2) in-cloud icing 3) hoar frost. Precipitation icing occurs in several forms such as freezing rain (glaze or rime), wet snow and dry snow. The different forms of precipitation icing differ in their density, shape and structure. In-cloud icing which usually happens to transmission lines located in mountainous areas is a process by which super-cooled water droplets in a cloud or fog freeze immediately upon impact on objects in the airflow. Finally, hoar frost forms when air with a dew point below freezing is brought to saturation by cooling and normally does not constitute a significant loading problem for transmission lines. Detailed information about the atmospheric conditions required for the formation of different types of atmospheric icing and their physical properties can be found in Farzaneh (2008).

Currently, ice loads are not directly measured at meteorological stations. Therefore, other means are usually used to obtain icing data which include (CAN/CSA-C22.3 2006, Farzaneh 2008):

- Direct measurements of ice thickness or ice weight formed on line conductors
- Measurement by devices that simulate ice accretion on conductors
- Estimation of ice thickness using conductor tension and or vertical component of the weight at insulator attachment point
- Estimation of ice thickness using the information about conductor tension and sag
- Application of ice accretion models and available meteorological data

Due to the lack of field data for a specific line, this study is based on the statistical information about reference ice thickness (t_R) provided by CAN/CSA-C22.3 (2006) and CEI/IEC 60826 (2003) which use an ice accretion model to predict equivalent radial ice thickness due to freezing rain or drizzle with a density of 900 kg/m^3 on a 25 mm conductor 10 m above ground. Equation 3-5 expresses the relation between ice weight per unit length of the conductor g (N/m) and uniform radial ice thickness t (mm).

$$g = 9.82 \times 10^{-3} \delta \pi t (d + \frac{t}{1000}) \quad (3\text{-}5)$$

where δ is the ice density (kg/m^3) and d is the conductor diameter (m). It is notable that, since the actual line conditions are usually different from the conditions under which the ice data is measured, the reference ice load, g_R, needs to be modified with a diameter factor K_d and height factor K_h as presented in appendix A. In addition, in reality, the shape of the ice on conductors is dependent upon a combination of factors present during the accretion stage including: wind speed and its variations, wind direction, wind flow turbulence, air temperature variations, rotational stiffness of conductors, and icing event duration. However, due to the variability of these parameters from one event to the other, and even during the same event, a cylindrical shape

with equivalent weight is usually assumed for design and for the purpose of this study (Farzaneh 2008).

Finally, ice accretion on structures increases the vertical loads applied on the structure and may govern the design of foundations and some tower elements (CEI/IEC 60826, 2003). In this study, the weight of ice is determined using the geometry of the tower elements and the relevant thickness of ice accretion.

It is noted that, unbalanced loads on transmission line supports, due to icing, can occur in a number of ways with varying degrees of severity. Some of the causes of unbalanced loads are:

- Unequal ice accumulation due to different exposures to wind
- Ice loads on unequal adjacent spans
- Ice shedding in one or several spans
- Galloping of conductors

For this reason, current design guidelines require the designer to investigate the reliability of transmission lines for unbalanced ice loads that generate torsion, longitudinal or transversal bending on towers (CAN/CSA-C22.3 2006, CEI/IEC 60826 2003 and EN 50341 2001). As mentioned in section 1.4, this study does not address failure scenarios due to dynamic loads imposed by galloping or ice shedding. However, chapter 6 presents fragility curves developed for wind loads and residual static unbalanced ice loads applied to conductors due to unbalanced ice accumulation and/or ice shedding. The configuration of the studied line is selected to address both cases of non-uniform ice accumulation due to different exposures to wind and unequal spans.

3.1.5 Combined wind and ice loadings

Operational experience with overhead transmission lines in cold climates indicates that combined wind and ice loads are often critical for the design of these structures. Ideally, statistics of wind speed during the presence of ice on conductors should be used to generate the combined loadings of ice and wind. Following this approach, a subcommittee of ANSI/ASCE7 compiled the 50 year return period extreme radial ice thicknesses combined with 3 second gust velocities map for the USA (ASCE 74, 2010, Jones et al. 2002). However, detailed data and observations on ice weight, ice shape and coincident wind, are not commonly available. Therefore, considering the low probability of joint occurrence of extreme wind and extreme ice, there is a tendency to reduce one or both wind speed and icing amounts to establish realistic combined loads. For this reason, CAN/CSA-C22.3 (2006) and CEI/IEC 60826 (2003) and CENELEC EN 50341 (2001) assume that maximum loads are most likely related to combinations involving a low probability value of one variable (associated with a return period of T) combined with high probability values of other variables (with reduced return periods). The two loading scenarios proposed by CAN/CSA-C22.3 (2006) and CEI/IEC 60826 (2003) include:

- A low ice probability (return period T) associated with the average of yearly maximum winds during icing presence which is assumed to be equal to 0.4 to 0.5 times the extreme wind speed.
- A low probability wind during icing (return period T) which is assumed to be equal to 0.6 to 0.85 times the extreme wind speed combined with 0.4 times the extreme ice load that corresponds to the average of the yearly maximum icing.

The two loading scenarios proposed by CENELEC EN 50341 (2001) include:

- An extreme ice load with a return period of T combined with a moderate wind load with a reduction factor equal to 0.4 which corresponds to a moderate wind speed taken as 0.55 to 0.65 times the extreme wind speed depending on the type of ice.
- A high wind speed corresponding to 0.7 to 0.85 times the extreme wind speed combined with a moderate ice load with a reduction factor equal to 0.35.

In addition, CEI/IEC 60826 (2003) proposes combining extreme icing with the maximum annual wind speed recorded during ice events and the consecutive periods where the air temperature is below 0°C. A maximum period of 72 h is recommended. Using data from 30 weather stations in Ontario, Krishnasamy and Kulendran (1996) report that a reduction factor between 0.45 and 0.75 for wind speed should be used for combined wind and ice load calculations. Goodwin et al. (1982) propose combining annual extreme ice with daily extreme wind velocities measured during or immediately following icing events. Savadjiev and Farzaneh (2003) suggest that wind speeds associated with icing events should be determined by reducing the extreme wind speed using a reduction factor between 0.4 and 0.7.

3.2 The extreme load model

One of the main probabilistic aspects that should be considered in the reliability analysis and fragility assessment of transmission lines is the randomness associated with the climatic loads. Besides, it is customary in the reliability-based design of transmission lines to associate extreme values of wind or ice loads with some selected return period. For this reason, extreme value distributions can be used to determine the extreme wind speeds or extreme uniform radial ice thicknesses that can be expected in a T-year return period (X_T). In other words, the aim is to define the form of the distribution and estimate the parameters, so that values of X_T can be

calculated. According to Palutikof et al. (1999), classical methods to calculate extreme wind speeds include: annual maxima, the method of independent storms (MIS), r-largest extremes with the generalized extreme value (GEV) distribution, and peak-over-threshold (POT) extremes with the generalized Pareto distribution (GPD). The great advantage of using the GEV distribution with annual maxima is that unlike other methods, it requires the user to make few decisions during the calculation of the distribution parameters. The decisions required to be made by other methods (for example, the value of r for the r-largest method, or the threshold value, for the MIS and POT methods) can have a strong impact on the estimated values of the distribution parameters. Besides, for extreme value theory to be applied successfully, the extremes must be independent and identically distributed which is more likely to be the case for the annual maxima method. For the r-largest and POT samples, additional steps are required to insure independence. However, the principal disadvantage of the annual maxima method is that only one value is selected per year, which reduces the amount of data available for analysis. As a result, Cook (1985) suggests that this method should not be used with data sets fewer than 10 years. According to CAN/CSA-C22.3 (2006), CEI/IEC 60826 (2003) and CENELEC EN 50341 (2001) guidelines for the loading of transmission lines, the distribution of annual maximum wind velocities or ice loads and ice thicknesses can be represented, with good approximation, by a type I GEV distribution, also known as the Gumbel distribution. The ASCE Manual 74 (2010) provides wind speed extreme values by fitting a Gumbel distribution to the series of annual maxima using the method of moments. Ice thickness extreme values are also presented from modeled freezing rain ice thicknesses, using the POT method and fitting a generalized Pareto distribution to the largest ice thicknesses (Jones and White 2002, Jones et al. 2002). Krishnasamy (1985) presented statistical parameters such as mean, coefficient of variation (COV), and

Gumbel extreme value constants for yearly extreme wind, ice and wind-on-ice data for 30 weather stations in Ontario, Canada. Savadjiev and Farzaneh (2003) proposed a probabilistic model for combined wind and ice loads on overhead power line conductors. They report that, the distributions of annual maxima of wind speed and those of wind speed associated with icing events converge rapidly toward the GEV distribution of type I (Gumbel) for the largest extremes. An and Pandey (2007) assigned a joint generalized extreme value distribution to r largest order statistics (r-LOS) of annual wind data. They analysed the data collected from 30 weather stations of Ontario, Canada and showed that the wind pressure follows a Gumbel distribution while the wind speed follows the GEV distribution. Chouinard et al. (1998) presented an event-based model for the distribution of glaze ice and of the associated wind speed based on observations up to 21 yr at 180 stations throughout the province of Quebec. They report that, for the statistical distribution of ice thickness, Pearson distribution is found to provide the best fit for the majority of the stations. While for the statistical distribution of associated wind speed, each of the distributions of extreme type I, extreme type II, Rayleigh, 3-p lognormal, and log Pearson type III provide a good fit for the data. Yang et al. (2013) derived joint probability distribution of ice and wind loads on both transmission lines and towers using the Generalized Pareto distribution and Copula functions.

It is noted that, when climatic data are available for a specific location, an extreme value analysis can be performed to determine the best probabilistic model that fits the data at hand. However, in this study, due to lack of data, the statistical information provided by CAN/CSA-C22.3 (2006) and CEI/IEC 60826 (2003) are used for modeling extreme wind speed during ice and extreme ice thickness which are based on the Gumbel distribution.

CHAPTER 4 MANUSCRIPT 1

Component Based Structural Reliability analysis of transmission line systems for climatic hazards Using Statistical Learning Theory

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Abstract

This paper describes a novel application of Statistical Learning Theory (SLT) to structural reliability analysis of transmission line systems considering the uncertainties of climatic variables such as wind speed, ice thickness and wind angle, and of the resistance of structural elements. The reliability analysis of such complex structural systems is hindered by the implicit nature of the limit state function. This problem can be related to statistical model selection, where the goal is to select a surrogate model of the finite element solver from several possible models that provides the value of the performance function. SLT, also known as Vapnik-Chervonenkis (VC) theory, provides analytic generalization bounds for model complexity control by applying the Structural Risk Minimization (SRM) inductive principle. After determining the performance function for each structural element of the transmission line using a regression type learning machine, Monte Carlo (MC) simulation is applied to calculate the

failure probability of tower elements, conductors, ground wires and insulators. The failure probabilities of towers are then estimated from the failure probabilities of their elements considering the correlation between failure events. Finally, the overall system failure probability of the line is determined from the correlated failure probabilities of its components. The calculated failure probabilities are compared with first and second order Ditlevson bounds to show the effect of failure correlation between different components. In order to quantify the relative importance of line components and provide the engineers with a practical decision tool, the paper presents the calculation of two types of component importance measures. The presented methodology can be used to achieve optimized design, and to assess upgrading strategies to increase the line capacity.

Keywords: Structural reliability analysis, Implicit performance function, Transmission lines, climatic hazards, Statistical learning theory, Monte Carlo simulation

4.1 Introduction

Electric transmission lines can be designed for different levels of reliability depending on the structure of the power grid and the consequences of a failure event. During the last decades, many organizations such as the International Electrical Commission (IEC), the Institute of Electrical and Electronics Engineers (IEEE) (Ghannoum 1983(a), 1983(b), 1984), the American Society of Civil Engineers (ASCE) and the Conseil International des Grands Réseaux Électriques (CIGRE) study committees (Ghannoum et al. 2006) introduced and applied the reliability/probability based design concepts to the transmission lines design. As a result, many reliability based guidelines such as the North-American Standard ASCE 74 (2010), the European Standard EN 50341 (2001), CEI/IEC 60826 (2003) and CAN/CSA-C22.3 (2006) were

developed. Since then, structural system reliability analysis of transmission lines has become a useful tool in designing new lines and in evaluating or upgrading existing lines (Dagher et al. 1991, Dagher and Lu 1993, Foschi 2004, Haldar 2006, Li et al. 2006). Such reliability models are also essential to evaluate icing hazards mitigation strategies (Hou et al. 2013, Hou et al. 2015, Liu and Peters 1999).

One challenge arising from the reliability analysis of transmission lines is the implicit nature of the limit state function for each structural component. This condition makes it difficult to implement gradient based reliability analysis methods (Haldar and Mahadevan 2000, Zhao and Ono 1999) which require to estimate the gradient of the performance function. In addition, simulation based methods (Karamchandani 1987, Melchers 1999) would be unfeasible and difficult to perform given the vast computational effort needed for repeated evaluations with the finite element solver of the line especially when dealing with low failure probabilities. Defining the performance function explicitly has traditionally been carried out by means of the Response Surface Method (RSM) (Bucher and Bourgund 1990, Rajashekhar and Ellingwood 1993). The principle of this method consists in the substitution of the implicit limit state function by approximate functions only at the neighbourhood of the design point using the Empirical Risk Minimization (ERM) principle. Although RSM has been successfully adapted for many engineering cases, it has some application problems including the high sensitivity of the value of the reliability index and the failure probability to the experimental design plan (Guan and Melchers 2001) which can be explained by the rigid and non-adaptive nature of the selected model (Hurtado 2004). In addition, the strict assumptions over the class of approximating functions imposed by the ERM inductive principle can lead to overfitting and poor prediction abilities (large model bias) for the meta model, even when the number of training samples is

fairly large (Cherkassky and Mulier 2007). All these shortcomings call for the application of advanced yet simple to use meta-modeling techniques for the component-based reliability analysis of complex structural systems with implicit limit states.

This paper proposes a novel probabilistic framework based on the concepts of Statistical Learning Theory (SLT) to address the problem of reliability analysis of complex structural systems with implicit limit states emphasizing on the reliability analysis of transmission lines subjected to weather-related hazards. Although the methodology is applied to a nine span segment of a typical 120 kV single circuit transmission line located in north America and subjected to extreme wind and ice storms, it can easily be modified and used for other structures and other natural and/or man-made external loads such as earthquakes, impacts, etc. SLT also known as Vapnik-Chervonenkis (VC) theory (Vapnik and Chervonenkis 1968) implements the Structural Risk Minimization (SRM) inductive principle to find a surrogate model of the finite element solver that has good generalization (prediction) properties and is regarded as a better alternative to RSM (Hurtado 2004). It provides analytical generalization bounds for model selection, which relate unknown prediction risk and known quantities such as the number of training samples, empirical error and a measure of model complexity called the VC-dimension (Cherkassky et al. 1999, Cherkassky and Mulier 2007). Therefore, it can overcome the problem of overfitting associated with the rigid model selection approach of ERM based methods. Cherkassky and Ma (2003), Cherkassky and Mulier (2007), Cherkassky et al. (1999), and Shao et al. (2000) compared the application of VC-generalization bound for model complexity control with other alternative model selection methods and demonstrated its superiority. SLT and VC-generalization bounds have been successfully used for model selection in solving practical problems such as motion estimation and tracking)2004) and traffic flow time series prediction

(2002). This paper aims to illustrate the practical application of VC-generalization bounds in the field of structural engineering and for reliability analysis of complex structural systems with implicit limit state functions such as electric transmission lines. In this paper, for each structural element, the optimal performance function is selected from the class of polynomial functions using VC-generalization bounds and Stepwise regression. Cross validation (CV) technique is used to determine the unbiased prediction error of the selected model. In addition, SLT is combined with Monte Carlo (MC) simulation to calculate the yearly failure probability of each element and component of the line as well as the overall yearly system failure probability. The methodology is applied to a part of an existing transmission line located in North America. In order to show the effect of failure correlation between different tower elements or different components of the line, the calculated failure probabilities are compared with first and second order Ditlevson bounds. In addition, to provide the engineers with a practical reliability-based decision tool, the paper also includes the calculation of two different types of component importance measures: (a) the B-reliability improvement potential importance and (b) the Bayesian reliability importance. These measures help to quantify the relative importance of components and can be used for various engineering decision making problems such as system upgrading and fault diagnosis and maintenance.

In general, the uncertainties on component demand can be classified as either epistemic or aleatory. Epistemic uncertainty is reducible and is associated with imperfect models for the underlying physics or to parameter uncertainty due to small sample sizes, while aleatory uncertainty is irreducible and associated with natural variability such as the shape and drag coefficients for ice deposits, and the variability of ice accumulation, wind speed and material properties. In this application, component demands are defined for given winds speeds and ice

accumulation combinations and the uncertainty in component demand is mainly epistemic and related to drag forces and the structural model. Due to lack of data on the latter, this study assumes that component demands are deterministic for given climatic conditions. Model uncertainty can be easily considered with the proposed methodology and could be included as more information becomes available in future studies.

The remainder of this paper is structured as follows. Section 4.2 presents a review of the concepts in model complexity control and SLT. Section 4.3 describes the transmission line section studied in this application. Section 4.4 discusses the methodology used for the reliability analysis and presents the estimated component and system failure probabilities. Section 4.5 presents the calculation of the importance measures. Section 4.6 demonstrates the determination of system design point. Finally, conclusions are drawn in Section 4.7.

4.2 Model complexity control and SLT concepts

The challenge of estimating an unknown model from available training data in order to use this model for prediction of future samples can be addressed by predictive learning. Figure 4-1 summarizes the general learning machine components. In the structural reliability analysis of a transmission line, the *Generator* of random input vectors **x**, is a MC simulation algorithm that produces the samples from the probability distribution of climatic variables. There is also a *Supervisor* that produces an output value *y*, for each input vector **x**. The latter corresponds to the finite element solver of the structure. Finally, the *Learning machine* applies the function $f(\mathbf{x},\omega)$, in which ω is the vector of unknown parameters, and produces an estimate of the output value, $\hat{y}$.

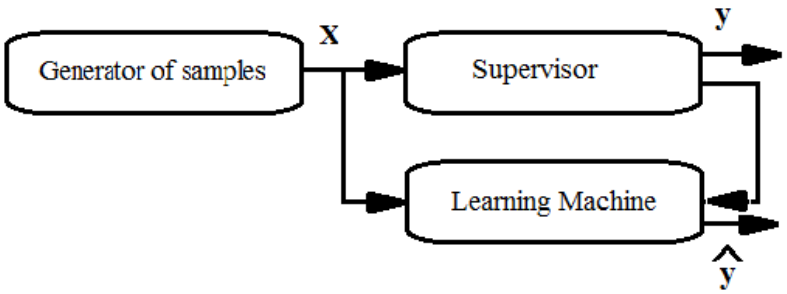


Figure 4-1. Fundamental statistical learning problem

There are two distinct types of error related to a learning machine: 1) the training error and 2) the generalization error. The former is the error in estimating the results corresponding to the given samples while the later is the error in assessing the outputs from samples not used in the training process. The quality of an approximation produced by a regression type learning machine is measured by the difference between the output produced by the *Supervisor* and the *Learning machine*, L(y, f(**x**,ω)). The expected value of the loss is called the *prediction risk functional* (Cherkassky and Mulier 2007).

$$E(\omega) = \int \mathrm{L}(y, f(\mathbf{x}, \omega))\, \mathrm{p}(\mathbf{x}, y)\, \mathrm{d}\mathbf{x}\mathrm{d}y \tag{4-1}$$

where $p(\mathbf{x}, y)$ is the unknown probability measure of the random variables **x** and y. Since this function cannot be calculated in its current format, an empirical risk is used on the basis of available data. For a regression problem, this function is the average of squared errors.

$$E_{emp}(\omega) = \frac{\sum_{i=1}^{n} (y - f(\boldsymbol{x}, \omega))^2}{n} \tag{4-2}$$

Under the ERM inductive principle, the optimal function f(**x**, ω) can be determined by minimizing the empirical risk. However, the goal of learning is to obtain a model providing minimal prediction risk; therefore the uncontrolled application of ERM based methods such as RSM can lead to overfitting the regression model to the training samples. This is due to the fact

that ERM imposes strict assumptions over the class of approximating functions. The discrepancy between the class of approximating functions and that of the actual governing function produces a high bias that impairs the generalization ability of the regressor. Using flexible-adaptive models based on SRM inductive principle, one can overcome this problem. SLT is the best currently available theory for flexible statistical estimation from finite samples (Cherkassky and Mulier 2007) which provides analytical generalization bounds for model selection. These bounds relate unknown prediction risk and known quantities such as the number of training samples, empirical error and a measure of model complexity called the VC-dimension. The suggested upper bound on prediction error for the regression problem which holds with the probability 1-η takes the form

$$E(\omega) = E_{emp}(\omega) \Big/ \left(1 - c\sqrt{a_1 \frac{h\left(\ln\left(\frac{a_2 n}{h}\right)+1\right) - \ln(\eta)}{n}}\right)_+ \tag{4-3}$$

where the VC dimension, h, is a characteristic of a set of functions, the parameter n is the number of training samples and c, a_1 and a_2 are constants. The sign "+" means the denominator should be positive. For a set of indicator functions, VC dimension is the maximum number of samples for which all possible binary labellings can be induced without error. In the case of linear real-valued functions, VC dimension equals the number of free parameters. The values of c, a_1 and a_2 are chosen based on the recommendations from Cherkassky and Mulier (2007) and Cherkassky et al. (1999). In addition, Vapnik (1995) recommends to choose the value of η equal to $\min(1,4/\sqrt{n})$ which is also adopted in the current study. Substituting these values in (4-3) gives the following upper bound on the prediction risk.

$$E(\omega) = E_{emp}(\omega) \Big/ \left(1 - \sqrt{p - p\ln(p) + \frac{\ln(n)}{2n}}\right)_+ \tag{4-4}$$

The generalization bounds (4-3) and (4-4) are particularly important for model selection under SRM inductive principle as they provide quantitative characterizations of the trade-off between the complexity of the approximating function and the quality of fitting the training data. In equation (4-4), the empirical error (numerator) decreases with increasing complexity, but the denominator also becomes very small causing the overall prediction risk to increase past an optimal VC dimension (Figure 4-2). Under SRM, the set S of approximating functions $f(\mathbf{x},\omega)$ consists of nested elements S_k ordered according to their VC dimension. For a given training data, SRM selects the function $f(\mathbf{x},\omega^*)$ which minimizes the empirical risk for the functions from element S_k,. Then, the prediction error is calculated for each element S_k using the bound provided by equation (4-4). The optimal structure element S_{opt} is the one providing minimum prediction error. This structure element has the optimal complexity (VC dimension).

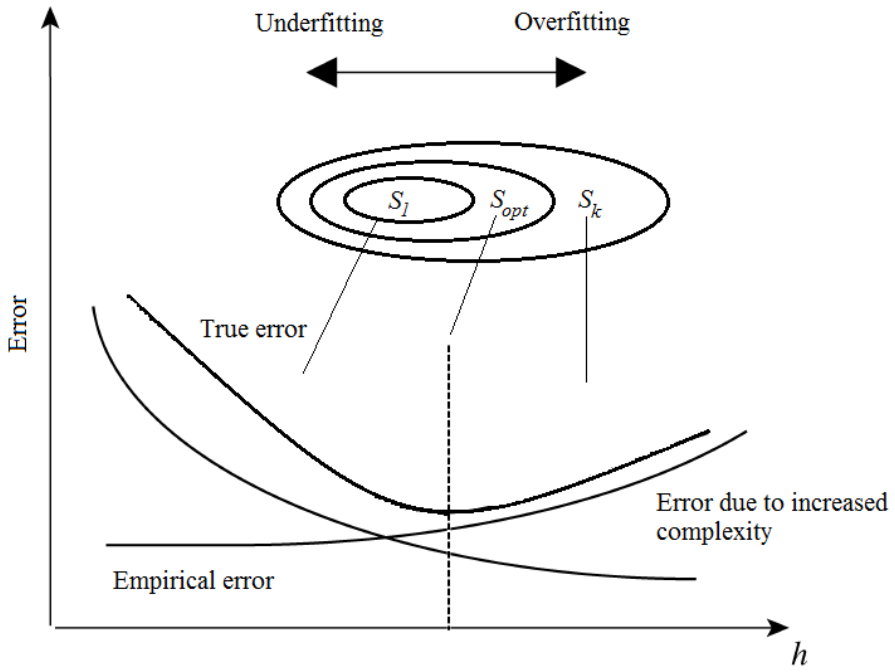


Figure 4-2. Model selection based on SRM inductive principle

It is notable that, since the prediction error determined from equation (4-4) is used for selecting the optimal meta-model, it is optimistic and cannot be used as an unbiased prediction error for

evaluating the selected model or comparing it with other meta-models. An honest estimate of the prediction risk can be found using a resampling procedure. For this purpose a K-fold CV technique is used based on discussions presented by Rodrıguez et al (2010). The method mimics the use of training and test sets. The training algorithm is repeated K times (K = 10 in this application) with a fraction 1/K of the sample points left out for testing purposes. The CV estimator is defined as the average of the errors on test folds. The procedure is meant to lower the variance of the prediction risk estimate (Bengio and Grandvalet 2004).

4.3 Transmission line description and modeling

The methodology is applied to a 2.6 km long 9 span segment of a typical 120 kV transmission line located in North America. Figure 4-3 shows the studied segment of the line. It consists of two line sections separated by angle/dead-end towers. It is noted that in this study, ten towers are included in the system reliability analysis due to the limited spatial extension of climatic hazards. According to CEI/IEC 60826 (2003) and CAN/CSA-C22.3 (2006), in a flat to rolling terrain and based on a span of 400 m, 1 to 5 towers are usually subjected to the maximum wind and the maximum combined ice and wind loads while 10 to 20 towers are usually subjected to the maximum ice loads.

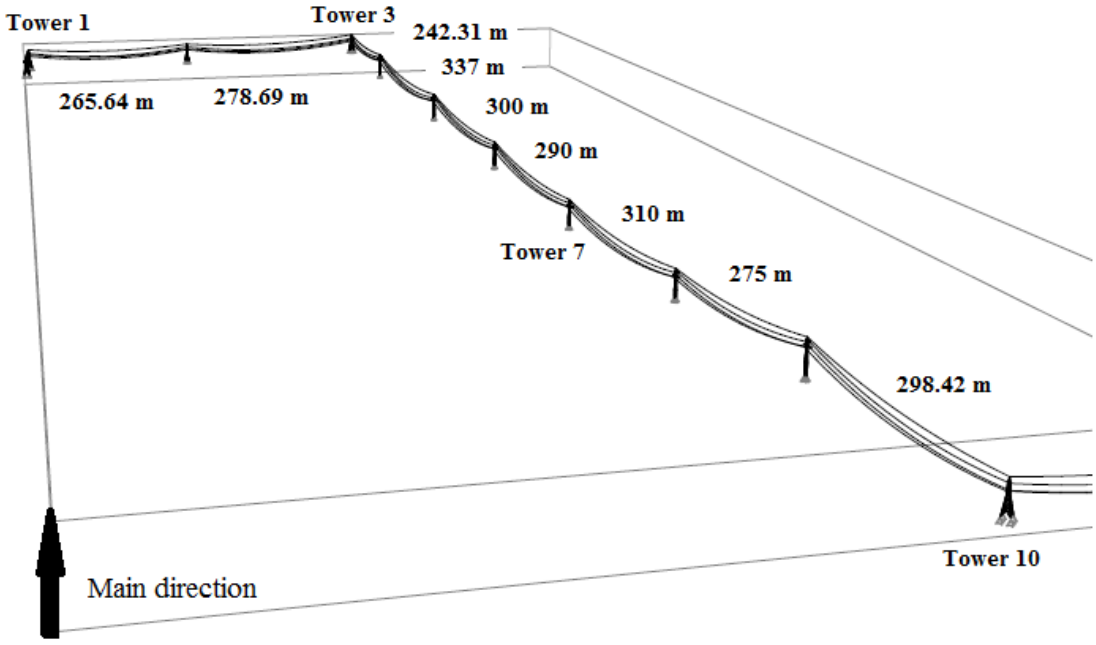


Figure 4-3. Nine span segment of the transmission line under study

The diameter and weight per unit length of conductors are 27.78 mm and 15.26 N/m, respectively, and the diameter and weight per unit length of ground wires are 12.57 mm and 7.44 N/m, respectively. Table 4-1 presents a description of towers within the studied line segment. HS and LS towers designate heavy suspension and light suspension towers, respectively while STR towers designate angle/dead-end towers. The configuration of the three types of towers are shown in figure 4-4.

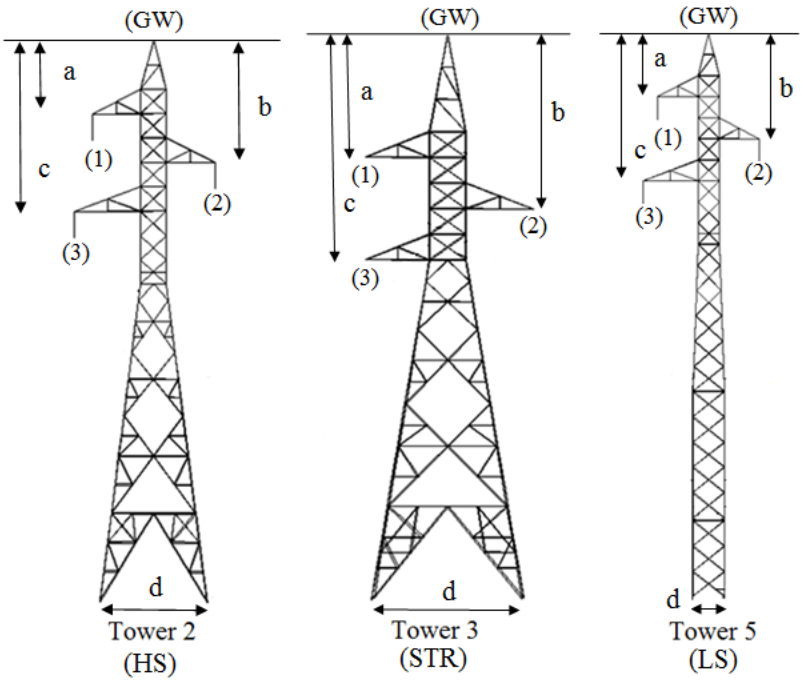


Figure 4-4. The configuration of tower types within the line

Table 4-1. Description of towers of the line segment

Tower Number	Tower Height(m)	a(m)	b(m)	c(m)	d(m)	Tower Type
1	30.5	6	8.5	11	8.06	STR
2	28.75	3.75	6.25	8.75	5.26	HS
3	27.5	6	8.5	11	7.08	STR
4	33.25	3.75	6.25	8.75	1.9	LS
5	33.25	3.75	6.25	8.75	1.9	LS
6	30.25	3.75	6.25	8.75	1.9	LS
7	30.25	3.75	6.25	8.75	1.9	LS
8	28.75	3.75	6.25	8.75	1.9	LS
9	33.25	3.75	6.25	8.75	1.9	LS
10	27.5	6	8.5	11	7.08	STR

A Visual Basic code is developed to prepare a three dimensional model of the line in the computer software SAP2000 (SAP2000, 1996). The model takes into account the flexibility of the towers and the resulting interactions between different components of the transmission line. The conductors and ground wires are modeled using cable elements with tension-only behaviour, the insulators are modeled with truss elements, and the overhead towers are modeled as frame-truss systems in that leg members with multi-bolted connections are modelled as frames with

rigid connections, and horizontal and diagonal members with single-bolted connections are modelled as truss elements with pinned connections. Wind, ice and wind on ice loads are applied on the wire system and the supports following CEI/IEC 60826 (2003). It is noted that the wind and wind on ice loads are calculated by taking into account the influence of the height, shape and surface properties of each member as well as the terrain category and the dynamic response of components. In addition, it is assumed that ice loads are applied to line components as a result of freezing rain events, and the density of ice is assumed to be equal to 900 kg/mm^3.

4.4 Methodology

Component-based structural system reliability analysis consists of two main steps. First, it is required to obtain an explicit approximation of the limit-state for each tower element, conductor, ground wire and insulator. It should be mentioned that foundations are not addressed in this study and may be considered in future studies when experimental data becomes available. Using the VC-generalization bound of equation 4-4, an explicit meta-model is selected for each tower element, conductor, ground wire and insulator that relates the structural demand to weather-related variables such as wind speed, wind direction and ice thickness. After determining an approximate close-form limit state function from the selected demand meta-model, MC simulations are performed to estimate the failure probability of each component or element. This step is very fast since the performance function is analytical. Second, the tower failure probabilities and the overall failure probability of the line can be calculated from the failure probabilities of tower elements and line components, respectively, considering the correlation between their failure events. A more detailed description of the required steps is presented in this section.

4.4.1. Weather statistics

The parameters of the joint distribution for concurrent wind speed and ice thickness were determined by assuming that the 50 year return period maximum ice thickness accumulated on a 25 mm conductor 10 meters above ground is 45 mm. The ice weight is then adjusted for height and cable diameter using the equations proposed in CAN/CSA-C22.3 (2006). Also, the 10 minute average maximum wind-on-ice speed at 10 meters of height above ground is assumed as 70% of the 50 year return period extreme wind speed and is 21.64 m/sec (CAN/CSA-C22.3, 2006). CAN/CSA-C22.3 (2006) presents climatic coefficients to derive the values of these variables for 25, 100, 150, 200, 400, and 500 years based on the 50 year design value for the Gumbel distribution. These coefficients are used to estimate the mean value and standard deviation for maximum wind speed and ice thickness variables. The corresponding values are 15.64 (m/sec) and 2 (m/sec) for wind speed during icing conditions and 32.2 (mm) and 4.1 (mm) for ice thickness, respectively. It is assumed that wind angle which is calculated with respect to the main direction of the line has a uniform distribution, while wind speed and ice thickness follow the Gumbel distribution (CAN/CSA-C22.3, 2006, CEI/IEC 60826, 2003). The temperature is assumed to be deterministic and equal to -5 °C during icing events (CAN/CSA-C22.3, 2006, CEI/IEC 60826, 2003). Figure 4-5 indicates the histograms of the simulated wind speeds, ice thicknesses and wind angles using Monte Carlo simulation for 10^8 events. Considering that the epistemic uncertainties in the evaluation of ice and wind induced component loads are neglected, the resulting values may be conservative but are coherent with the assumption of CAN/CSA-C22.2 (2006). Note that for an actual line, site specific data on wind speed and ice accumulation could be used to obtain more accurate estimates of atmospheric hazards in comparison to code derived distributions (Chouinard et al. 1998).

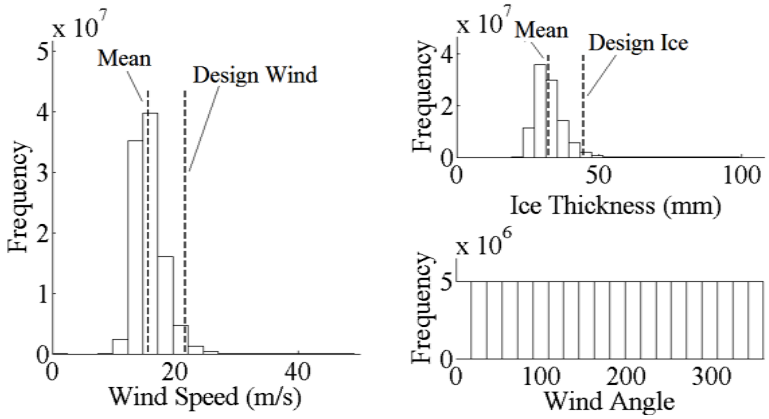


Figure 4-5. Histograms of simulated climatic variables

4.4.2. Defining the limit state function

In order to calculate the failure probability of each structural component of the line, it is required to describe its performance by a computational model called the performance or limit state function and denoted as G(**x**). The limit state function is normally a function of all the intervening random variables, some of which may affect the structural demand denoted by D, while the others may influence the load carrying capacity, denoted by C. Therefore, G(**x**) can be written as Equation 4-5.

$$G(\mathbf{x}) = C(\mathbf{x_c}) - D(\mathbf{x_d}) \quad (4\text{-}5)$$

where $\mathbf{x_c}$ includes all the random variables related to the capacity and $\mathbf{x_d}$ includes all the random variables related to the demand. The probability of non-performance is the probability of the event $G < 0$ or $D > C$, whereas the reliability corresponds to the probability of the event $G > 0$ or $D < C$.

In order to obtain an analytical function of G(**x**) in this application, it is sufficient to determine an approximate substitute function for $D(\mathbf{x_d})$, where $\mathbf{x_d}$ includes climatic variables such as wind speed, wind angle and ice thickness. This is done by fitting a polynomial function based on

selected points from the distribution of climatic variables. The goal is to have a learning machine that can predict accurately this function with the simplest model possible. Table 4-2 shows the range of wind speed, ice thickness and wind angle used for the full factorial design of weather combinations for which structural analysis is performed by SAP2000. The training sample size of 1287 is chosen so that the upper generalization bound of equation (4-4) holds with the probability of 90% ($\eta=\min(1,4/\sqrt{n})$ as suggested by Vapnik (1995)). Although higher probability levels can be achieved by increasing the sample size, the additional time and computational effort required for analysing the line with SAP2000 promotes the idea of keeping a balance between the target confidence level and the sample size.

Table 4-2. Range of values for climatic variables in the full factorial design for analysis with SAP2000®

Climatic Variable	Wind Speed (m/s)	Ice Thickness (mm)	Wind Angle (Degree)
Lower Bound	0	0	0
Upper Bound	50	80	360
Step Size	5	10	30

Equation 4-4 provides an upper limit for the complexity of polynomial functions to be tested in terms of VC-dimension or degree of the polynomial. Each candidate polynomial of a given degree is fitted to training samples and the optimal polynomial function is identified by minimizing the prediction error calculated with the VC-generalization bounds. In order to exclude non-significant variables and further reduce the complexity of the model, a stepwise regression method is used to fit each candidate function to the training samples. Stepwise regression is a variable selection process that selects an optimal subset of variables according to a certain criterion. During the procedure, a sequence of regression models is constructed iteratively by adding or removing variables at each step (Miller 2002, Montgomery and Runger 2003). Stepwise regression is used as a variable selection method because of its simplicity, computational efficiency and good prediction ability of the resulting model.

The estimated accuracy of the model relative to the training data may not reflect its true accuracy. A better estimate of the accuracy is achieved by using the CV technique. For this purpose, the data set from the structural analysis of the line using SAP2000 is randomly partitioned into 10 subsets of similar size. Each subset is used in turn as a test set for models produced on the basis of the remaining data which is called the training set. The CV estimator is defined as the average of the errors on all subsets.

The adjusted coefficient of multiple determination ($\bar{R}^2$) is calculated based on equation 4-6 to express the quality of fit between the regression model and the sample data.

$$\bar{R}^2 = 1 - \frac{MSE}{MST} = 1 - \left(\frac{n-1}{n-p}\right)(1 - R^2) \tag{4-6}$$

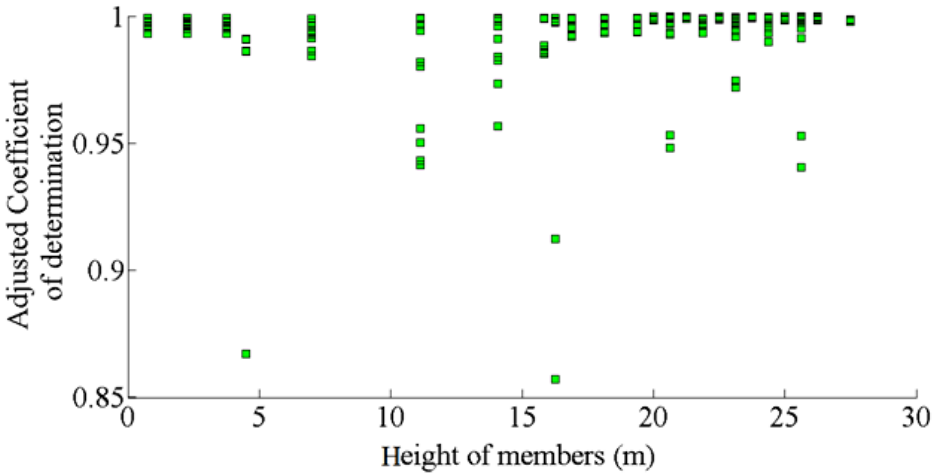
where *MSE* is the error mean square, *MST* is the total mean square of variation in observations, *n* is the total number of samples, *p* is the number of parameters in the model and R^2 is the coefficient of multiple determination. $\bar{R}^2$ is a useful tool in evaluating the regression models as it guards against overfitting by penalizing the analyst for adding terms to the model (Montgomery and Runger 2003).

To evaluate the ability of the final selected model to predict future demands, the prediction coefficient of multiple determination ($\bar{R}_p^2$) is calculated using equation 4-7.

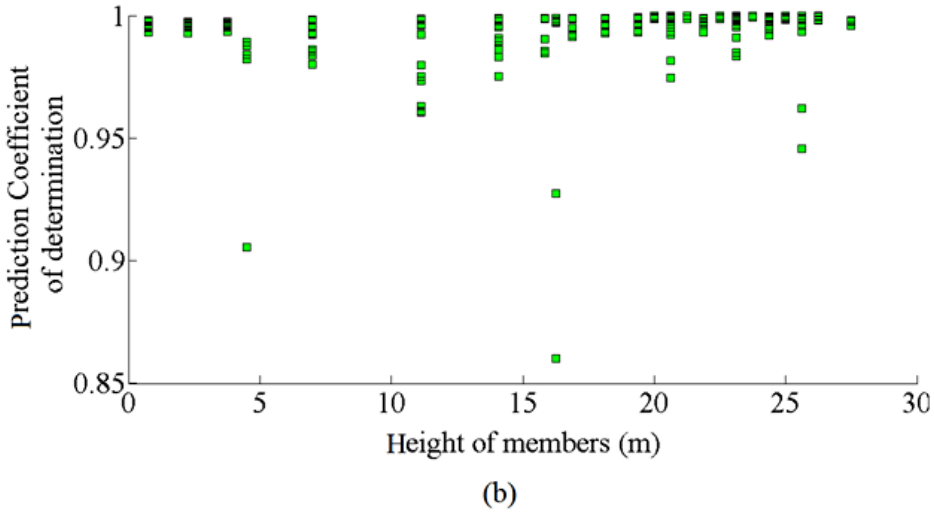
$$R_p^2 = 1 - \frac{\sum_{i=1}^{n_{te}} (y_{ite} - \hat{y}_{itr})^2}{\sum_{i=1}^{n_{te}} (y_{ite} - \bar{y}_{te})^2} \tag{4-7}$$

where n_{te} and $\bar{y}_{te}$ are respectively the number and mean of the observed demands (y_{ite}) in the test subset and $\hat{y}_{itr}$ are the predictions of the demands for the test subset using the model fitted to the training subset. Figure 4-6 presents $\bar{R}^2$ and $\bar{R}_p^2$ for the demands for each member of tower 2 as a

function of its elevation in the tower. In this figure $\bar{R}_p^2$ is the average prediction coefficient of determination for all 10 test subsets. It is inferred from the results that the proposed approach provides a good fit of the demand model to SAP2000 results for most line members as the values of $\bar{R}^2$ and $\bar{R}_p^2$ for 99 percent of members are greater than 0.95. Members with lower values of coefficient of multiple determination have negligible contributions to the overall structural failure probability because they are mostly horizontal truss members with low demands compared to member capacities.



(a)



(b)

Figure 4-6. (a) Adjusted coefficient of determination (b) prediction coefficient of determination for demands for each member of tower 2

4.4.3. Structural component demands and capacities

After determining the performance function for each structural component of the transmission line, structural demands are simulated by randomly sampling from the distributions for the climatic variables and substituting into the selected response function for that component. Figures 4-7 to 4-9 compare the resulting forces for the most critical component of tower 2 (element 204 as shown in figure 4-10) obtained from the structural analysis using SAP2000 and the surrogate regression model.

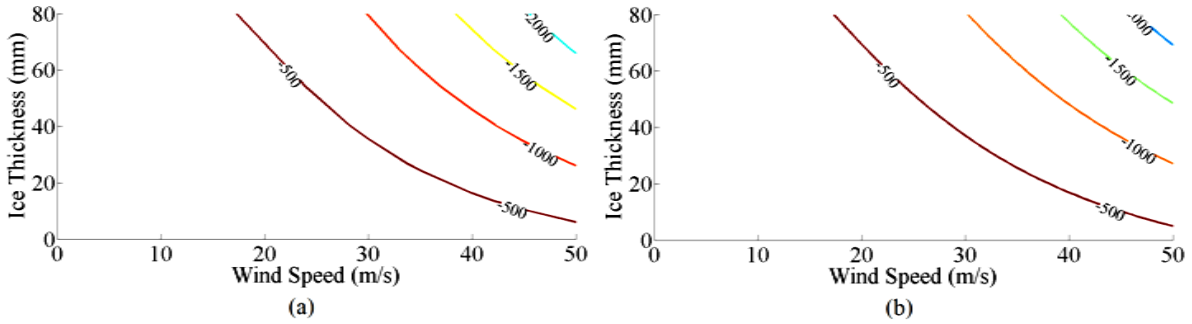


Figure 4-7. Axial forces (kN) for the most critical component of tower 2 as a function of wind speed and ice accumulation (wind angle=0) obtained from (a) structural analysis using SAP2000 (b) the surrogate regression model

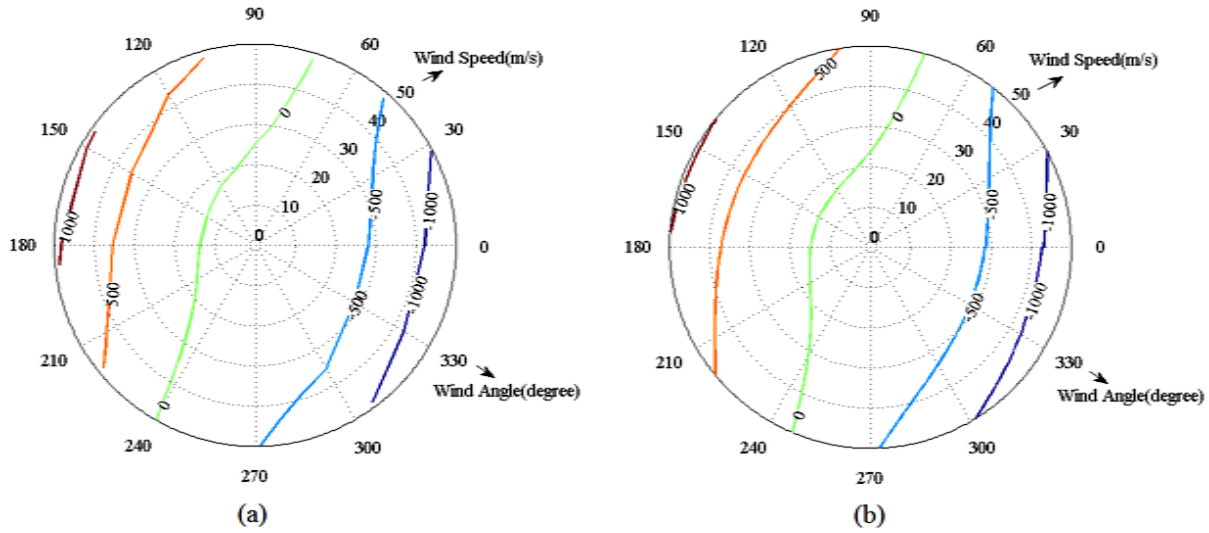


Figure 4-8. Axial forces (kN) for the most critical component of tower 2 as a function of wind speed and wind direction (ice thickness=41 mm) obtained from (a) structural analysis using SAP2000 (b) the surrogate regression model

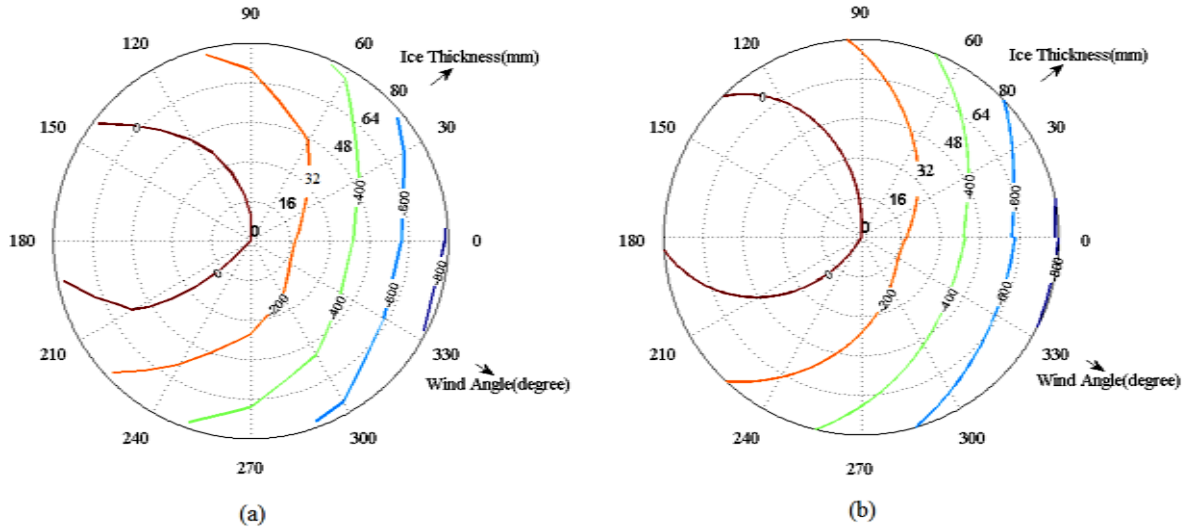


Figure 4-9. Axial forces (kN) for the most critical component of tower 2 as a function of ice thickness and wind direction (wind speed=20 m/sec) obtained from (a) structural analysis using SAP2000 (b) the surrogate regression model

The capacity for each member of the towers is calculated based on ASCE10-97 (2003). The capacity of tension members is assumed to be equal to the minimum of the bearing capacity of the connection, the shear capacity of the bolts and the tension capacity of the member. Similarly, the capacity of the compression members is assumed to be equal to the minimum of the bearing capacity of the connection, the shear capacity of the bolts and the compression capacity of the member. The limitation of this assumption is that it implies perfect correlation between different failure modes of each tower member. It is assumed that the capacity of overhead tower components have a lognormal distribution with a coefficient of variation of 10% which is consistent with assumptions used in CEI/IEC 60826 (2003).

The nominal tensile strength of conductors and ground wires are 142 kN and 139 kN, respectively. They are assumed to have a lognormal distribution with a coefficient of variation of 3% (CEI/IEC 60826, 2003). Three groups of insulators with nominal capacities of 70 kN (towers 2, 6, 7), 140 kN (towers 4-5, 8-9) and 240 kN (towers 1, 3, 10) are used in the studied line

section. It is assumed that the capacity of insulators have a lognormal distribution with a coefficient of variation of 5% (CEI/IEC 60826, 2003). In addition, following CEI/IEC 60826 (2003), it is assumed that the probability of each element of the line not meeting the nominal strength is 10%.

4.4.4 Estimating component and system failure probabilities

Using Monte Carlo simulation, the yearly probability of failure of each conductor, ground wire, insulator or member of the overhead towers is calculated as

$$P_f = Pr(\mathrm{G}(\mathbf{x}) \leq 0) \cong Pr(\mathrm{f}(\mathbf{x}) \leq 0) \cong \frac{N_f}{N} \tag{4-8}$$

where G(**x**) is the limit state or performance function, f(**x**) is the surrogate function selected based on SLT concepts, N is the total number of simulations of climatic variables ,and N_f is the number of events for which the structural demand exceeds the structural capacity of the studied component ($\mathrm{f}(\mathbf{x}) \leq 0$). In this study, considering the small failure probabilities of some components or tower elements, 10^8 simulations are used to estimate failure probabilities in the order of 10^{-6} with a target coefficient of variation of 0.1 as suggested by Lemaire (2013). Table 4-3 summarizes the estimates of failure probabilities for conductors of the line section, while table 4-4 summarises the failure probabilities of insulators. The failure probability of ground wires and most insulator strings are very low such that not a single failure is detected in 10^8 simulations. Due to the small failure probability of these members compared to other members of the line, their contribution to the system failure can be neglected.

Table 4-3. Failure probabilities of conductors and ground wires

	Failure Probability		
	Conductor 1*	Conductor 2*	Conductor 3*
Span 1	0.072×10^{-3}	0.073×10^{-3}	0.078×10^{-3}
Span 2	0.076×10^{-3}	0.074×10^{-3}	0.083×10^{-3}
Span 3	0.251×10^{-3}	0.219×10^{-3}	0.200×10^{-3}
Span 4	0.291×10^{-3}	0.261×10^{-3}	0.233×10^{-3}
Span 5	0.274×10^{-3}	0.248×10^{-3}	0.220×10^{-3}
Span 6	0.268×10^{-3}	0.240×10^{-3}	0.211×10^{-3}
Span 7	0.247×10^{-3}	0.221×10^{-3}	0.196×10^{-3}
Span 8	0.271×10^{-3}	0.244×10^{-3}	0.215×10^{-3}
Span 9	0.256×10^{-3}	0.233×10^{-3}	0.209×10^{-3}

* Conductors are numbered based on the elevation of their attachment point with 1 being the highest elevation as shown in figure 4-4.

Table 4-4. Failure probabilities of insulator strings

	Failure Probability		
	Insulator. 1	Insulator. 2	Insulator. 3
Tower 2	6.1×10^{-7}	8.4×10^{-7}	3.5×10^{-8}
Tower 7	5.5×10^{-6}	4.4×10^{-6}	3.5×10^{-6}
Tower 8	1.3×10^{-6}	9.3×10^{-7}	6×10^{-7}

* Insulators are numbered based on the elevation of their attachment point with 1 being the highest elevation as shown in figure 4-4.

For the calculation of their failure probability, the towers are represented as series systems. This assumption is conservative and implies that the tower fails once any one of its structural components fails. In addition, due to common material sources and external loads, failure modes of different tower members are correlated. In this study, it is assumed that this correlation is mainly due to the same climatic conditions, and the capacity of tower members are simulated independently. Higher correlations would increase the probability of simultaneous failure of multiple members for given combinations of wind speed, ice thickness and wind angle and would reduce the overall probability of failure of the line system. Table 4-5 identifies the 10 most critical components of three towers (one of each type) and their corresponding failure probabilities as well as the overall failure probability of each tower. The locations of these components are shown in Figure 4-10.

Table 4-5. Failure probabilities of 10 most critical components of three types of towers

Tower 2 (HS)		Tower 3 (STR)		Tower 5 (LS)	
Comp.	Failure Prob.	Comp.	Failure Prob.	Comp.	Failure Prob.
204	1.2×10^{-3}	172	2.9×10^{-5}	34	2.6×10^{-3}
207	1.0×10^{-3}	171	2.2×10^{-5}	35	2.6×10^{-3}
408	0.9×10^{-3}	174	2.2×10^{-5}	36	2.5×10^{-3}
406	0.5×10^{-3}	17	2.0×10^{-5}	33	2.1×10^{-3}
405	0.3×10^{-3}	173	1.6×10^{-5}	12	1.9×10^{-3}
409	0.3×10^{-3}	1	1.4×10^{-5}	1	1.9×10^{-3}
3	0.3×10^{-3}	136	1.0×10^{-5}	7	1.7×10^{-3}
22	0.2×10^{-3}	237	0.8×10^{-5}	56	1.7×10^{-3}
403	0.2×10^{-3}	188	0.7×10^{-5}	192	1.6×10^{-3}
402	0.2×10^{-3}	177	0.6×10^{-5}	198	1.6×10^{-3}
System	2.7×10^{-3}	System	1.1×10^{-4}	System	9×10^{-3}

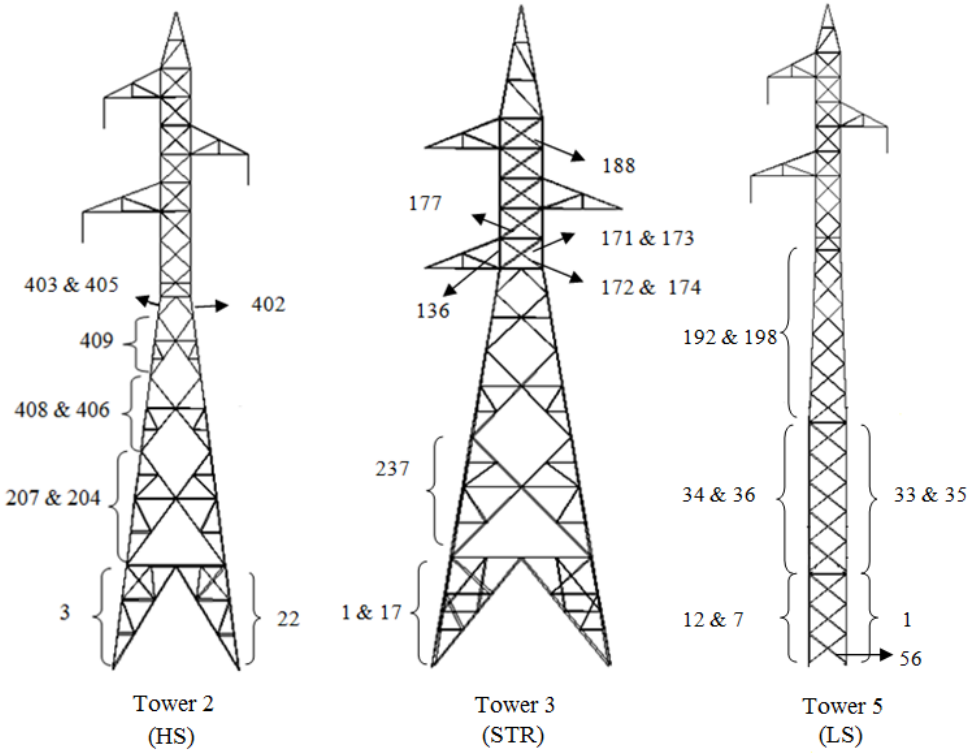


Figure 4-10. Location of most critical components in each tower

The results show that, as expected, critical tower elements are mainly leg members located at the lower part of towers, where aggregate demands concentrate. In addition, suspension towers (Type HS and LS) have higher failure probabilities compared to other components of the line.

This is consistent with the fact that current design procedures are based on a specific strength coordination in which the suspension towers are designed as weakest links in the transmission line system (CAN/CSA-C22.3, 2006, CEI/IEC 60826, 2003). It should be mentioned that the failure probability of foundations are not addressed in this study and may be considered in future studies when experimental data becomes available.

The calculation of the overall failure probability of the studied transmission line is obtained in a similar way as that of each tower. The failure probability of the whole line section is obtained as the ratio of the number of simulated climatic combinations that result in the failure of at least one component of the line to the total number of simulations. The resulting failure probability of the line section is 0.0154, which corresponds to a reliability index of 2.16.

System failure probability of series systems such as the towers and the line section can also be estimated using Ditlevsen Bounds. Upper and lower bounds are established on the basis of component failure probabilities and all pairwise component intersection failure probabilities as indicated in equations 4-9 and 4-10, respectively (Ditlevsen 1979).

$$Pr(F) \geq Pr(F_1) + \sum_{i=2}^{m} \max\left\{Pr(F_i) - \sum_{j=1}^{i-1} Pr(F_i \cap F_j), 0\right\} \quad (4\text{-}9)$$

$$Pr(F) \leq \sum_{i}^{m} Pr(F_i) - \sum_{i=2}^{m} \max_{j<i} Pr(F_i \cap F_j) \quad (4\text{-}10)$$

where the failure probabilities of *m* components are arranged in the order of increasing reliability to produce narrower bounds. In this study, the pairwise intersections of events are calculated using MC simulation. In addition, perfectly correlated and independent systems are two special cases which result in linear bounds (lower bound and upper bound respectively).

Table 4-6. Comparison of calculated failure probabilities using MC simulation and Ditlevsen bounds

	MC Simulation	Linear Bounds [Lower Bound-Upper Bound]	Second Order Bounds [Lower Bound-Upper Bound]
Tower 1	1.6×10^{-5}	5.3×10^{-6} - 2.5×10^{-5}	1.3×10^{-5} - 1.8×10^{-5}
Tower 2	2.7×10^{-3}	1.2×10^{-3} - 9.0×10^{-3}	1.9×10^{-3} - 3.6×10^{-3}
Tower 3	1.1×10^{-4}	2.9×10^{-5} - 2.4×10^{-4}	6.2×10^{-5} - 1.5×10^{-4}
Tower 4	8.7×10^{-3}	3.3×10^{-3} - 6.4×10^{-2}	4.7×10^{-3} - 1.7×10^{-2}
Tower 5	9.0×10^{-3}	2.6×10^{-3} - 7.8×10^{-2}	4.0×10^{-3} - 2.0×10^{-2}
Tower 6	3.9×10^{-3}	1.1×10^{-3} - 3.0×10^{-2}	1.6×10^{-3} - 8.6×10^{-3}
Tower 7	3.5×10^{-3}	1.0×10^{-3} - 2.9×10^{-2}	1.5×10^{-3} - 8.2×10^{-3}
Tower 8	4.4×10^{-3}	1.7×10^{-3} - 3.1×10^{-2}	2.6×10^{-3} - 7.8×10^{-3}
Tower 9	6.9×10^{-3}	1.5×10^{-3} - 4.7×10^{-2}	3.9×10^{-3} - 1.4×10^{-2}
Tower 10	1.1×10^{-4}	2.2×10^{-5} - 2.6×10^{-4}	5.5×10^{-5} - 1.6×10^{-4}
Line section	1.54×10^{-2}	9.0×10^{-3} - 3.9×10^{-2}	1.2×10^{-2} - 1.6×10^{-2}

Table 4-6 compares the calculated failure probabilities using MC simulation with linear and second order Ditlevsen bounds. The results presented in this table highlight the effect of having correlation between failure events of different tower members. As mentioned previously, higher correlations would increase the probability of simultaneous failure of multiple members for given climatic conditions and would reduce the overall probability of failure of the line. It is noted that for the studied line, the estimated second order Ditlevsen bounds for the system failure probability are narrower and more practical and realistic compared to the first order Ditlevsen bounds as they are calculated by taking into account the pairwise correlation between component failures. The failure probability obtained from MC simulation is more accurate and lies within both first order and second order bounds because it implicitly accounts for all correlations between members due to the common climatic variables.

4.4.5. Target reliability of transmission lines

For the reliability assessment of transmission lines, it is required to specify the target reliability level. According to ISO 2394 (1998), the target reliability of a structure should be selected based on the consequence and nature of failure, the economic losses, the social inconvenience and the amount of expense and effort required to reduce the probability of failure. Therefore, the acceptable level of reliability for a structure is determined by solving an optimization problem between the consequences of failure and the costs of safety measures. In other words, the objective is to minimize the total service-life cost.

CEI/IEC 60826 (2003) and CAN/CSA-C22.3 (2006) propose three reliability levels for transmission lines based on their importance within the supply network.

1) All lines should at least have a reliability level characterized by setting the 50 year return period loads equal to the strengths with an exclusion limit of 10% (level 1).
2) Lines below 230 kV which constitute the principal source of electric supply to a region or lines above 230 kV should be designed for a reliability level characterized by setting the 150 year return period loads equal to the strengths with an exclusion limit of 10% (level 2).
3) Lines above 230 kV which constitute the principal source of electric supply to a region should be designed for a reliability level characterized by setting a 500 year return period loads equal to the strengths with an exclusion limit of 10% (level 3).

According to CEI/IEC 60826 (2003) and CAN/CSA-C22.3 (2006), these reliability levels result in an almost constant yearly reliability between (1-1/T) and (1-1/2T), independent of the shape of the load and strength curves and the coefficients of variation of each variable. Therefore, the

studied 120 kV transmission line should have a level 1 reliability with an expected annual system failure probability between 0.01 and 0.02. It is noted that the obtained annual system failure probability of 0.0154 lies between the expected range and the line satisfies the reliability specifications proposed by the aforementioned design standards.

4.5 Calculating component importance measures

In order to measure the relative importance of a component in a system, many alternative importance measures can be used (Kuo and Zhu 2012b, 2012c). It should be mentioned that different perspectives on the same system can lead to different views about which factors make one component more important than another. Therefore, no single type of importance measure is universal. In this study, two different reliability importance measures are used for each component of the line section: the improvement potential importance and the Bayesian reliability importance measures.

4.5.1. The improvement potential importance measure

When limited resources are available for upgrading a transmission line system, it is best to put the resources into the components whose upgrade could bring the greatest improvement in system performance and reliability. The problem is to decide which components should be further improved to gain the greatest improvement in system reliability at a limited cost or effort. The relative importance of components in such a case could be ranked according to the improvement potential importance which is designed to determine which components merit the most additional research and development to improve overall systems reliability at minimum cost or effort (Kuo and Zhu 2012).

As indicated in equation 4-11, the improvement potential importance ($I_{ip}(i)$) of component *i* in system Φ_s is equal to the increase of system reliability (or decrease of system failure probability) when component *i* is perfect (with reliability of one). It represents the maximum potential improvement in systems reliability that can be obtained by improving the reliability of component *i*.

$$I_{ip}(i) = \Pr\{\Phi_s(x) = 1 | x_i = 1\} - \Pr\{\Phi_s(x) = 1\} \quad (4\text{-}11)$$

where $\Phi_s(x)$ and x_i are indicator functions for the system and component *i* with values of 0 and 1 representing failure and perfectly reliable, respectively. In this paper, in order to produce a more understandable measure, a normalized version of $I_{ip}(i)$ is obtained by dividing its value by the system failure probability. It is noted that further improvement of components of very high reliability may be very costly. Therefore, cost constraints and other factors may be involved to limit the optimal action of a system upgrade. However, since cost information usually is rather difficult to find, in order to make final decisions on how to allocate resources for reliability improvement, it seems worthwhile to look for costs of improvement balanced against the improvement in system reliability only for those components with large values of the improvement potential importance (Kuo and Zhu 2012).

4.5.2. The Bayesian reliability importance measure

In system reliability analysis, it is of interest to identify which component is more likely to cause a system failure, and to use this information to define component importance relative to system failure. When a system fails, a systems analyst needs to find the components that caused the system failure and then repair them to restore the system as soon as possible. Therefore, the analyst has to assign priorities to the components in the system and then check and repair the

components in the order of these priorities. Using the Bayesian reliability importance, these priorities can be assigned so that the systems analyst can generate a repair checklist to sequence the components that have likely failed and caused or contributed to the system failure (Kuo and Zhu 2012). This is particularly important for transmission lines due to their spatial extension and accessibility limitations and costs especially during and after extreme winter storms. To facilitate the process, the components with large Bayesian reliability importance measure values are checked first in the fault diagnosis (Kuo and Zhu 2012).

The Bayesian reliability importance of component *i*, $I_{Bay}(i)$, is defined as the probability that component *i* has failed, given that the system Φ_s failed.

$$I_{Bay}(i) = \Pr\{x_i = 0 | \Phi_s(x) = 0\} \quad (4\text{-}12)$$

4.5.3. Results of importance measures

Table 4-7. Bayesian reliability importance and improvement potential importance for components of the line section

	Bayesian Reliability Importance	Proposed check list (1)	Normalized Improvement Potential Importance (%)	Proposed check list (2)
Tower 1	0.0010	11	0	10
Tower 2	0.1751	8	12.9	2
Tower 3	0.0074	9	0.01	8
Tower 4	0.5677	2	15.3	1
Tower 5	0.5863	1	10.4	3
Tower 6	0.2553	6	0.42	6
Tower 7	0.2266	7	0.17	7
Tower 8	0.2835	5	0.85	5
Tower 9	0.4480	3	7.6	4
Tower 10	0.0074	10	0.01	9
Wire System	0.0356	4	0	11
Insulators	0.0005	12	0	12

Table 4-7 presents the aforementioned importance measures for different components of the line section. Two check lists are proposed based on the Bayesian reliability importance and the improvement potential importance to help utility owners make well founded decisions for two

cases of fault diagnosis after a climatic hazard or upgrading a transmission line with limited resources, respectively. It is inferred that in a series system, arranging transmission line components based on their Bayesian reliability importance measure provides the same order of important components as arranging them based on their failure probability. In other words, the component that has higher failure probability contributes more to the failure of the system. However, normalized improvement potential importance measure provides a different order of important components. This is due to the fact that it considers the correlation of each component with other components of the line as well as its failure probability. This is the reason why tower 2, which is a HS type suspension tower in section 1, seems less important than most of LS type suspension towers in section 2 based on Bayesian reliability importance (ranked 8 in checklist (1)), while it offers higher potential improvement in systems reliability (ranked 2 in checklist (2)). The obtained importance measures offer valuable information that can be used to optimize fund allocation and determine the most efficient retrofit strategy to enhance the overall reliability of the line.

4.6 Determining the design point

Figure 4-11 indicates the climatic combinations that result in the failure of the line section. In this figure the star represents the design point which is the combination of wind speed, wind angle and ice thickness that results in the failure of system and has the highest likelihood. In order to obtain the design point, all combinations of climatic variables that result in the failure of the line section are investigated to determine which has the highest value of the joint probability density function of wind speed, wind angle, ice thickness and capacity. Assuming that all these

variables are independent, the joint probability density function can be calculated from equation 4-13.

$$f_{W_S W_a IR}(w_s, w_a, i, r) = f_{W_S}(w_s) \times f_{W_a}(w_a) \times f_I(i) \times f_R(r) \quad (4\text{-}13)$$

where W_S, W_a, I and R represent wind speed, wind angle, ice thickness and capacity, respectively. $f_{Ws,}$ f_{Wa}, f_I and f_R are the marginal probability density functions and f_{WsWaIR} is the joint probability density function of these variables. It is noted that, the assumption of independence between wind speed (and direction) and ice accumulation is consistent with those used in deriving design criteria for combined wind and ice loads in Canadian and international design codes (CAN/CSA-C22.3 2006, CEI/IEC 60826 2003) and has also been validated empirically by several researchers (El-Fashny 2002, Yang et al. 2013). To clarify, the distribution for ice accumulation is derived for the maximum ice accumulation during an ice storm event and the distribution of wind is obtained from the sample of maximum winds coincident with the presence of ice on the structure. Ice accumulation is dependent on the average wind speed during the ice storm but not with the maximum wind speed that is coincident with the presence of ice on the structures.

The design point for the system in this case corresponds to a combination of wind speed of 82 km/h (22.74 m/s), an ice thickness of 33 mm and an orientation of 82 degrees (perpendicular to the axis of the long segment of the line section). These values can be compared to the values used in design for combined wind and ice loads. According to CEI/IEC 60826 (2003) and CAN/CSA-C22.3 (2006), two loading combinations should be considered for the design of transmission lines under combined wind and ice loads. The first combination corresponds to low ice probability (50 year return period) associated with the average of yearly maximum winds

during icing presence. For the location of the studied transmission line, the corresponding values are 45 mm and 56 km/h (15.64 m/s), respectively. The second combination corresponds to low probability wind during icing (50 year return period) associated with the average of the maximum yearly icing. These values are 78 km/s (21.64 m/s) and 32.2 mm, respectively. It is noted that the most probable failure point estimated from the reliability analysis of the line is similar to the design combination of low probability wind during icing and high probability ice suggested by current standards. It is inferred that the studied line most likely fails under low amounts of ice accompanied by high winds blowing perpendicular to the main direction of the line as shown in figure 4-3.

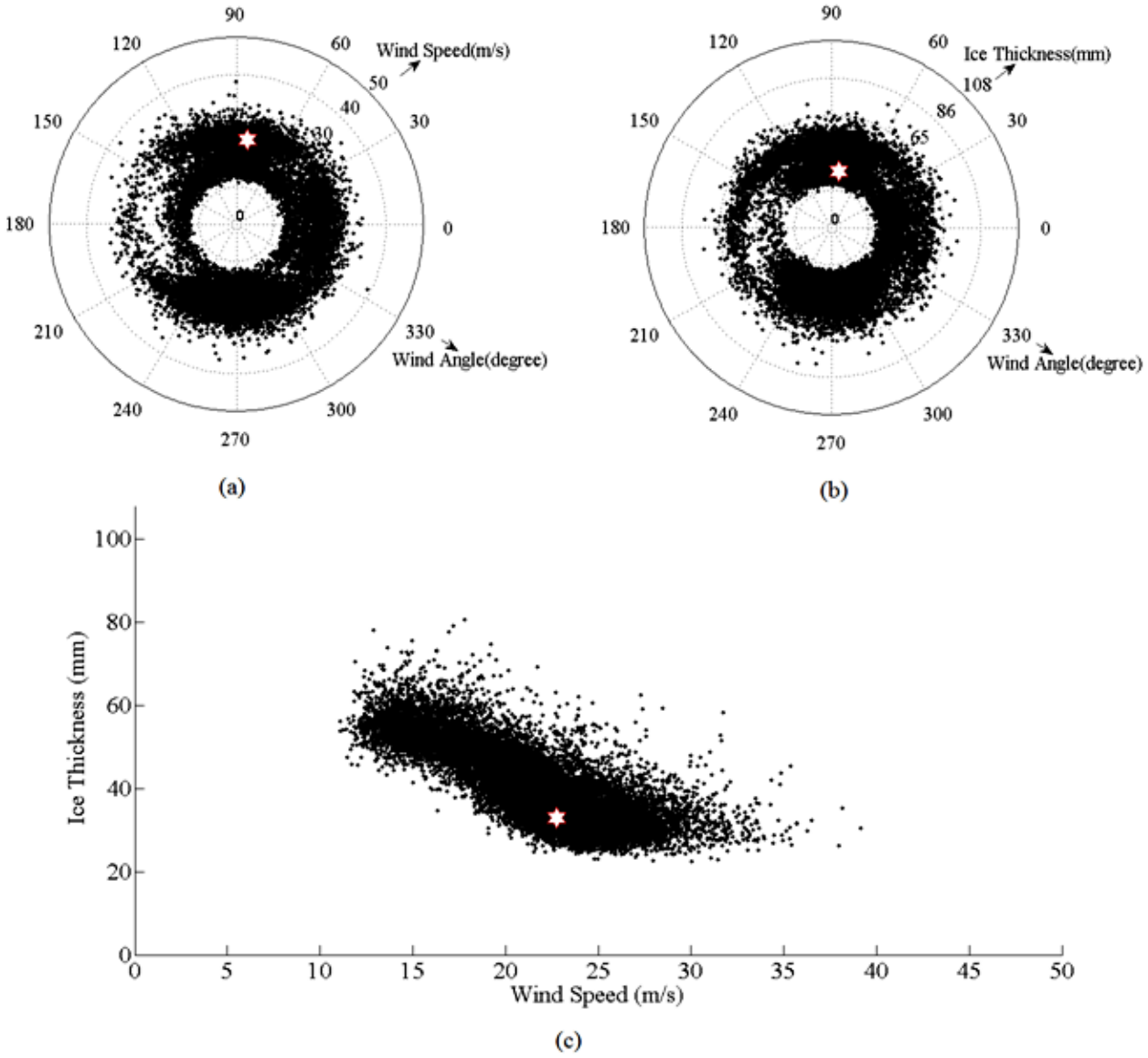


Figure 4-11. Climatic combinations that result in failure of the line section and the corresponding design point as a function of (a) wind speed and wind angle (b) ice thickness and wind angle (c) wind speed and ice thickness.

4.7 Conclusion

This paper describes the application of Statistical Learning Theory (SLT) to component based structural reliability analysis of transmission line systems considering the uncertainties of climatic variables such as wind speed, ice thickness and wind angle, and of the resistance of structural elements. Considering the implicit nature of component limit state functions, the optimal performance function is selected from the class of polynomial functions using VC-

generalization bounds and Stepwise regression. The unbiased prediction errors determined using CV technique indicate good prediction properties for the selected surrogate model of the finite element solver. Component and system failure probabilities are calculated using MC simulation and considering the correlation between failure of different components. The results show higher failure probabilities for suspension towers compared to other components of the line. The overall annual failure probability of the line is 0.0154 which corresponds to a reliability index of 2.16. The relative importance of different components are further quantified and components are ranked based on the improvement potential importance measure and the Bayesian reliability importance measure. The proposed checklist based on the Bayesian reliability importance measure suggests that tower 5 has the highest contribution to system failure and is the most critical component from a fault diagnosis perspective while the checklist based on the improvement potential importance measure indicates that tower 4 offers the highest potential system reliability improvement and is the most important component for fund allocation purposes when upgrading the line with limited resources.

The proposed probabilistic framework provides transmission line owners with detailed information about their transmission lines that can be used in asset condition assessment while overcoming the shortcomings of traditional structural reliability analysis methods. The obtained information enables the asset manager to take intelligent and well-founded decisions regarding maintenance, refurbishment, upgrading and replacement of overhead lines. The proposed probabilistic framework can also be used to develop component and system fragility curves for transmission lines under different scenarios of unbalanced ice loads which is the subject of a future study by the authors. In addition, future research remains to be done on developing a system based probabilistic framework for reliability assessment and fragility analysis of

transmission lines. The system based approach allows developing system fragility curves without requiring information about component fragilities and is particularly useful and computationally feasible in fragility analysis of complex structural systems such as transmission lines including hundreds of components.

Finally, the failure probabilities presented in this study are based on code derived distributions. For site specific studies, the proposed methodology can be adapted to distributions that are more representative of local conditions. In addition, future refinements in modelling capabilities on component loads can also be incorporated in order to reduce epistemic uncertainty.

CHAPTER 5 MANUSCRIPT 2

Component Based and System Based Fragility Analysis of Transmission Line Systems for Multiple Climatic Hazards

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Abstract

Component based and system based probabilistic frameworks are developed and compared for structural performance assessments of transmission lines under multiple climatic hazards. Analytical component and system fragility curves are developed for given intensity measures of wind speed considering its angle of incidence and ice thickness. The multi-hazard fragility surface and the wind and ice hazards at a specific site are then combined to estimate the system annual failure probability. For the component based approach, Statistical Learning Theory (SLT) is used to provide analytical generalization bounds for determining the optimal demand meta model for each structural component of the transmission line. This method implements the Structural Risk Minimization (SRM) inductive principle to find a surrogate model of the finite element solver that has good prediction properties. For the system based approach, a Venn Predictor (VP) is developed based on the Support Vector Machine (SVM) classifier to produce calibrated probability estimates. This method allows developing system fragility curves without

requiring information about component fragilities and is particularly useful in fragility analysis of complex structural systems such as power networks. The results indicate that both approaches can be successfully applied for fragility assessment and reliability analysis of transmission line systems.

Keywords: Fragility analysis, Transmission lines, Climatic hazards, Statistical learning theory, Support vector machines, Venn machines

5.1 Introduction

Previous experience with electric transmission lines shows that these structures may face extreme climatic events during their service life which can damage line sections and consequently affect power supply to customers. The ice storm of January 1998 in Eastern Canada (Jones and Mulherin 1998), the December 1999 windstorm in France (Kirschen 2001), the heavy snowstorm in Poland in November 2004 (Sundell et al. 2006), and the Gurdun wind storm in Sweden and Latvia in January 2005 (Johansson et al. 2006) are just a few recent examples of extreme climatic events that caused severe damages to transmission and distribution lines and highlighted the potential vulnerability of transmission lines.

The aftermath of these natural disasters and the potential for higher losses in the future motivates the need for a probabilistic framework to evaluate the vulnerability of transmission lines considering all uncertainties inherent in the prediction of their performance. The fragility analyses integrate finite element and reliability analyses to assess structural performance considering wind speed, its angle of incidence, and ice accumulation. The fragility is expressed as the conditional probability of failure as a function of wind speed, wind angle and ice thickness. Structural fragilities can be used in decision analysis for updating existing

transmission lines and to optimize the design of new lines. Component and system level fragility curves can also be used to optimize and prioritize inspections and structural assessments in the aftermath of a climatic event. Finally, by incorporating fragilities with climatic hazard functions at a specific location, structural reliabilities can be estimated and used for risk mitigation and decision analysis.

Researchers have developed multiple techniques for estimating fragility curves for various types of engineering structures subjected to different types of hazards. These techniques can be generally classified in two main categories: 1) component based fragility analysis methods, and 2) system based fragility analysis methods. Component based fragility analysis methods require fragility information for each component of a structural system. System fragility curves can then be developed by combining the fragility information of these components. Gardoni et. al. (2003), Choi et. al. (2004), Mackie and Stojadinovic (2005), Nielson and DesRoches (2007), Padgett and DesRoches (2008) and Ramanathan et. al. (2010, 2012) applied this approach for developing seismic fragility curves for various types of bridges. Ham et. al. (2009) used Monte-Carlo simulation technique to develop component and system fragility curves for industrial buildings subjected to extreme typhoon winds. Shafieezadeh (2014) developed age-dependent fragility curves for wood poles in power distribution networks against extreme wind hazards. Ang et. al. (1996) used connectivity and power flow analysis to develop seismic fragility curves for electric power transmission systems from the fragility curves of substations while neglecting the failure probability of network links (transmission lines). This category of methods provides useful information about the relative importance of different components of a system and their contribution to the overall failure of the system. However, developing system fragility curves from fragility information of components can be computationally extensive and time consuming

particularly in case of complex structural systems such as transmission lines comprising hundreds of components. For these reasons, system based fragility analysis methods are often more efficient. Shinozuka et. al (2000a) developed system seismic fragility curves for bridge structures by using the maximum likelihood procedure to treat each event of bridge damage as a realisation from a Bernoulli experiment. In another study, Shinozuka et al (2000b) applied a nonlinear static procedure for the development of system fragility curves for bridges. Xie et. al (2012) developed seismic fragility curves for electric transmission towers using nonlinear buckling analysis and Monte-Carlo Simulation (MCS). Although system based fragility analysis methods provide system level fragility functions in less time compared to component based methods, especially for complex structural systems, they do not provide any information about component fragilities and their contribution to the system failure. Therefore, selecting a method for fragility assessment or reliability analysis depends on the type and complexity of the structure and the objectives for which probabilistic analyses are performed.

Recently, the application of machine learning and surrogate models in the fragility analysis and reliability assessment of structures has gained popularity since they can replace computationally expensive finite element solvers of the structure with approximate statistical models for the response of structures. These methods can be used at both component and system levels in the form of regression models or classifiers. Surrogate models have traditionally been derived by means of the Response Surface Method (RSM) (Bucher and Bourgund 1990, Guan and Melchers 2001, Rajashekhar and Ellingwood 1993, Towashiraporn 2004). Although RSM has been successfully applied to many engineering problems, it has some limitations since it is based on the Empirical Risk Minimization (ERM) principle (Guan and Melchers 2001, Hurtado 2004). The strict assumptions over the class of approximating functions imposed by the ERM inductive

principle can lead to overfitting and poor prediction abilities (large model bias) for the meta model, even when the number of training samples is fairly large (Cherkassky and Mulier 2007). This explains the high sensitivity of the value of reliability index and failure probability resulting from the RSM method to the experimental plan reported by Guan and Melchers (2001). Statistical Learning Theory (SLT), also known as Vapnik-Chervonenkis (VC) theory, on the other hand overcomes the limitations of RSM by implementing the Structural Risk Minimization (SRM) inductive principle (Cherkassky and Mulier 2007, Cherkassky et al. 1999). Flexible methods based on SRM inductive principle reduce the model bias by adapting the model complexity to the training samples at hand (Cherkassky and Mulier 2007). Hurtado (2004) compared the application of traditional RSM method with various classifiers (classical Bayesian approach, Fisher Discrimination, Logistic Regression, Probabilistic Neural Networks, the Nearest Neighbours, Classification Trees, Multi-Layer Perceptrons (MLP) and Support Vector Machines (SVM)) to the problem of reliability analysis of complex structures and concluded that MLPs and SVMs are the most appropriate methods in this regard because 1) they can estimate the limit state function on the basis of a few samples, 2) they use flexible and adaptive models and 3) they can overcome the curse of dimensionality. He also reported the superiority of SVMs over MLPs due to: 1) the availability of the equations for the margins that facilitates the selection of useful training samples; 2) the search of the model parameters in a constrained quadratic optimization programme which is easier to solve than the nonlinear unconstrained minimization required by MLP.

This paper presents a novel application of statistical learning theory for the development of component and system based fragility curves for transmission lines subjected to atmospheric icing hazards. The component based method is a regression problem and adopts VC

generalization bounds for selecting the demand models, while the system based method is a probabilistic classification problem and applies a probabilistic support vector machine (PSVM) technique called Venn-machines. Next, the fragility curves are coupled with the climatic hazard function at a specific location to estimate the overall structural reliability of the line. In general, the uncertainties on component demand can be classified as either epistemic or aleatory. Epistemic uncertainty is reducible and is associated with imperfect models for the underlying physics or to parameter uncertainty due to small sample sizes, while aleatory uncertainty is irreducible and associated with natural variability such as the shape and drag coefficients for ice deposits, and the variability of ice accumulation, wind speed and material properties. In this application, component demands are defined for given winds speeds and ice accumulation combinations and the uncertainty in component demand is mainly epistemic and related to drag forces and the structural model. Due to lack of data on the latter, this study assumes that component demands are deterministic for given climatic conditions. Model uncertainty can be easily considered with the proposed methodology and could be included as more information becomes available for future studies.

In the following, the concepts for model complexity control using SLT, SVM and Venn-machines are reviewed in Sections 5.2, 5.3 and 5.4, respectively. Section 5.5 describes the studied transmission line section. Section 5.6 presents the methodology for fragility curve development using both component based and system based approaches. Section 5.7 presents a system reliability analysis to evaluate the overall system failure probabilities. And finally, conclusions are presented in Section 5.8.

5.2 VC-based model complexity control

Predictive learning addresses the challenge of estimating an unknown model from available training data. The resulting model can be used for prediction of future observations. There are two types of errors that are relevant for learning machines: the training error (empirical error) which corresponds to the estimated error based on given samples, and the generalization error which is the error in assessing the outputs from samples not used in the training process. The goal of learning is to obtain a model with minimal differences relative to the unknown underlying function. However, the prediction risk cannot be obtained exactly since its calculation requires knowing the probability distribution of the sample space (Cherkassky and Mulier 2007). ERM based methods such as RSM estimate the unknown prediction risk as the empirical error or the average error for the training data. In other words, under the ERM inductive principle the optimal approximating function $f(\mathbf{x}, \omega)$ can be determined by minimizing the empirical error. Here, ω is a vector of parameters that define the function. Since ERM imposes strict assumptions over the class of approximating functions, the discrepancy between the meta model and the actual governing function can lead to overfitting and poor generalization of the approximating function.

To prevent overfitting, flexible-adaptive models based on the SRM inductive principle are used. SLT provides analytical generalization bounds for model selection based on SRM (Cherkassky and Mulier 2007). The upper bound for the prediction error for a regression problem is as follows (Cherkassky and Mulier 2007, Cherkassky et al. 1999, Vapnik 1995),

$$E(\omega) = E_{emp}(\omega) \Big/ \left(1 - \sqrt{\frac{h}{n} - \left(\frac{h}{n}\right)\ln\left(\frac{h}{n}\right) + \frac{\ln n}{2n}}\right)_{+} \tag{5-1}$$

where n is the number of training samples, E_{emp} is the empirical error and h is a measure of model complexity called the VC-dimension. The VC dimension is a characteristic of a set of functions. For a set of indicator functions, VC dimension is the maximum number of samples for which all possible binary labellings can be induced without error. In the case of linear real-valued functions, VC dimension equals the number of free parameters.

Equation 5-1 provides a quantitative characterization of the trade-off between the complexity of the approximating function and the goodness-of-fit for the training data. Under SRM, the set S of approximating functions $f(\mathbf{x},\omega)$ consists of nested elements S_k ordered according to their VC dimension. For a given training data set, SRM selects the function $f(\mathbf{x},\omega^*)$ which minimizes the empirical error for the functions for element S_k,. Next, the prediction error is calculated for each element S_k using the upper bound provided by equation 5-1. The optimal element S_{opt} with optimal complexity (VC dimension) is the one providing minimum prediction error.

Since the minimum value of the prediction error found from model selection is biased, a resampling procedure using a 10-fold cross validation (CV) (Rodrıguez et al. 2010) is used to obtain an unbiased estimate of the prediction error. The training algorithm is applied for each of the training samples where the test data set is left out. The CV estimator of the prediction error is defined as the average of the errors for the test folds. The procedure is efficient in lowering the variance of the prediction error estimate (Bengio and Grandvalet 2004).

5.3 SVM-based model selection and learning

SVM, proposed by Vapnik (1995, 1998), is a supervised learning method based on SRM inductive principle that can be used for regression or classification problems. It creates a mapping between the inputs and outputs of a training sample. Since SVM has excellent learning

abilities with small amounts of data, it can overcome the curse of dimensionality and has good generalization. In consequence, the procedure has gained much popularity in structural engineering applications. Many researchers have applied the method for structural health monitoring and damage detection purposes (Noori 2010, Sharma 2008, Shimada and Mita 2005, Yang et al. 2005). The application of SVM for reliability analysis was first introduced by Rocco and Moreno (2002). Hurtado and Alvarez (2003) proposed an SVM approach for reliability analysis of structures modeled with Stochastic Finite Elements (SFE). Tang et. al. (2007) combined the SVM with importance sampling for structural reliability analysis, while Li et al. (2006) and Zhao (2008) used the SVM-based First Order Reliability Method (FORM) and the SVM-based MCS method for this purpose.

Given a binary classification problem with a training data set $(x_1,y_1),(x_2,y_2),...,(x_n,y_n)$ where $x_i \in R^n$ and $y_i \in \{+1,-1\}$ is the class label for x_i. The SVM technique finds the optimal hyper-plane that separates one class (+1) from the other (-1). The formulation of SVM is dependent on the characteristics of the training samples which can be categorized as linearly separable or linearly non-separable.

The training set is linearly separable when a linear function matches the sign for the class of all training samples. In this case, the optimal separating hyper-plane is the one which maximizes the distance between the hyper-plane and the nearest sample points of each class. This optimization problem is presented in equation 5-2.

$$Min(\frac{1}{2}\mathrm{w}^T.\mathrm{w}) \qquad \text{subjected to} \quad y_i(\mathrm{w}^T.\phi(\mathrm{x}_i)+b) \geq 1 \qquad (5\text{-}2)$$

where w is the weight vector, b is the bias and $\phi: \chi \rightarrow \mathcal{F}$ is the feature function that maps the input vectors into a high dimensional feature space. It is difficult to directly solve this problem

since the constraints are quite complex, therefore, it is usually expressed into its dual form which corresponds to finding parameters α_i that maximize the function $L(\alpha)$

$$L(\alpha) = \sum_{i=1}^{n} \alpha_i - 1/2 \sum_{i,j=1}^{n} \alpha_i \alpha_j y_i y_j \phi(\mathrm{x}_i)^T . \phi(\mathrm{x}_j) \quad \text{subjected to} \quad \sum_{i=1}^{n} y_i \alpha_i = 0, \quad \alpha_i \geq 0 \qquad (5\text{-}3)$$

The optimal hyper-plane is then obtained from

$$f(x) = \sum_{i=1}^{n} \alpha_i y_i \phi(\mathrm{x}_i)^T . \phi(x) + b \qquad (5\text{-}4)$$

Determination of the bias factor *b* is solved as a simple one dimensional problem. There is no need to compute the feature functions $\phi(\mathrm{x})$ since the dot product $\phi(\mathrm{x})^T . \phi(\mathrm{x}')$ can be represented by a kernel function $\mathrm{K}(\mathrm{x},\mathrm{x}')$. The selection of the type of kernel function corresponds to the selection of the class of functions used for feature construction. For instance, for the degree-γ polynomial functions, the kernel is $\mathrm{K}(\mathrm{x},\mathrm{x}') = (\mathrm{x}^T . \mathrm{x}' + 1)^{\gamma}$, while for the Gaussian Radial Basis Function (RBF), the kernel function is $\mathrm{K}(\mathrm{x},\mathrm{x}') = \exp(-||\mathrm{x} - \mathrm{x}'||^2 / 2\gamma^2)$, where $\gamma > 0$ is a parameter that controls the width of the Gaussian function.

For the case of linearly non-separable samples, SVM implements a margin-based loss function such that most of the training data can be separated perfectly (with zero empirical loss) and the rest of the training data falls within the margin and can only be explained with some level of uncertainty (non-zero loss). The best decision boundary is the one which minimizes the total empirical loss for the samples that lie within the margin while achieving maximum separation (margin) between correctly classified training samples (Cherkassky and Mulier 2007). For this purpose, slack variables (ξ_i) are introduced as shown in equation 5 to allow miss-classification of difficult or noisy data points (Vapnik 1995).

$$Min(\frac{1}{2}\mathrm{w}^T.\mathrm{w} + C\sum_{i=1}^{n}\xi_i) \qquad \text{subjected to } \; y_i(\mathrm{w}^T.\phi(\mathrm{x_i}) + b) + \xi_i - 1 \geq 0 \quad , \quad \xi_i \geq 0 \tag{5-5}$$

where $\xi_i = \max(1 - y_i f(x_i, w), 0)$ and *C* is a user selected parameter that controls the trade-off between complexity and the proportion of non-separable samples. This formulation is called the soft-margin SVM, and was introduced by Cortes and Vapnik (1995). The dual formulation of equation 5-5 is

$$L(\alpha) = \sum_{i=1}^{n}\alpha_i - 1/2\sum_{i,j=1}^{n}\alpha_i\alpha_j y_i y_j \phi(\mathrm{x_i})^T.\phi(\mathrm{x_j}) \quad \text{subjected to} \sum_{i=1}^{n} y_i\alpha_i = 0, \quad 0 \leq \alpha_i \leq \mathrm{C} \tag{5-6}$$

The examples $\mathrm{x_i}$ for which $\alpha_i > 0$ are called the support vectors which are the points that are on the margin, or within the margin when a soft-margin SVM is used.

The quality of SVM models is measured in terms of the prediction error which varies as a function of the kernel function, the kernel parameter (γ) and the penalty parameter (C). Therefore, the problem of model selection includes selecting the parameters that form the model with lowest prediction error. This objective is achieved by using a grid search strategy in combination with (10-fold) cross validation (Chang and Lin 2001). The best kernel function is also the one that results in the lowest unbiased prediction error.

5.4 PSVM/Venn Machine

The outputs from the standard SVM analysis is the value of *sign*(f(x)), where f is the decision function presented in equation 5-4. Therefore, SVM is in effect a non-probabilistic classifier. Probabilistic extensions of the classifier have been proposed by several researchers, examples are Platt's method (Platt 1999), SVM binning (Drish 2001) and SVM with Isotonic Regression

(Zadrozny and Elkan 2002). Venn machines, proposed by Vovk et. al. (2003, 2005), is another method that provides predictions with a confidence interval. Zhou et. al. (2011) compared the existing methods and demonstrated the superiority of Venn machines since the probability estimates are well calibrated, reliable and unbiased under the independent and identically distributed (i.i.d) assumption (Vovk et al. 2003).

Consider a training set $\{(x_1,y_1),(x_2,y_2),...,(x_{n-1},y_{n-1})\}$ where x_i is an object and y_i is its class (label). A Venn Predictor (VP) estimates the probability that a new object x_n belonging to each class $Y_j \in \{Y_1, \cdots, Y_c\}$. Under the VP framework, each one of the possible classifications Y_j is assigned to x_n and all examples (pairs of objects and labels) $\{(x_1,y_1),...,(x_n,Y_j)\}$ are assigned into a number of categories based on a taxonomy. For a SVM-based VP, the taxonomy is defined from the output of the SVM decision function (equation 5-4). The examples are considered to be in the same category if their decision values are in the same interval defined for a given number of categories. Once all examples are partitioned into categories, the empirical probability of each classification for the category τ which contains (x_n,Y_j) is calculated based on equation 5-7.

$$p^{Y_j}(Y_k) = \frac{|\{(x^*, y^*)\epsilon\tau : y^* = Y_k\}|}{|\tau|} \tag{5-7}$$

After assigning all possible classifications to x_n, a K×K matrix of probabilities is derived where K=|Y| (Generally, for fragility assessment of structures the two labels of survival and failure are considered and K=2). The quality of each column of this 2×2 matrix is its minimum entry. The best column, j_{best}, is the one with the highest quality. The VP outputs the prediction $\hat{y}_n = Y_{j\,best}$, and the maximum and minimum probabilities obtained from equation 5-7 for each label Y_k, define the interval for the probability of the new example belonging to Y_k.

5.5 Description of the transmission line section

In this study, both component-based and system-based approaches are applied to develop component and system fragility curves for a 2.6 kilometre long 9 span segment of a typical 120 kV transmission line located in North America. Figure 5-1 shows the studied segment of the line. It consists of two line sections separated by angle/dead-end towers. It is noted that in this study, ten towers are included in the system reliability analysis due to the limited spatial extension of climatic hazards. According to CEI/IEC 60826 (2003) and CAN/CSA-C22.3 (2006), in a flat to rolling terrain and based on a span of 400 m, 1 to 5 towers are usually subjected to maximum wind and maximum ice and wind loads while 10 to 20 towers are usually subjected to maximum ice loads. The diameter and weight per unit length of conductors are 27.78 mm and 15.26 N/m, respectively, and the diameter and weight per unit length of ground wires are 12.57 mm and 7.44 N/m, respectively.

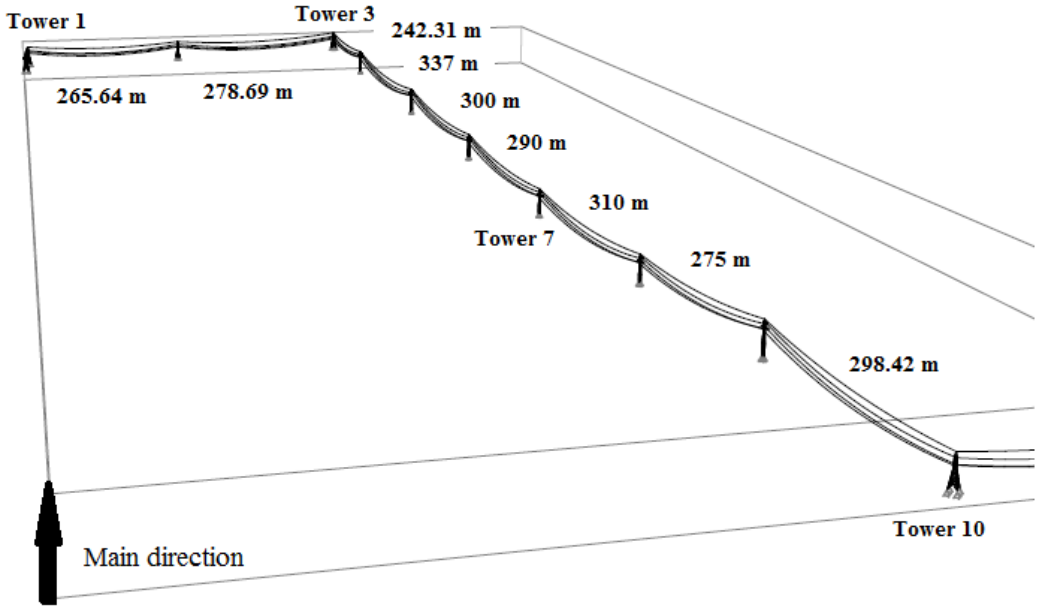


Figure 5-1. Nine span segment of the transmission line under study

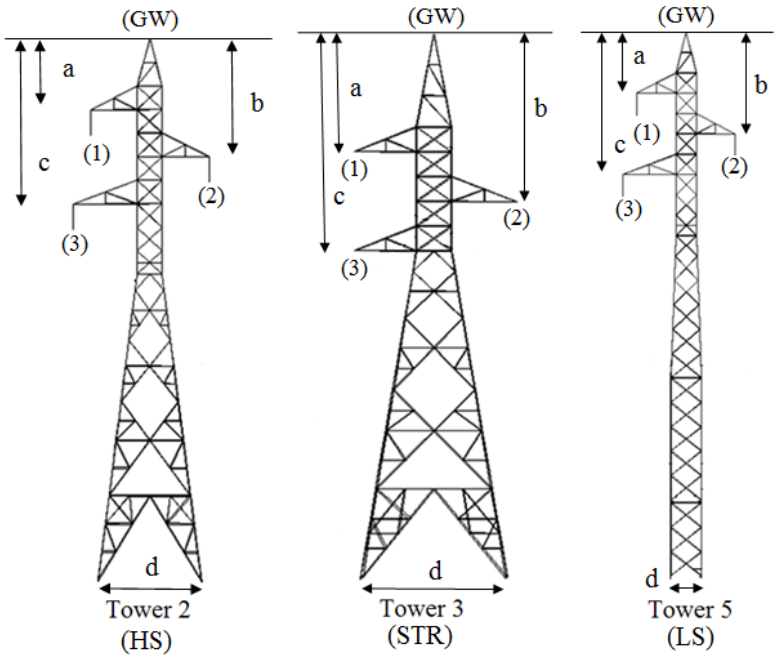


Figure 5-2. The configuration of tower types within the line

Table 5-1 presents a description of the towers within the studied line segment. HS and LS towers are heavy suspension and light suspension towers, respectively, while STR towers are angle/dead-end towers. Figure 5-2 shows the configuration of three types of towers used within the studied line section.

Table 1. Description of towers of the line segment

Tower Number	Tower Height(m)	a(m)	b(m)	c(m)	d(m)	Tower Type
1	30.5	6	8.5	11	8.06	STR
2	28.75	3.75	6.25	8.75	5.26	HS
3	27.5	6	8.5	11	7.08	STR
4	33.25	3.75	6.25	8.75	1.9	LS
5	33.25	3.75	6.25	8.75	1.9	LS
6	30.25	3.75	6.25	8.75	1.9	LS
7	30.25	3.75	6.25	8.75	1.9	LS
8	28.75	3.75	6.25	8.75	1.9	LS
9	33.25	3.75	6.25	8.75	1.9	LS
10	27.5	6	8.5	11	7.08	STR

A detailed three dimensional model of the line is developed with the computer software SAP2000® (1996) in order to capture the interactions between different components of the transmission line. The conductors and ground wires are modeled using cable elements with tension-only behaviour and the overhead towers are modeled as frame-truss systems in that leg members with multi-bolted connections are modelled as frames with rigid connections, and horizontal and diagonal members with single-bolted connections are modelled as truss elements with pinned connections. The structural system is analysed under static ice and quasi-static wind loads based on CEI/IEC 60826 (2003). It is noted that the wind and wind on ice loads are calculated taking into account the influence of the height, shape and surface properties of each member as well as the terrain category and the dynamic response of components. In addition, it is assumed that ice loads are applied to line components as a result of freezing rain events, and the density of ice is assumed to be equal to 900 kg/mm^3. The analysis takes into account the geometrically nonlinear behaviour of cables, and the structural components are assumed to have linear elastic material properties. In addition, in order to keep the model simple and computationally affordable, the randomness in component behaviour is not considered in this model which results in a deterministic demand for given climatic conditions.

5.6 Methodology for developing fragility curves

5.6.1. Component based approach

In order to develop component fragility functions, an explicit approximation of the limit-state function, G(x), is obtained for each component of the line. For this purpose, the approximating function, D(x_d), relates structural component demands to climatic variables (wind speed, ice thickness and wind angle). This is done by fitting a polynomial function to the results obtained

from SAP2000 model for a selected set of climatic conditions (1287 samples). Each candidate degree of polynomial is fitted to training samples and the optimal polynomial function is the one which minimizes the prediction error based on VC-generalization bounds (equation 5-1). A stepwise regression is used to select statistically significant predictors. The goodness-of-fit of the surrogate model is evaluated by calculating the adjusted coefficient of multiple determination ($\bar{R}^2$) and the prediction coefficient of multiple determination ($\bar{R}_p^2$) using equations 5-8 and 5-9.

$$\bar{R}^2 = 1 - \frac{MSE}{MST} = 1 - \left(\frac{n-1}{n-p}\right)(1 - R^2) \tag{5-8}$$

$$R_p^2 = 1 - \frac{\sum_{i=1}^{n_{te}} (y_{ite} - \hat{y}_{itr})^2}{\sum_{i=1}^{n_{te}} (y_{ite} - \bar{y}_{te})^2} \tag{5-9}$$

In equation 5-8, *MSE* is the mean square error, *MST* is the total mean square of variation in observations, *n* is the total number of samples, *p* is the number of parameters in the model and R^2 is the coefficient of multiple determination. $\bar{R}^2$ is a useful tool in evaluating the quality of fit between the regression model and the sample data since it guards against overfitting by penalizing the analyst for adding terms to the model (Montgomery and Runger 2003).

In equation 5-9, n_{te} and $\bar{y}_{te}$ are respectively the number and mean of the observed demands (y_{ite}) in the test subset and $\hat{y}_{itr}$ are the predictions of the demands in the test subset using the fitted model to the training subset. The prediction coefficient of multiple determination ($\bar{R}_p^2$), evaluates the ability of the final selected model to predict future demands. As indicated in figure 5-3, the calculated $\bar{R}^2$ and $\bar{R}_p^2$ values are close to 1 which confirms a good prediction ability for the surrogate demand models selected.

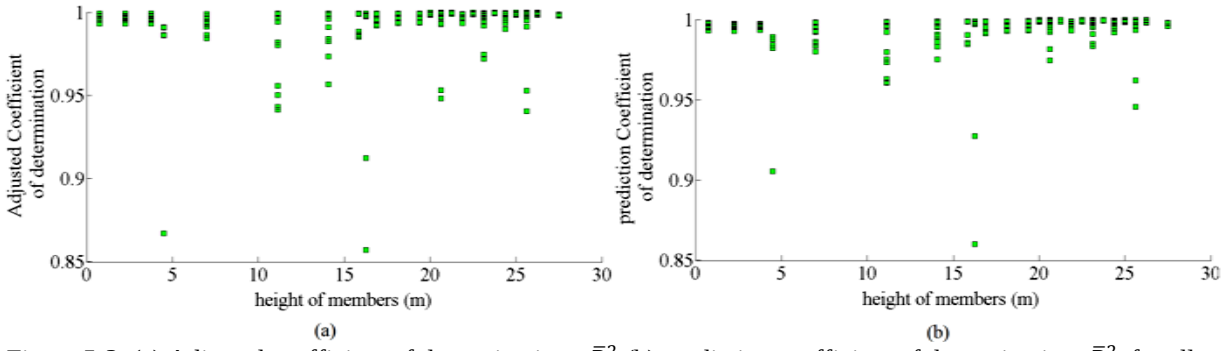


Figure 5-3. (a) Adjusted coefficient of determination, $\bar{R}^2$ (b) prediction coefficient of determination, $\bar{R}_p^2$, for all members of tower 2

Given $D(x_d)$, the approximate limit-state function can be defined as,

$$G(\mathbf{x}) = C - D(\mathbf{x_d}) \tag{5-10}$$

The probability of failure is the probability of the event $G(x) < 0$ or $D(x) > C$. The un-factored nominal capacity, C, for each member of the overhead towers is calculated based on ASCE10-15 (2015). The capacity of tension members is assumed to be equal to the minimum of the bearing capacity of the connection, the shear capacity of the bolts and the tension capacity of the member, and the capacity of the compression members is assumed to be equal to the minimum of the bearing capacity of the connection, the shear capacity of the bolts and the compression capacity of the member. It is noted that this assumption implies perfect correlation between different failure modes of each tower member. It is assumed that the capacity of overhead tower components have a lognormal distribution with a coefficient of variation of 10% (CEI/IEC 60826, 2003).

The nominal rated tensile strength of conductors and ground wires are 141.7 KN and 139 KN, respectively. They are assumed to have a lognormal distribution with a coefficient of variation of 3% (CEI/IEC 60826, 2003). Three groups of insulators with nominal capacities of 70, 140 and 240 KN are used in the studied line section. It is assumed that the capacity of insulators have a

lognormal distribution with a coefficient of variation of 5% (CEI/IEC 60826, 2003). In addition, following CEI/IEC 60826 (2003), it is assumed that the probability of each element of the line not having its nominal strength is 10%.

A component fragility curve is the conditional probability of failure of a component given a specific combination of climatic variables. Since model uncertainties are not considered in this study, each combination of climatic variables results in a deterministic load in each tower member, insulator, conductor and ground wire which is compared to the corresponding probability distribution function of capacity to determine the conditional probability of failure. In order to develop fragility curves for the towers and for the overall line, both are idealized as series systems. This assumption is conservative and implies that the system fails once any one of its components fails. It is noted that the distribution of component capacities are assumed to be independent between members. Figures 5-4 and 5-5 present the resulting fragility curves for suspension tower 7 and the overall system, respectively. It is inferred that the line is most vulnerable when wind blows perpendicular to the longest section of the line since this direction is the most detrimental direction for most of the suspension towers which are the weakest components of the line.

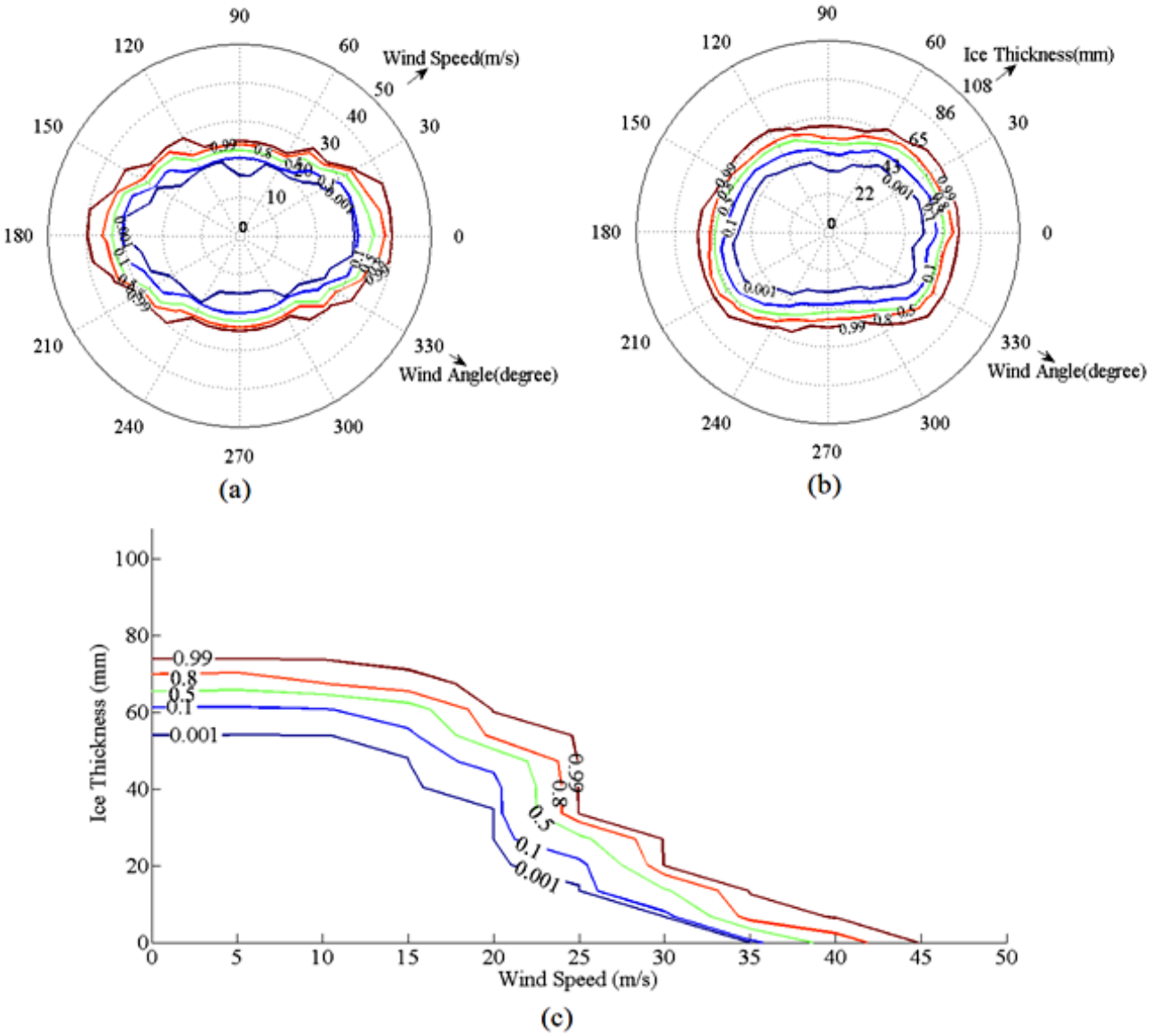


Figure 5-4. Fragility curves for tower 7 as a function of (a) average 10 minute wind speed and wind angle with 45 mm of ice (b) ice thickness and wind angle with 20 m/s average 10 minute wind speed (c) average 10 minute wind speed and ice thickness given wind angle=90.

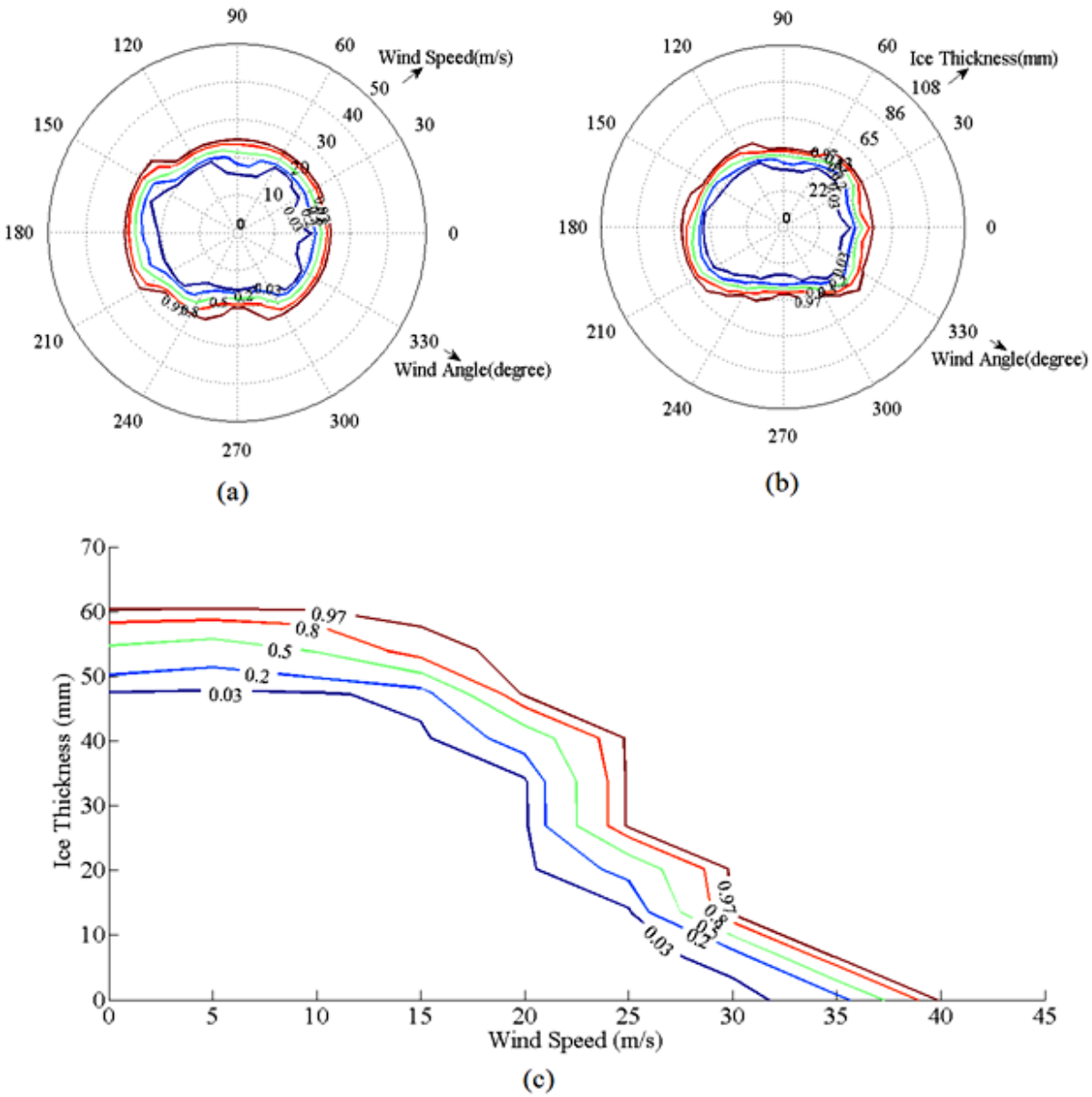


Figure 5-5. Fragility curves for the studied line as a function of (a) average 10 minute wind speed and wind angle with 45 mm of ice (b) ice thickness and wind angle with average 10 minute 20 m/s wind speed (c) average 10 minute wind speed and ice thickness given wind angle=90.

5.6.2. System based approach

In the system based approach, a binary survival-failure vector is constructed for the transmission line by comparing the component demands obtained from structural analysis and the component capacities. The system based approach also adopts the series system assumption where failure of any one of the transmission line components is indicative of the system failure. In order to

account for the randomness of component capacities, MCS with 10^4 samples is used. The system condition is "failure" if it fails in at least 50% of the simulations. The resulting binary vector (936 samples) is then used to train and validate a SVM classifier using the 10-fold cross-validation technique. It should be noted that since the binary survive-failure output vector is less sensitive to the wind direction variable in comparison with the wind speed and ice thickness input variables, a SVM model is trained for each wind direction separately.

As described in section 5.3, the choice of SVM parameters such as the kernel function, γ and C affects the quality of the SVM model. Therefore, these parameters should be tuned properly to result in a model with lowest prediction error. Using a double re-sampling technique (10-fold cross validation), the sample points are randomly divided into 10 equal sized folds. At each iteration, one fold is used as test set, one fold is used as validation set and the remaining folds are used to train the SVM model. Parameters γ and C are selected to minimize the validation error. The optimal kernel type is the one that minimizes the test error.

In this study both polynomial and RBF kernels are used for SVM training. Figure 5-6 presents the average validation error when the wind angle is equal to 90 degrees and the polynomial kernel function is used. From these results, $(C,\gamma)=(1,4)$ corresponds to the simplest possible model with the minimum validation error. The average unbiased prediction error corresponding to these parameters is calculated using the test set and is equal to 0.0089.

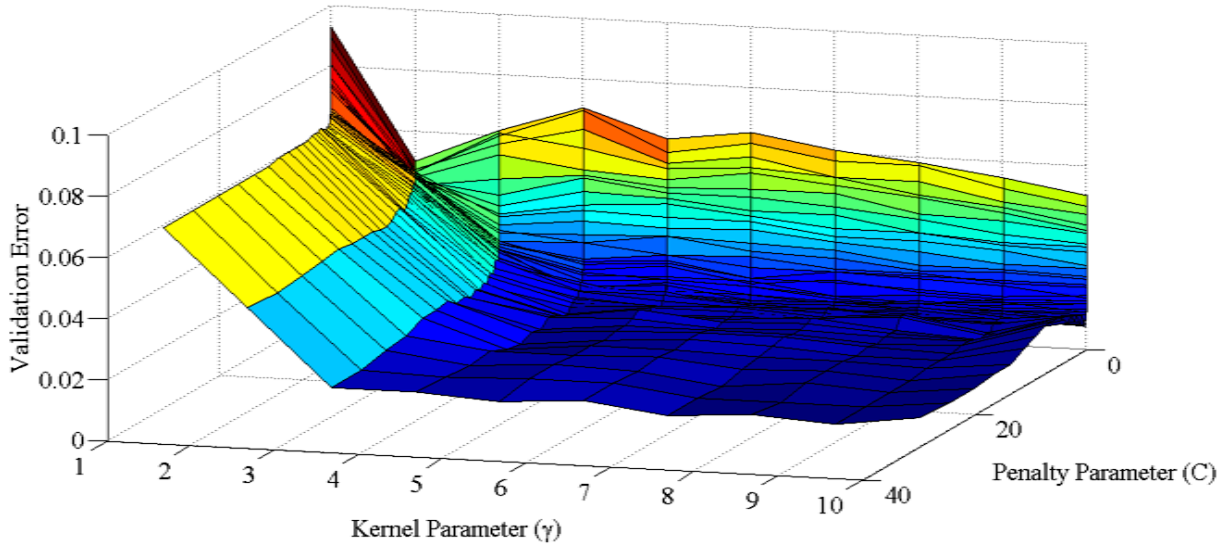


Figure 5-6. Average validation error given that wind angle is equal to 90 degrees and Polynomial kernel function is used

The performance of the SVM classifiers with polynomial and RBF kernels are evaluated and compared by computing the number of correctly classified "failure" examples (true positives, *tp*), the number of correctly classified "survival" examples (true negatives, *tn*), and the number of examples that are incorrectly assigned to the "failure" class (false positives, *fp*) or that are incorrectly assigned to the "survival" class (false negatives, *fn*). Figure 5-7 shows the confusion matrix developed for binary classification using these four parameters.

		Actual	
		P	N
Predicted	P'	True Positive	False Positive
	N'	False Negative	True Negative

Figure 5-7. Confusion matrix for binary classification

Table 5-2 shows five performance measures calculated from the confusion matrix which are mostly used for evaluating binary classifications (Sokolova and Lapalme 2009). The error rate

evaluates a classifier by its percentage of incorrect predictions using equation 5-11. The complement of the error rate is called the accuracy rate.

$$\text{Error rate} = 1 - \text{Accuracy rate} = 1 - \frac{tp + tn}{tp + tn + fp + fn} \quad (5\text{-}11)$$

Sensitivity, also known as the true positive rate or recall, summarizes how well the model detects failure. It is the probability that a failure event is correctly classified as failure by the meta model (equation 5-12). Specificity measures how well the model identifies survival events. It is the percentage of survival events correctly classified as survival (equation 5-13). Precision is the percentage of events identified as failure by the model who actually have failed (equation 5-14).

$$\text{Sensitivity} = \frac{tp}{tp + fn} \quad (5\text{-}12)$$

$$\text{Specificity} = \frac{tn}{tn + fp} \quad (5\text{-}13)$$

$$\text{Precision} = \frac{tp}{tp + fp} \quad (5\text{-}14)$$

Finally, the F-measure (equation 5-15) takes into account both sensitivity and precision; and is considered as a better metric for classification algorithms than accuracy (Lewis and Gale 1994).

$$F_\beta = \frac{(1 + \beta^2) \times \text{precision} \times \text{sensitivity}}{\beta^2 \times \text{precision} + \text{sensitivity}} \quad (5\text{-}15)$$

where β controls the relative weights assigned to precision and sensitivity. When β is greater than 1 sensitivity has a higher weight than precision, while when β is less than 1, more weight is

assigned to precision relative to sensitivity. For the purpose of this study, equal importance is given to both precision and sensitivity (β=1).

As indicated in Table 5-2, SVMs with Polynomial kernels have a superior performance compared to SVMs with RBF kernels for our data. Therefore, the SVM-based VP is constructed using polynomial kernels.

Table 5-2. Comparison between different methods

Method	Underlying Algorithm	Unbiased prediction Error rate	Sensitivity	Specificity	Precision	F-measure
SVM	Polynomial	0.0097	99.63%	98.17%	98.76%	99.19%
SVM	RBF	0.0198	98.99%	96.65%	97.66%	98.31%
Venn Machine	Polynomial SVM	0.0108	99.25%	98.44%	98.96%	99.1%

The additional parameter of the SVM-based VP, the number of categories, can be determined using 10-fold CV method as illustrated in figure 5-8.

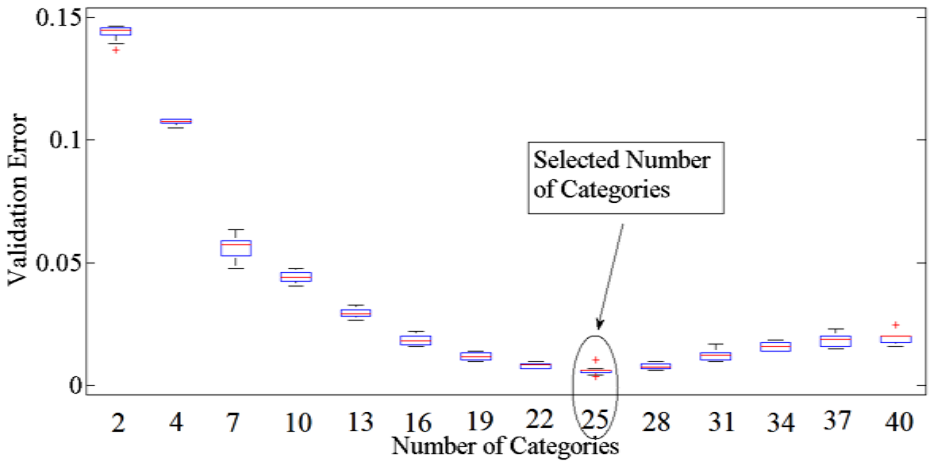


Figure 5-8. Validation error calculated using 10-fold cross validation technique given that wind angle is equal to 90 degrees and the underlying algorithm is polynomial SVM

The VP model with the selected number of categories provides the lowest average validation error and also the smallest variation of the error (narrowest box plot). The unbiased error rate, sensitivity, specificity, precision and F-measure for the SVM-based VP are also presented in Table 5-2. The results show good prediction abilities for the SVM-based VP and validate its application for the fragility assessment and reliability analysis of transmission lines.

Figure 5-9 demonstrates how VP is used to calculate lower and upper failure probabilities for a given climatic condition. As previously described in section 5.4, the objective is to estimate the probability that a new object (point of interest) belongs to each class $Y_j \in \{\text{Survival}, \text{failure}\}$. Therefore, two possible outcomes are studied: 1) The new object belongs to the "Survival" class (Figure 5-9.a) 2) The new object belongs to the "Failure" class (Figure 5-9.b). For each case, all examples are assigned into 25 categories (selected from Figure 5-8). And a 2×2 matrix of probabilities is derived from equation 5-7. Using the VP, the predicted lower and upper failure probabilities for the given climatic conditions (wind speed=23 (m/s), ice thickness=30 (mm) and wind angle=90) are 0.211 and 0.237, respectively. And the new object belongs to the "Survival" class.

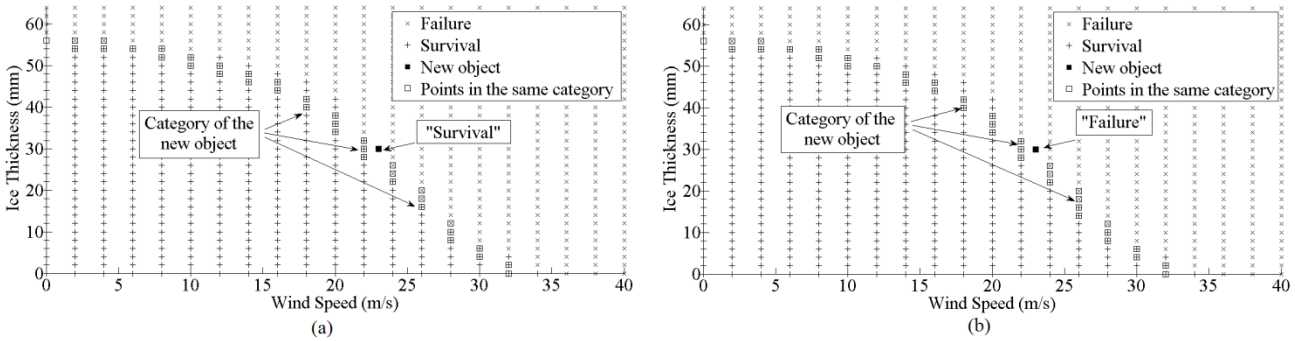


Figure 5-9. Calculating conditional failure probability of the line given that wind speed=23 (m/s), ice thickness=30 (mm) and wind angle=90 (Degree) under (a) survival assumption (b) failure assumption.

Using the aforementioned methodology, the lower and upper bound fragility curves for the studied line section are developed and presented in figures 5-10 to 5-12.

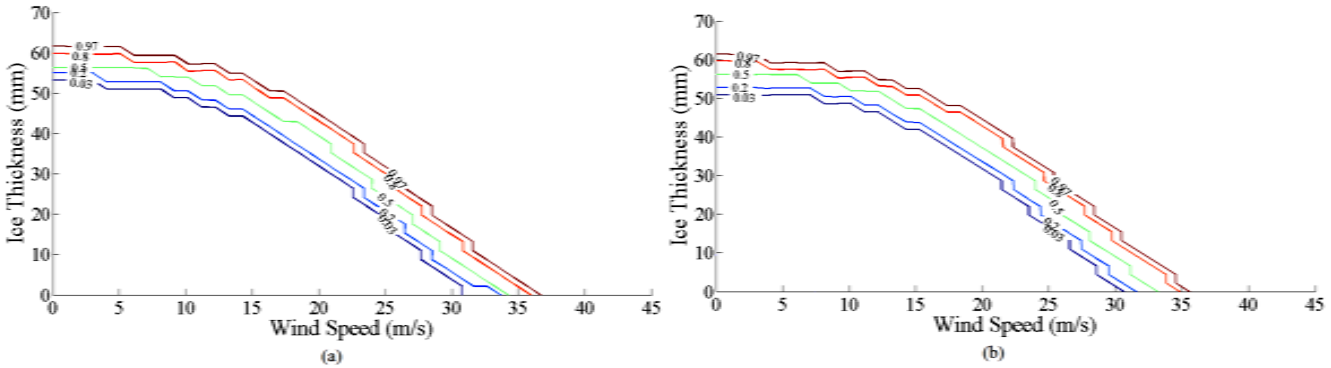


Figure 5-10. Fragility curves for the studied line as a function of average 10 minute wind speed and ice thickness given wind angle=90 (a) lower bound (b) upper bound.

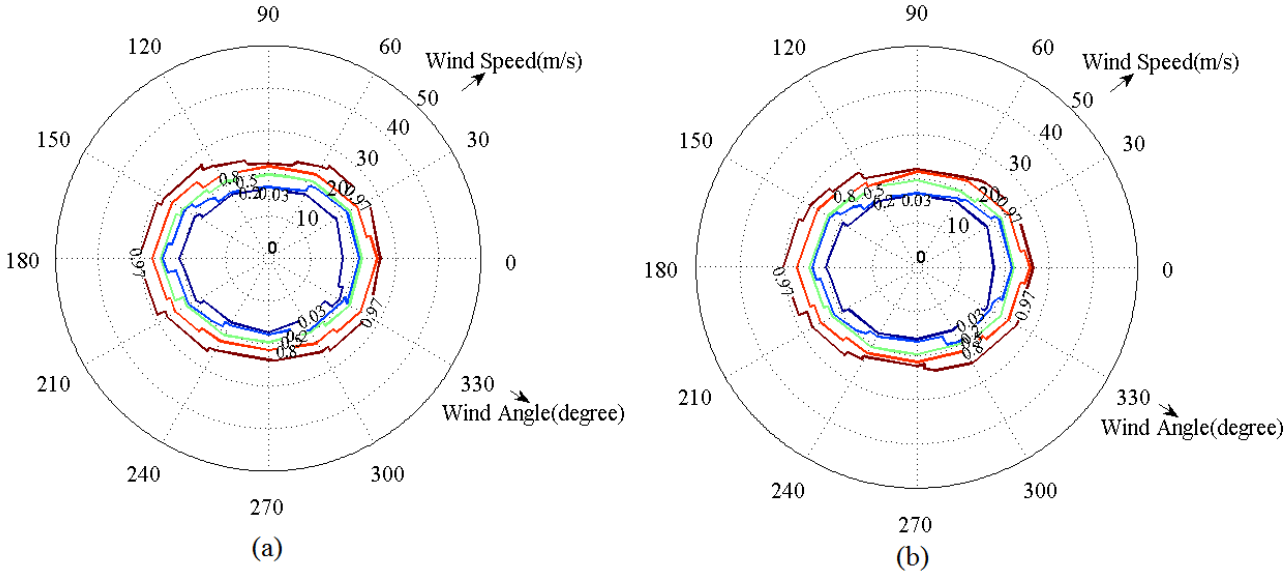


Figure 5-11. Fragility curves for the studied line as a function of average 10 minute wind speed and wind angle with 45 mm of ice (a) lower bound (b) upper bound.

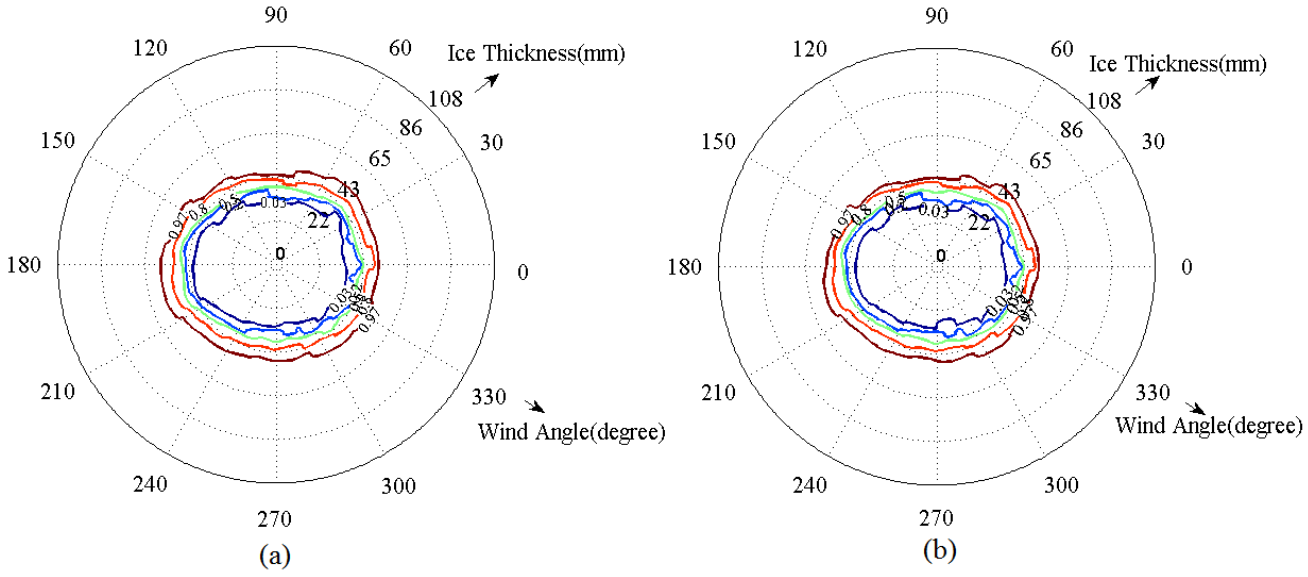


Figure 5-12. Fragility curves for the studied line as a function of ice thickness and wind angle with 20 m/s average 10 minute wind speed (a) lower bound (b) upper bound.

5.7 System reliability analysis

Next the fragility curves are combined with the distribution of climatic variables at a specific location to calculate the overall system failure probability. For this purpose, the parameters of the joint distribution for concurrent wind speed and ice thickness are determined by assuming that the 50 year return period ice thickness accumulated on a 25 mm conductor and average 10 minute wind-on-ice speed at 10 meters of height above ground are 45 (mm) and 21.64 (m/sec), respectively. CAN/CSA-C22.3 (2006) presents climatic coefficients to derive the values of these variables for 25, 100, 150, 200, 400, and 500 years based on the 50 year design value. These coefficients are used to estimate the mean value and standard deviation for wind speed and ice thickness variables. The corresponding values are 15.64 (m/sec) and 2 (m/sec) for wind speed during icing conditions and 32.2 (mm) and 4.1 (mm) for ice thickness, respectively. It is assumed that wind angle, which is calculated with respect to the main direction has a uniform distribution, while wind speed and ice thickness follow the Gumbel distribution. The temperature is assumed

to be deterministic and equal to -5 °C. Considering that the epistemic uncertainties in the evaluation of ice and wind induced component loads are neglected, the resulting values may be conservative but coherent with the assumption of CAN/CSA-C22.2 (2006). Note that for an actual line, site specific data on wind speed and ice accumulation could be used to obtain more accurate and less conservative estimates of atmospheric hazards in comparison to code derived distributions (Chouinard et al. 1998).

The failure probability of the transmission line can be calculated using different techniques which are further elaborated in sections 5.7.1 and 5.7.2.

5.7.1. Component based approach

Applying the component based approach, 10^8 structural demands are simulated for each component of the line by randomly sampling from the distributions for the climatic variables and substituting into the selected response function for that component. The simulated demands are then paired with the simulated capacities for each component. The failure probability of each component of the line is estimated as the ratio of the number of events for which the structural demand exceeds the structural capacity to the total number of simulations. The system failure probability is the number of climatic events that cause at least one component failure divided by the total number of simulations. It is noted that this technique accounts for the correlation between component failures due to the common source of external loading. Table 5-3 shows the estimates of the failure probabilities, the Bayesian reliability importance and the normalized improvement potential importance (%) for different components of the line and the whole system. It should be mentioned that the failure probability of foundations are not addressed in this study and may be considered in future studies when experimental data becomes available.

Table 5-3. Failure probabilities and reliability indices of line components and the system using the component based approach.

	Failure Probability	Bayesian Reliability Importance	Normalized improvement Potential Importance (%)
Tower 1	1.60×10^{-5}	0.0010	0
Tower 2	2.70×10^{-3}	0.1751	12.9
Tower 3	1.10×10^{-4}	0.0074	0.01
Tower 4	8.70×10^{-3}	0.5677	15.3
Tower 5	9.00×10^{-3}	0.5863	10.4
Tower 6	3.90×10^{-3}	0.2553	0.42
Tower 7	3.50×10^{-3}	0.2266	0.17
Tower 8	4.40×10^{-3}	0.2835	0.85
Tower 9	6.90×10^{-3}	0.4480	7.6
Tower 10	1.10×10^{-4}	0.0074	0.01
Wire system	5.40×10^{-4}	0.0356	0
Insulators	7.00×10^{-6}	0.0005	0
Transmission line	1.54×10^{-2}	_	_

The Bayesian reliability importance, $I_{Bay}(i)$, identifies which component is more likely to cause a system failure and is defined as the probability that component *i* fails, given that the system Φ_s fails (Kuo and Zhu 2012b, 2012c).

$$I_{Bay}(i) = \Pr\{x_i = 0 | \Phi_s(x) = 0\} \tag{5-16}$$

The improvement potential importance, $I_{ip}(i)$, represents the maximum potential improvement in systems reliability that can be obtained by improving the reliability of component *i*. The normalized version of $I_{ip}(i)$ is obtained by dividing its value by the system failure probability.

$$I_{ip}(i) = \Pr\{\Phi_s(x) = 1 | x_i = 1\} - \Pr\{\Phi_s(x) = 1\} \tag{5-17}$$

The results show that suspension towers (Type HS and LS) have higher failure probabilities compared to other components of the line. This is consistent with the fact that current design procedures are based on a specific strength coordination in which the suspension towers are designed as weakest links in the transmission line system (CEI/IEC 60826, 2003, CAN/CSA-C22.3, 2006).

In addition, table 5-3 indicates that the improvement potential importance measure provides a different order of important components compared to the Bayesian reliability importance measure. For example, tower 2, which is a HS tower in section 1, seems less important than most of LS towers in section 2 based on Bayesian reliability importance, while it offers higher potential improvement in system reliability. This is due to the fact that the improvement potential importance considers the correlation of each component with other components of the line as well as its failure probability, while the Bayesian reliability importance only takes into account the component failure probabilities.

It is noted that the proposed component based approach provides valuable information that can be used to optimize fund allocation and determine the most efficient retrofit strategy to enhance the overall reliability of the line section. However, since it requires selecting demand meta models for each component of the line, its computation time and effort increases as the number of components increases. Hence, this method is not computationally affordable for condition assessment of power networks with hundreds of kilometres of transmission lines.

5.7.2. System based approach

For the system based approach, the system failure probability is calculated using two methods:

a) SVM-based MCS:

As described in section 5.3, the SVM classifier directly develops the limit state surface of the transmission line and classifies each new object (climatic condition) to the "survival" or "failure" classes based on its location with respect to the limit state surface. The failure limit state surface of the transmission line obtained by the polynomial SVM model is shown in Figure 5-13. In this

figure, the points above the limit state surface which correspond to high wind speeds and/or ice thicknesses belong to the "failure" class, while the points below the limit state surface belong to the "survival" class. This figure demonstrates that the line is most vulnerable when the wind direction is perpendicular to the main direction which is consistent with the results obtained from the system fragility curves shown previously (figures 5-5, 5-10 to 5-12).

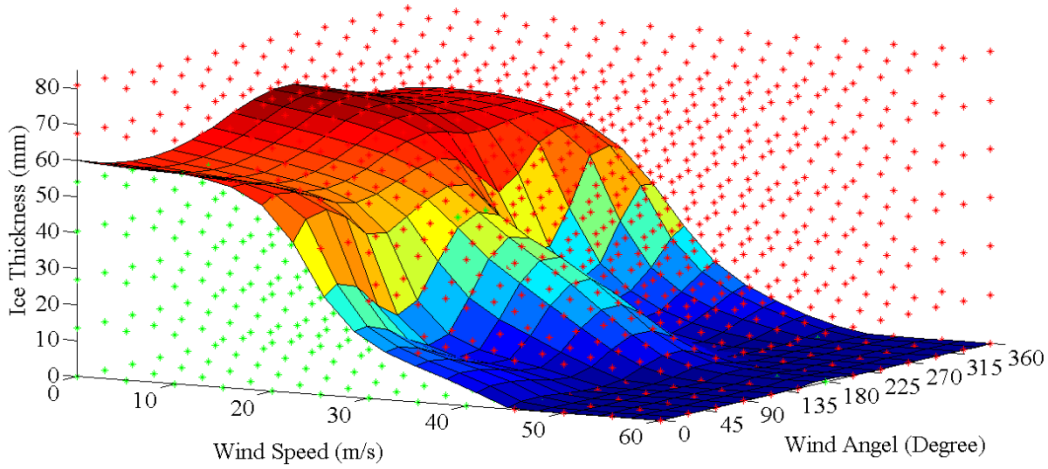


Figure 5-13. The system failure surface obtained by the polynomial SVM model

Using the MCS, 10^4 climatic variables are simulated and substituted into the polynomial SVM, and the resulting labels (failure or survival) are used to determine the failure probability as the number of failure events to the total number of simulations. Figure 5-14 presents the simulated failure and survival events separated by the SVM model given that wind is blowing perpendicular to the main direction. In this figure, the points above the limit state belong to the "failure" class, and the points below the limit state belong to the "survival" class. The circles correspond to the support vectors which are determined from equation 5-6.

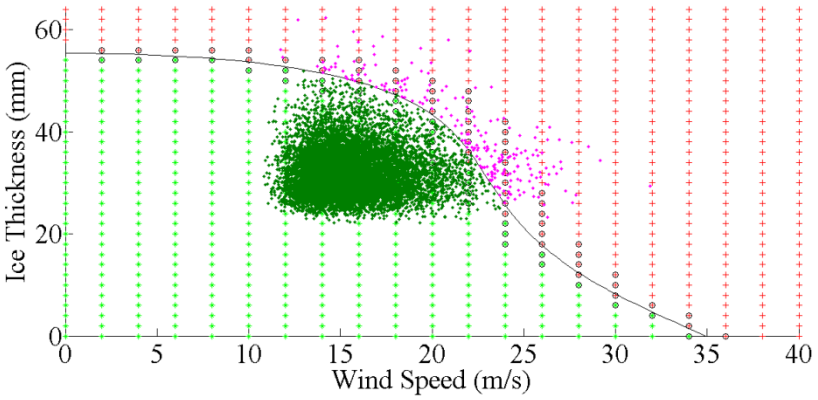


Figure 5-14. The generated system survival and failure events using polynomial SVM model given that wind angle=90

b) VP-based MCS:

As mentioned in section 5.4, the VP model not only classifies each new object into one of the "survival" or "failure" classes, it also estimates the probability that the new object belongs to each class. Therefore, both of these outputs can be used in combination with MCS to estimate the system failure probability. First, the simulated climatic variables (10^4 samples) are substituted into the polynomial SVM-based VP, and the resulted labels (failure or survival) are used to determine the failure probability as the number of failure events to the total number of simulations. Also, the average of the probabilistic outputs are presented as lower and upper bounds for the system failure probability. The calculated system failure probabilities and the corresponding reliability indices are presented in table 5-4. The resulting system failure probabilities from both system based methods (table 5-4) and the component based method (table 5-3) show good compatibility for the studied line.

Table 5-4. System failure probability and reliability index using the system based approach

Method	Underlying Algorithm	Failure probability	Reliability index
SVM (a)	Polynomial Kernel	0.0137	2.21
Venn Machine (b)	Polynomial SVM	0.0149	2.17

	[0.0133-0.0317]	[1.86-2.22]

The time and effort required for the proposed system based approach (SVM or VP) is independent of the number of components within the line, because it directly determines the failure limit state surface of the whole system (Figure 5-14) which makes it an efficient solution for overall reliability assessment of systems comprising numerous components. In condition assessment of large systems such as power networks, this methodology can efficiently be applied to each subsection of the system. Once the vulnerable subsections are identified within the power network, the most efficient retrofit strategy can be determined using the component based approach for each weak subsection.

5.8 Conclusion

This paper presents a component based and a system based approach for fragility assessment and reliability analysis of transmission lines considering the uncertainties of climatic variables such as wind speed, ice thickness and wind angle, and of the resistance of structural elements. For the component based approach, VC-bound and stepwise regression are used to select the optimal demand model for each structural component from the class of polynomial functions. The quality of the surrogate model is investigated by calculating the adjusted coefficient of multiple determination, $\bar{R}^2$, and the prediction coefficient of multiple determination, $\bar{R}_p^2$, using a 10-fold cross validation technique. The results indicate good prediction abilities for the selected surrogate models. Using this approach, both component and system fragility curves are developed for a typical 120 kV transmission line located in North America. The developed fragility information are further combined with the distribution of climatic variables at a specific

location to calculate component and system failure probabilities. The relative importance of different components of the transmission line and their contribution to the overall failure of the system is quantified by the Bayesian reliability importance measure and the improvement potential importance measure. These information enables power network asset managers to take intelligent and well-founded decisions regarding maintenance, refurbishment, upgrading and replacement of overhead lines.

In the system based approach, system fragility curves are developed by introducing a binary classification problem. Then, a SVM-based VP is developed to produce calibrated probability estimates for this classification problem. The prediction ability of both SVM and VP models are thoroughly investigated by estimating the prediction error, sensitivity, specificity, precision and F-measure. The results show good prediction abilities for the polynomial SVM-based VP developed for the studied transmission line. Unlike the component based approach, the computational effort and time required for the system based approach is independent of the number of components because it directly determines the failure limit state surface of the whole system. Therefore, this approach is particularly useful and computationally efficient for developing system fragility curves and assessing the reliability of complex structural systems such as power networks without requiring detailed component fragility information.

Both approaches proposed in this study can be successfully applied for fragility assessment and reliability analysis of other complex structural systems such as distribution lines and transportation networks subjected to various hazards such as earthquakes, tornados, etc. The proposed probabilistic framework can also be used to develop component and system fragility curves for transmission lines under different scenarios of unbalanced ice loads which is the subject of a future study by the authors. Although this paper addresses the uncertainties of

climatic variables, and of the resistance of structural elements, further research needs to be done to quantify the associated model uncertainties.

CHAPTER 6 MANUSCRIPT 3

Fragility Assessment of Transmission Lines Subjected to Unbalanced Ice Loads Using Statistical Learning Theory

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Abstract

The main objective of this paper is to develop component and system fragility curves for transmission lines subjected to unbalanced ice loads resulting from non-uniform ice formation or ice shedding. The developed fragility curves provide useful information about the conditional failure probability of transmission lines and their components given specific climatic conditions, which can be used to evaluate existing transmission lines and to optimize the design of new ones. The secondary objective of this paper is to study the effect of various design parameters such as wind speed and direction and ice loading proportion on the developed fragility curves. To achieve the aforementioned objectives, Statistical Learning Theory (SLT) is applied to select the best function that describes the structural demand in each component. In other words, in order to reduce the computational effort required for fragility analysis of transmission lines, the implicit limit state function of each element is replaced by an approximate polynomial function that has good prediction properties. The results are presented in the form of fragility curves for 3 different unbalanced loading scenarios of longitudinal, transverse and torsional loadings. The results show

that the proposed methodology can serve as a practical tool to study and compare the vulnerability of transmission line components and systems under various climate loading scenarios. For the studied transmission line, the results indicate that suspension towers are more vulnerable to unbalanced loads than tension towers. This study also concludes that wind speed, wind direction and ice loading proportion have notable effects on the transmission line fragility curves. and that for each climatic condition, the relative importance of different ice loading scenarios may change based on the order of the most critical components.

Keywords: Fragility analysis, Transmission lines, Unbalanced ice loads, Statistical learning theory

6.1 Introduction

Extreme ice and snow storms are identified as the primary cause of transmission line failure especially in cold regions (Jiazheng et al 2008, Klinger 2011, Komorski 2013). Large amounts of ice accumulation not only impose excessive vertical loads on conductors and towers, but also can result in non-uniform loads which can occur either during ice accretion due to significant changes in elevation, span or exposure, or during ice shedding. The amount of ice built up on a wire is significantly dependent on wind speed and wind direction with maximum ice thickness forming on wires perpendicular to the wind direction. This can result in unbalanced ice formation of as much as 70% on adjacent spans in some cases for example where there is a change in the direction of the line (ASCE 74, 2010). In addition, concurrent winds can exacerbate the imbalance by increasing the differential tension loads in adjacent conductors or ground wires. As a result, current design guidelines require the designer to investigate adequate reliability of transmission lines for unbalanced ice loads that generate torsion, longitudinal or

transversal bending on towers (CAN/CSA-C22.3 2006, CEI/IEC 60826 2003 and EN 50341 2001). Although some researchers have studied the effect of various factors such as structural flexibility, non-uniform ice loading proportion, ice position, span length and conductor properties on the resulting unbalanced longitudinal tension of transmission lines (Fleming et al 1978, Hou et al 2012, Mozer et al 1977, Yang et al 2012), no comprehensive probabilistic framework is available to develop component and system fragility curves for transmission lines subjected to unbalanced ice loads considering the uncertainties involved in predicting their performance. Such fragility information gives a better understanding of the behavior of various components of the line under different scenarios of icing. Fragility curves can further help in the assessment of serviceability conditions of transmission lines and their components in the aftermath of a climatic event, and they are an essential element for risk mitigation and decision analysis.

This study is part of an ongoing Project to develop a practical probabilistic framework for fragility assessment and reliability analysis of transmission lines subjected to extreme climatic events (Rezaei et al, "Component Based Structural Reliability analysis of transmission line systems for climatic hazards Using Statistical Learning Theory", working paper, Rezaei et al, "Component Based and System Based Fragility Analysis of Transmission Line Systems for Multiple Climatic Hazards", working paper). The framework is based on the Statistical Learning Theory (SLT) also known as Vapnik-Chervonenkis (VC) theory (Cherkassky et al 1999, Cherkassky and Mulier 2007) to find a surrogate demand model for each structural component of the line that can substitute the finite element solver (SAP 2000), and has good generalization (prediction) properties. The proposed approach is superior to other probabilistic methods such as the simulation based methods (Karamchandani 1987, Melchers 1999) and the Response Surface

Method (RSM) (Bucher and Bourgund 1990, Rajashekhar and Ellingwood 1993) because it is computationally more efficient, and overcomes the problem of overfitting associated with the rigid and non-adaptive model selection approach of Empirical Risk Minimization (ERM) based RSM (Guan and Melchers 2001, Hurtado 2004). The methodology is demonstrated for a typical transmission line located in North America, and the results are presented in the form of component and system fragility curves for 3 different unbalanced loading scenarios of longitudinal, transverse and torsional loadings. The configuration of the selected transmission line is defined such that various scenarios of non-uniform ice accumulation due to different exposures and ice shedding can be studied. This study also investigates the effect of design parameters such as wind speed and direction and ice loading proportion on the fragility curves. In general, the uncertainties on component demand can be classified as either epistemic or aleatory. Epistemic uncertainty is reducible and is associated with imperfect models for the underlying physics or to parameter uncertainty due to small sample sizes, while aleatory uncertainty is irreducible and associated with natural variability such as the shape and drag coefficients for ice deposits, and the variability of ice accumulation, wind speed and material properties. In this application, component demands are defined for given winds speeds and ice accumulation combinations and the uncertainty in component demand is mainly epistemic and related to drag forces and the structural model. Due to lack of data on the latter, this study assumes that component demands are deterministic for given climatic conditions. Model uncertainty can be easily considered with the proposed methodology and could be included as more information becomes available for future studies.

The remainder of this paper is structured as follows. Section 6.2 presents the methodology used for developing fragility curves based on a set of structural analyses that span the sample space

using SAP2000. Section 6.3 describes the transmission line section studied as a case study. Section 6.4 presents the developed fragility curves for both suspension and tension towers considering the effect that line exposure to wind has on non-uniform ice accumulation. The results are compared to fragility curves developed by assuming that ice accumulates uniformly on all conductors regardless of their angle relative to the wind. It is noted that the latter case is usually considered in the design of transmission lines where towers are designed to perform under extreme climatic loads while wind is assumed to blow perpendicular to the line (CEI/IEC 60826, 2003). Section 6.5 presents the fragility curves for unbalanced loads due to different ice shedding scenarios, and further investigates the effect of different design parameters such as wind speed, wind direction and ice loading proportion on the fragility curves. System fragility curves for the transmission line are presented in section 6.6 and compared to component fragility curves for different climatic conditions. Finally, Section 6.7 includes the conclusions of this study and some recommendations for future research.

6.2 Fragility assessment methodology

6.2.1 Background

Fragility curves are central to probabilistic safety assessment of transmission lines subjected to extreme climatic conditions. Conceptually, they are useful for decoupling the uncertainty associated with structural performance and the uncertainty associated with the climatic variables that generate loads. The fragility functions describe the conditional probability of failure of components or of a system given a set of climatic variables. The probability that a transmission line exposed to a climatic hazard exceeds its performance limit state can be expressed as equation (6-1).

$$P[G(\mathbf{x}) < 0] = \sum P[G(\mathbf{x}) < 0 | D = d].P[D = d] \quad (6\text{-}1)$$

where $P[G(\mathbf{x}) < 0|D = d]$ is the conditional probability of non-performance for a given climatic condition or the fragility function, and $P[D = d]$ is the hazard. $G(\mathbf{x})$ is the performance or limit state function which is normally a function of all the intervening random variables, some of which may affect the structural demand denoted by D, while the others may influence the load carrying capacity, denoted by C. Therefore, $G(\mathbf{x})$ can be written as equation (6-2).

$$G(\mathbf{x}) = C(\mathbf{x_c}) - D(\mathbf{x_d}) \quad (6\text{-}2)$$

where $\mathbf{x_c}$ includes all the random variables related to the capacity and $\mathbf{x_d}$ includes all the random variables related to the demand. The probability of non-performance is the probability of the event $G < 0$ or $D > C$. The fragility function provides useful information about the robustness of the line and allows the analysis of the performance of the line for various scenarios for the distribution of the climatic conditions. The fragility function provides useful information about the capability of the line to withstand a specific demand. It is particularly necessary for engineering decision-making in situations where the hazard curve is not available or is costly to define such as the case of unbalanced ice loads on adjacent spans of transmission lines.

The first step for developing fragility curves is to determine the limit state function of each structural component of the line explicitly. This requires the estimation of demands as a function of the climatic conditions, which is achieved with a finite element model (SAP 2000) of the transmission line. Given the complexity of the structure, the computational cost of multiple structural evaluations required for estimating the fragility curves is prohibitive and a surrogate model to the full structural model is more efficient. Therefore, this study applies the concept of

SLT for estimating surrogate demand functions, D ($\mathbf{x_d}$), for tower elements and line components. Section 6.2.2 presents more information about model selection based on SLT.

6.2.2 Model selection based on SLT

SLT is the best currently available theory for flexible statistical estimation from finite samples, which provides analytical generalization bounds for model selection (Cherkassky and Mulier, 2007). As shown in equation (6-3) (Cherkassky and Mulier 1998, Cherkassky et al. 1999, Vapnik 1995), VC generalization bounds relate the unknown prediction error, $E(\omega)$, to the number of training samples, *n*, empirical error, $E_{emp}(\omega)$, and a measure of model complexity called the VC-dimension, *h* .

$$E(\omega) = E_{emp}(\omega) \Big/ \left(1 - \sqrt{\left(\frac{h}{n}\right) - \left(\frac{h}{n}\right) * \ln\left(\frac{h}{n}\right) + \frac{\ln(n)}{2n}}\right)_{+} \tag{6-3}$$

The VC dimension, *h*, is a characteristic of a set of indicator functions, which corresponds to the maximum number of samples for which all possible binary labelings can be induced without error. In the case of linear real-valued functions, *h* is equal to the number of free parameters in the model.

SLT is based on the Structural Risk Minimization (SRM) principle under which the set *S* of approximating functions $f(\mathbf{x},\omega)$ consists of nested elements S_k ordered according to their VC dimension. For a given training data, SRM selects the function $f(\mathbf{x},\omega^*)$ which minimizes the empirical risk for the functions from element S_k,. Then, the prediction error is calculated for each element S_k using the bound provided by equation (6-3). The optimal structure element S_{opt} is the one providing minimum prediction error. This structure element has the optimal complexity (VC dimension). In this study, the optimal demand functions for each tower element/ line component

are selected from the class of polynomial functions. In addition, in order to exclude non-significant variables and further reduce the complexity of the model, a stepwise regression method is used to fit each candidate function to training samples. Stepwise regression is a variable selection process that selects an optimal subset of variables according to a certain criterion. During the procedure, a sequence of regression models is constructed iteratively by adding or removing variables at each step (Miller 2002, Montgomery and Runger 2003).

6.2.3 Meta-model validation based on re-sampling techniques

Assessing model accuracy is a crucial aspect in the meta-modeling framework. Surrogate demand models without sufficient accuracy will result in unrealistic failure probabilities for the transmission line and its components. This highlights the importance of a systematic model validation procedure which shows the reliability and accuracy of the selected demand model. Various re-sampling techniques have been developed to deal with this issue such as cross-validation (Stone, 1974), bootstrapping (Efron 1979, Efron 1983, Efron and Tibshirani 1997), subsampling (Xu and Liang 2001), and nested resampling as a combination of the former approaches. Many researchers have compared the advantages and limitations of different re-sampling techniques and have provided recommendations about the appropriate method to be used for a specific problem (Bischl et al. 2012, Burman 1989, Molinaro et al. 2005, Simon 2007)

In this study, since the prediction error provided by equation 3 is used to select the optimal demand model from different candidate polynomial functions, it is biased and does not provide a good measure for the prediction accuracy of the selected model. Considering the suggestions of Molinaro et al. (2005), Bischl et al. (2012) and Rodrıguez et al. (2010) and aiming to minimize

the computational effort required for determining the best demand model for each structural component of the line, a 5-fold cross-validation technique is used to divide the analysis data from SAP2000 into training and test subsets. The demand model is developed on the training subset and the error is evaluated on the test subset. The training algorithm is repeated 5 times with a fraction 1/5 of the sample points left out for testing purposes. The selected model is validated by calculating the adjusted coefficient of multiple determination ($\bar{R}^2$) and the prediction coefficient of multiple determination ($\bar{R}_p^2$) using equations 6-4 and 6-5. A good model will have $\bar{R}^2$ and $\bar{R}_p^2$ values near 1.

$$\bar{R}^2 = 1 - \frac{MSE}{MST} = 1 - \left(\frac{n-1}{n-p}\right)(1 - R^2) \tag{6-4}$$

where *MSE* is the mean square error, *MST* is the total mean square of variation in observations, *n* is the total number of samples, *p* is the number of parameters in the model and R^2 is the coefficient of multiple determination. The adjusted coefficient of multiple determination which expresses the quality of fit between the regression model and the sample data is a useful tool in evaluating the regression models as it guards against overfitting by penalizing the analyst for adding terms to the model (Montgomery and Runger 2003).

$$\bar{R}_p^2 = 1 - \frac{\sum_{i=1}^{n_{te}} (y_{ite} - \hat{y}_{itr})^2}{\sum_{i=1}^{n_{te}} (y_{ite} - \bar{y}_{te})^2} \tag{6-5}$$

In equation 6-5, n_{te} and $\bar{y}_{te}$ are the number and mean of the observed demands (y_{ite}) in the test subset and $\hat{y}_{itr}$ are the predictions of the demands in the test subset using the fitted model to the training subset. The prediction coefficient of multiple determination evaluates the ability of the final selected model to predict future demands. It is noted that in this study, $\bar{R}_p^2$ is the average prediction coefficient of determination for all 5 test subsets.

6.2.4 Defining the strength of components and limit states

The limit states considered for determining the strength of tower members include: 1) failure of members in tension 2) failure of members in compression due to yielding and global or local buckling 3) failure of bolts in shear 4) failure of connections under bearing stresses (ASCE10-97, 2003). It is assumed that the capacity of tower elements have a lognormal distribution with a coefficient of variation of 10% (CEI/IEC 60826, 2003).

The nominal tensile strength of conductors and ground wires is specified as 141.7 kN and 139 kN, respectively, and is assumed to follow a lognormal distribution with a coefficient of variation of 3% (CEI/IEC 60826, 2003). The studied line section has three types of insulators with nominal tensile strengths of 70, 140 and 240 kN and follow lognormal distributions with a coefficient of variation of 5% (CEI/IEC 60826, 2003). It is noted that according to CEI/IEC 60826 (2003), the nominal properties of components of transmission lines are specified for a 10% exclusion limit.

6.2.5 Developing component and system fragility curves

As mentioned previously, a component fragility curve presents the conditional probability of failure of a component given a specific combination of climatic variables. In this study, in order to keep the model simple and computationally practical, the uncertainties in material properties are only taken into account while determining the component capacities, and the structural model of the line is assumed to be deterministic. Therefore, the component fragility curve for each tower member, insulator, conductor and ground wire is developed by comparing the resulting deterministic load from each combination of climatic variables with the corresponding probability distribution function of the component capacity. In order to develop system fragility

curves for towers as well as the overall line, both are idealized as series systems. This assumption is conservative and implies that the system fails once any one of its components fails. It is noted that the correlation between failures of different components due to the same source of external loading is taken into account when developing system fragility curves for the towers and the line.

6.3 Case study

6.3.1 Description of the transmission line

The methodology is applied to a typical 120 kV transmission line located in North America with 9 spans and a total length of 2.6 km (Figure 6-1). It consists of two line sections separated by angle/dead-end towers. A total number of ten towers are included in the case study to consider for the effect of limited spatial extension of climatic hazards on the failure probability of transmission line systems. According to CEI/IEC 60826 (2003) and CAN/CSA-C22.3 (2006), in a flat to rolling terrain and based on a span of 400 m, 1 to 5 towers are usually subjected to maximum wind and maximum ice and wind loads while 10 to 20 towers are usually subjected to maximum ice loads. The configuration of the selected line section also allows for investigating the fragility of the line for both scenarios of non-uniform ice accumulation due to different exposures and ice shedding.

A description of towers within the studied line segment is presented in Table 6-1. As shown in Figure 6-2, HS and LS towers are heavy suspension and light suspension towers, respectively while STR towers are angle/dead-end towers. The diameter and weight per unit length of conductors are 27.78 mm and 15.26 N/m, respectively, and the diameter and weight per unit length of ground wires are 12.57 mm and 7.44 N/m, respectively.

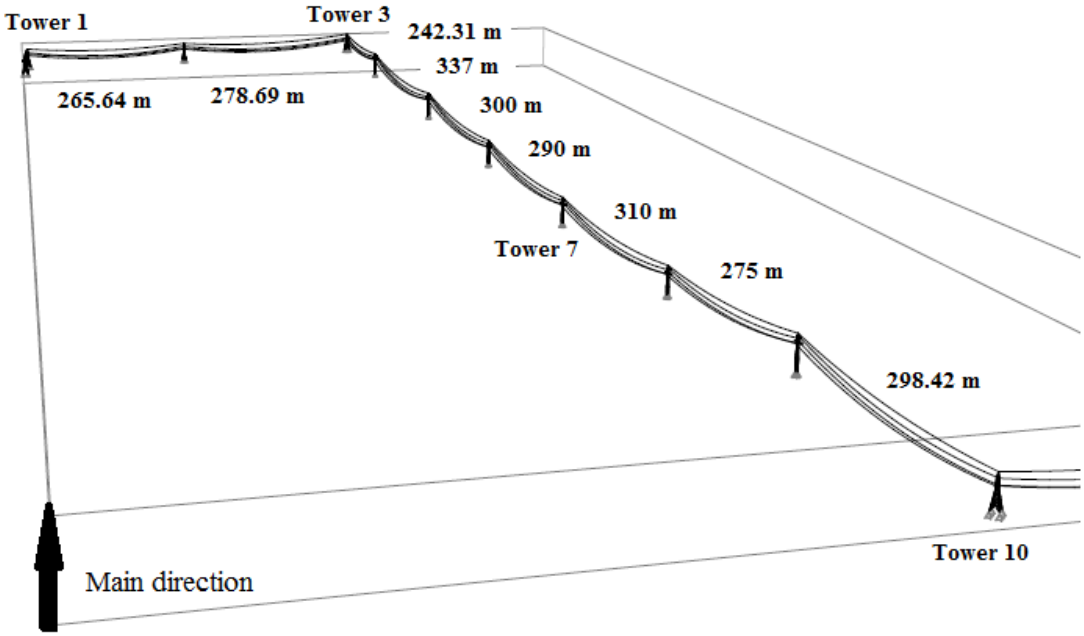


Figure 6-1. Nine span segment of the transmission line under study

Table 6-1. Description of towers of the line segment

Tower Number	*Tower Height (m)*	*a (m)*	*b (m)*	*c (m)*	*d (m)*	*Tower Type*
1	*30.5*	*6*	*8.5*	*11*	*8.06*	*STR*
2	*28.75*	*3.75*	*6.25*	*8.75*	*5.26*	*HS*
3	*27.5*	*6*	*8.5*	*11*	*7.08*	*STR*
4	*33.25*	*3.75*	*6.25*	*8.75*	*1.9*	*LS*
5	*33.25*	*3.75*	*6.25*	*8.75*	*1.9*	*LS*
6	*30.25*	*3.75*	*6.25*	*8.75*	*1.9*	*LS*
7	*30.25*	*3.75*	*6.25*	*8.75*	*1.9*	*LS*
8	*28.75*	*3.75*	*6.25*	*8.75*	*1.9*	*LS*
9	*33.25*	*3.75*	*6.25*	*8.75*	*1.9*	*LS*
10	*27.5*	*6*	*8.5*	*11*	*7.08*	*STR*

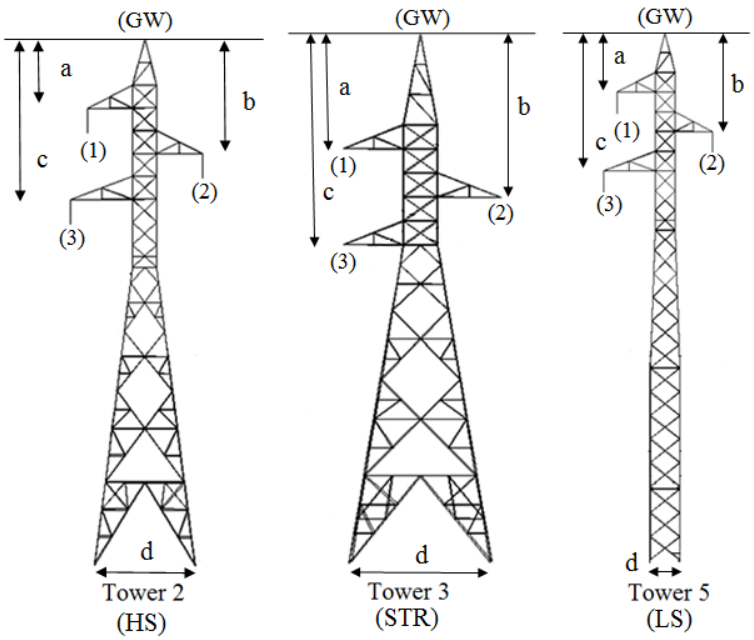


Figure 6-2. The configuration of tower types within the line

6.3.2 Modeling of the transmission line

A three dimensional model of the line is developed in Sap2000. The model takes into account the nonlinear behaviour of the wire system as well as the flexibility of the towers and the resulting interactions between different components of the transmission line. The overhead towers are modeled as frame-truss systems in that leg members with multi-bolted connections are modelled as frames with rigid connections, and horizontal and diagonal members with single-bolted connections are modelled as truss elements with pinned connections. The conductors and ground wires are modeled using cable elements with tension-only behaviour, and the insulators are modeled with truss elements. Ice loads are applied on the wire system and the supports according to CEI/IEC 60826 (2003). It is assumed that ice loads are applied to line components as a result of freezing rain events, and the density of ice is assumed to be equal to 900 kg/mm^3. Wind on ice loads are also calculated taking into account the influence of the height, shape and surface

properties of each member as well as the terrain category and the dynamic response of components (CEI/IEC 60826, 2003).

6.3.3 Structural analysis of the transmission line and validation of the demand meta-model

In order to study the effect of various design parameters on the fragility curves, the selected transmission line is analyzed under various combinations of climatic variables as shown in table 6-2.

Table 6-2. Range of values for climatic variables for which the transmission line is analyzed with SAP 2000

	Ice accretion scenario		Ice shedding scenario	
	Range	Step	Range	Step
Wind speed (m/s)	0 - 30	10	0 - 30	10
Wind angle	0 - 90	15	0, 30, 45, 60, 90	-
Ice thickness (mm)	0 - 80	5	0 - 80	5
Temperature (°C)	-5	-	-5	-

The fragility functions are presented as a function of ice thickness. For this purpose, for each scenario, the SLT demand models are trained and validated based on 17 samples using a 5-fold cross validation technique. Figures 6-3 and 6-4 present typical examples of the calculated $\bar{R}^2$ and $\bar{R}_p^2$ respectively for the demands on each member of tower 3 as a function of its elevation in the tower. The high values of $\bar{R}^2$ indicate that the SLT model provides a close fit to the data, while the $\bar{R}_p^2$ measures the predictive ability of the model for combinations not considered in the training set. In this figure $\bar{R}_p^2$ is the average prediction coefficient of determination for all 5 test subsets. The results show the adequacy of the selected model based on both the training samples and the test samples.

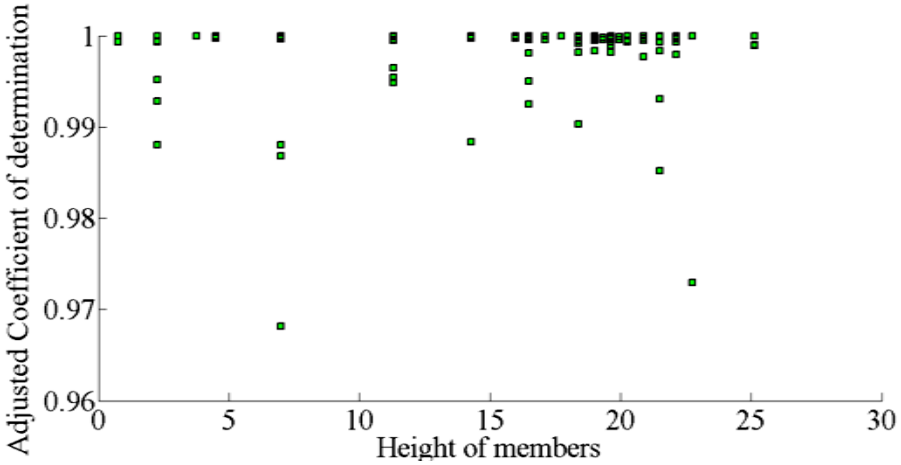


Figure 6-3. Adjusted coefficient of determination for each member of tower 3 as a function of its elevation in the tower

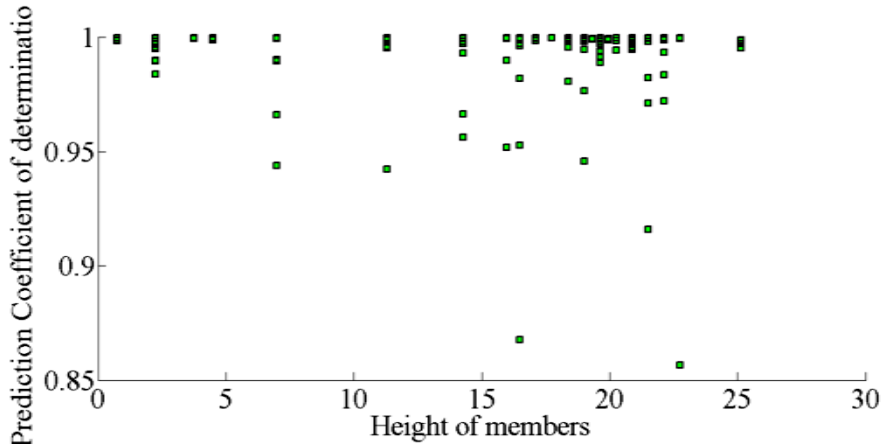


Figure 6-4. Prediction coefficient of determination for each member of tower 3 as a function of its elevation in the tower

6.4 Fragility curves for non-uniform ice accumulation

Current transmission line design guidelines such as the European Standard EN 50341 (2001), CEI/IEC 60826 (2003) and CAN/CSA-C22.3 (2006) specify the design ice loads in terms of equivalent uniform ice thickness which is applied uniformly on all surfaces of the structure. However, the amount of ice accretion is dependent on the amount of wind-blown raindrops and can vary significantly depending on the wind speed and direction. Because ice loads are difficult to measure and little data is available on equivalent uniform ice thickness from natural ice accretion on transmission lines, many researchers have developed mathematical models to

simulate the ice built up from available meteorological information (Chaine and Castonguay 1974, Jones 1996, 1998, Jones et. al 2002, Jones and Egelhofer 1991, Makkonen 2000, MRI 1977). These ice models can be divided into two main categories. 1) simple analytical models that rely on strong assumptions or empirical results which do not consider details of the icing process 2) heat-balance numerical models that account for the physical details of the icing process (Makkonen and Lozowski 2008). Although icing models from the second category provide more accurate estimates compared to the simple models, they are more complex and computationally expensive and rely on accurate estimation of many climatic parameters which are not usually available. For this reason, in this paper, in order to study the effect of wind speed and direction on ice accumulation, the Simple model proposed by Jones (1998) is used which requires only precipitation and wind measurements to determine the amount of accreted ice and assumes that all the impinging water freezes (equation 6-6).

$$R_{eq} = \sum_{j=1}^{N} \frac{1}{\rho_i \pi} \left\{ (P_j \rho_0)^2 + (3.6 V_j \omega_j \sin[\varphi_1 - \varphi_2])^2 \right\}^{1/2} \quad (6\text{-}6)$$

where P_j, ρ_0 , ρ_i , φ_1 and φ_2 are the precipitation rate (mm in the jth hour), the density of water (1 gr/cm^3), the density of glaze ice (0.9 gr/cm^3), the wire direction and the wind direction, respectively. V_j is the wind speed (m/s) and $\omega_j = 0.067 P_j^{0.846}$ (Best 1949) is the liquid water content (gr/m^3) of the rain-filled air in the jth hour. Figure 6-5 shows the effect of wind speed and direction on the amount of ice accretion using equation 6-6.

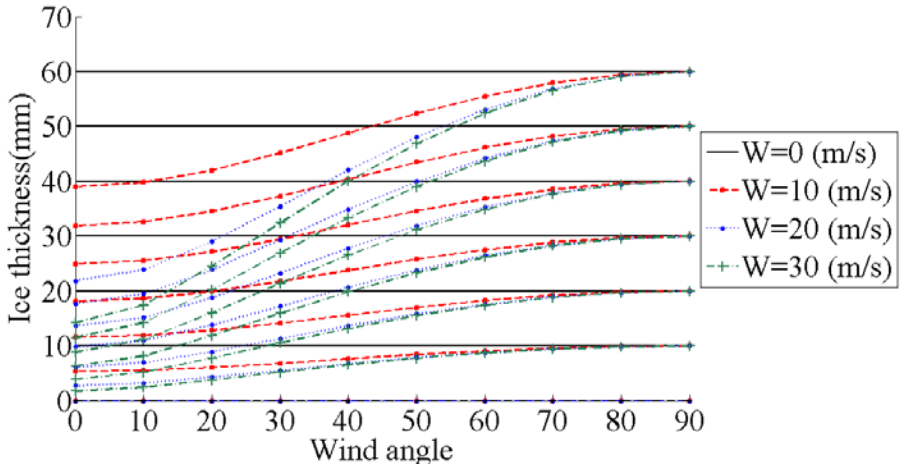


Figure 6-5. Effect of wind speed and wind direction on the amount of ice accretion using the Jone's simple model

It is inferred from this figure that when no wind is blowing, ice forms uniformly on all conductors regardless of their direction, however this is not the case for other values of wind speed. For example, when wind speed is 20 m/s, and 40mm of ice is formed on conductors perpendicular to the wind direction, only 14 mm of ice forms on conductors which are parallel to the wind direction.

Figure 6-6 shows the developed fragility curves for tension tower 3 assuming that maximum amount of ice accretes uniformly on all conductors regardless of their exposure to wind. It is noted that this case is usually considered in the design of transmission lines where towers are designed to perform under extreme climatic loads while wind is assumed to blow perpendicular to the line. On the other hand, figure 6-7 presents the developed fragility curves for tension tower 3 considering the influence of line exposure to wind on non-uniform ice accumulation. In these figures, wind angle is estimated with respect to the main direction of the line as shown in figure 6-1. In figure 6-7, the horizontal axis represent the maximum amount of ice that can form on conductors which corresponds to ice formation on conductors that are perpendicular to wind

direction. As shown in figure 6-5, lower amount of ice forms on other conductors depending on their direction with respect to the wind direction.

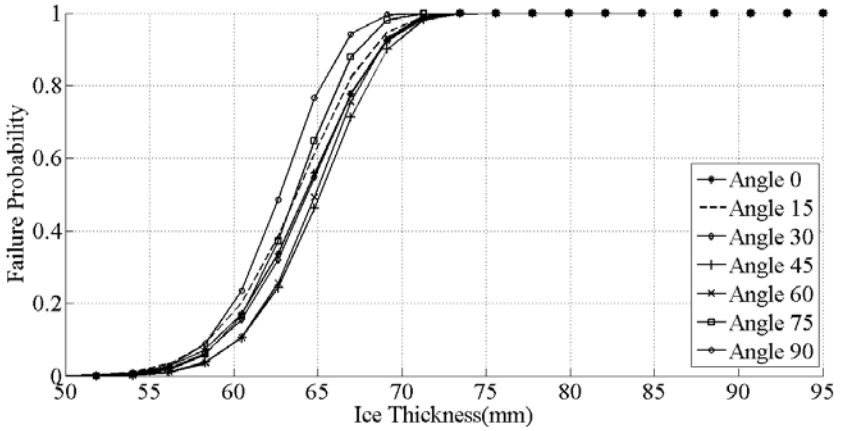


Figure 6-6. Fragility curves for tension tower 3 assuming that maximum amount of ice accretes uniformly on all conductors regardless of their exposure to wind (wind speed=20 m/s)

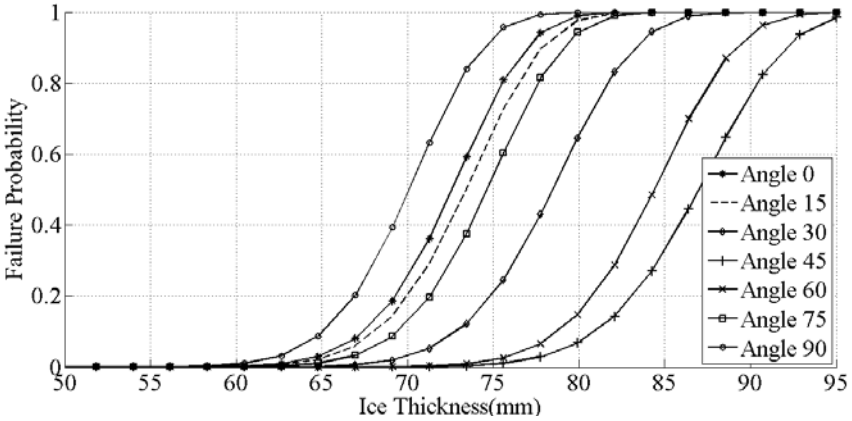


Figure 6-7. Fragility curves for tension tower 3 considering the influence of line exposure to wind on non-uniform ice accumulation (wind speed=20 m/s)

Comparing the results from figures 6-6 and 6-7, it is inferred that for angle towers, neglecting the non-uniform ice accumulation due to significant changes of line exposure results in non realistic fragility curves for all directions of wind and generally higher failure probabilities.

Figures 6-8 and 6-9 show the developed fragility curves for suspension tower 5 under the same two conditions. Comparing the fragility curves in figures 6-8 and 6-9, it is inferred that when neglecting the effect of non-uniform ice accumulation due to line exposure, only the fragility

curve for the case when wind is blowing perpendicular to the line section is realistic and all other fragility curves show non realistic and higher failure probabilities. The most significant difference is for the case when wind is parallel to the line. As expected for this case, figure 6-9 shows zero failure probability for the studied suspension tower while figure 6-8 shows an increased failure probability with increasing amounts of ice. The reason is that figure 6-8 is neglecting the fact that this amount of ice cannot actually form on this tower and its surrounding conductors due to their lower exposure to the wind-blown rain drops.

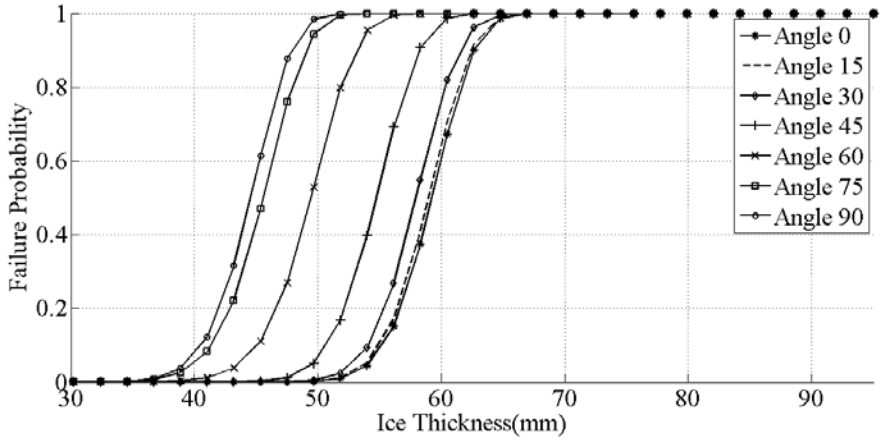


Figure 6-8. Fragility curves for suspension tower 5 assuming that maximum amount of ice accretes uniformly on all conductors regardless of their exposure to wind (wind speed=20 m/s)

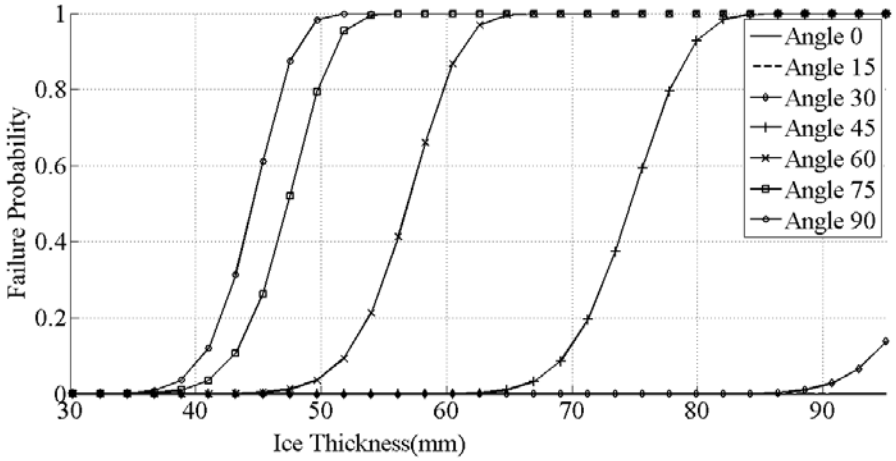


Figure 6-9. Fragility curves for suspension tower 5 considering the influence of line exposure to wind on non-uniform ice accumulation (wind speed=20 m/s)

In addition, the results indicate that suspension towers are more vulnerable than tension towers for all climatic conditions. This is consistent with the fact that current design procedures are based on a specific strength coordination in which suspension towers are designed as weakest links in the transmission line system in order to minimize the impact of a potential failure (CAN/CSA-C22.3, 2006 and CEI/IEC 60826, 2003).

6.5 Fragility curves for various ice shedding scenarios

In this section, fragility curves are developed for various ice shedding scenarios. As mentioned previously, current design guidelines suggest that in addition to uniform ice loads, transmission lines should withstand unbalanced ice loads that generate torsion, longitudinal or transversal bending on towers. It is suggested to apply non-uniform ice loads on up to 3 consecutive spans for these scenarios of unbalanced ice loads (EN 50341, 2001, CEI/IEC 60826, 2003 and CAN/CSA-C22.3, 2006). In this study, 6 loading scenarios are considered: uniform, transverse, 2 longitudinal and 2 torsion scenarios. Figure 6-10 shows how loads are applied on the wire system for each scenario. The solid lines represent the wires that are loaded more, while the dash lines represent the wires which are loaded less.

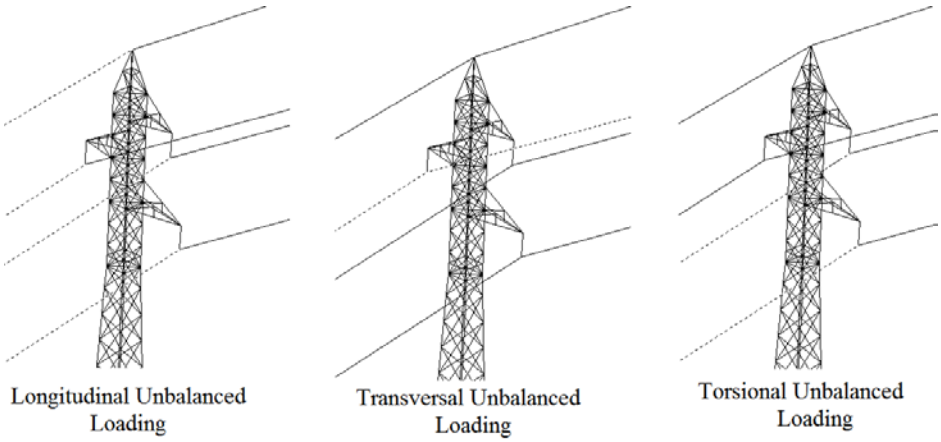


Figure 6-10. Ice shedding scenarios

6.5.1. The effect of wind direction

Figure 6-11 shows the developed fragility curves for suspension tower 5 given that Wind speed is 20 m/s and ice loading proportion is 100/50. In this figure, L, Tr and To represent Longitudinal bending, Transversal bending and Torsion, respectively. Indices 1 and 2 correspond to an ice shedding of 50% from three spans before and after the studied tower, respectively, while other spans are subjected to 100% of the reference ice loads. The fragility curves are presented for three wind angles of 0, 45 and 90 degrees with respect to the main direction of the line. It is noted that when wind angle is 0, fragility curves are shifted to the far right side of figure 6-11 and result is zero failure probabilities for the selected range of ice thickness. The results indicate that for each wind angle, ice shedding from conductors of three spans after the studied tower (L2 and TO2 scenarios) is a more critical loading scenario compared to ice shedding from conductors of three spans before the studied tower (L1 and TO1 scenarios) . This is due to the fact that ice shedding from the spans before the tower results in lower longitudinal loads on the tower since the effect of higher ice loads applied on the spans after the tower can be reduced by the longitudinal displacement of several suspension insulators. However, this is not the case when higher ice loads are applied to the spans near an angle tower where the insulators are in series with the conductor. The length of spans before and after each tower is another factor affecting the resulting vertical and longitudinal loads applied to suspension towers. For instance, in adjacent spans, ice shedding from the larger span is less critical than ice shedding from the smaller one. Therefore, ice shedding from the spans before suspension tower 4 is more critical than shedding from the spans after the tower. Is summary, the most critical ice shedding scenario is the one that results in higher vertical and/or longitudinal loads on towers and conductors. And for actual lines where various span lengths and tower types are used, determining the most

critical ice shedding scenario is complicated and highlights the need for a thorough fragility analysis.

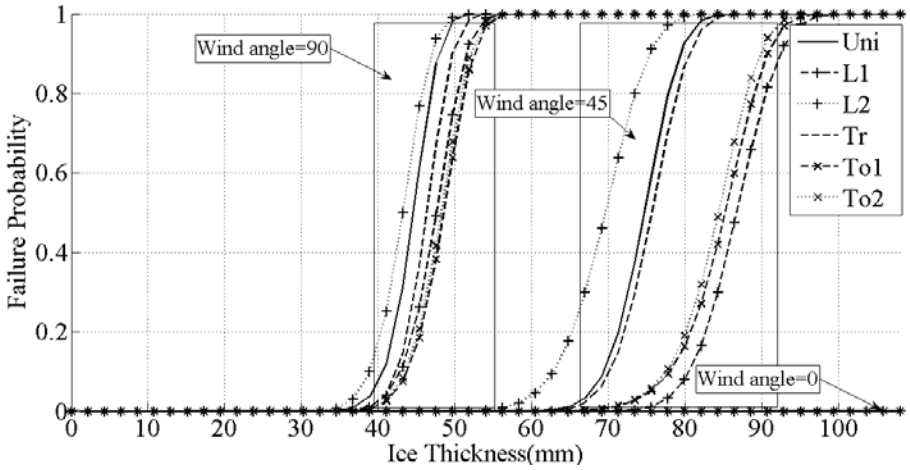


Figure 6-11. Fragility curves for suspension tower 5 for different ice shedding scenarios given that Wind speed=20 m/s, Ice loading proportion=100/50

Figures 6-12 and 6-13 show the fragility curves for tension tower 3 when wind angle is 0 and 90 degrees, respectively. It is inferred from these figures that when wind angle is 0 with respect to the main direction of the transmission line, more ice forms on previous spans of the studied tower which are perpendicular to the wind direction compared to the spans after the tower which are parallel to the wind direction. Therefore, ice shedding from all conductors of the spans before the tower (L1 scenario) is the least critical scenario. While ice shedding from all conductors after the tower (L2 scenario) has similar effects on the tower as the case of no shedding which is the most critical scenario of ice loading for the studied tower. On the other hand, when wind angle is 90 degrees with respect to the main direction, following the same concepts indicates that ice shedding from the spans after tower 3 (L2 scenario) which are perpendicular to the wind direction is the least critical ice shedding scenario.

In summary, it is inferred from figures 6-11 to 6-13 that wind direction has a significant effect on the fragility curves of both tension and suspension towers, and the relative importance of

different ice loading scenarios may change for different wind directions based on the order of most critical components that contribute to the failure of the tower.

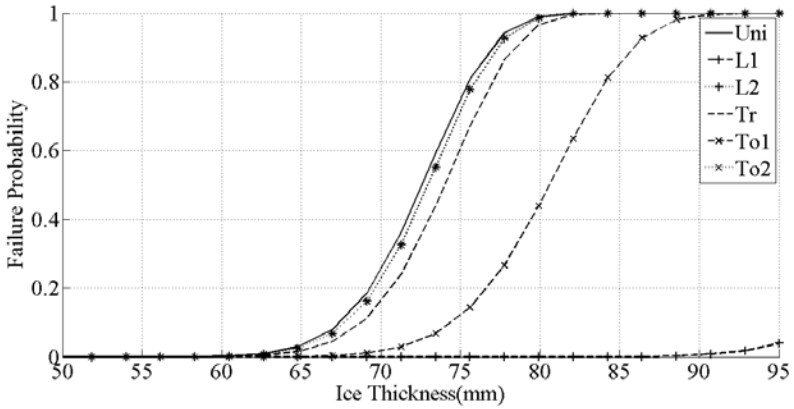


Figure 6-12. Fragility curves for tension tower 3 for different ice shedding scenarios given that Wind speed=20 m/s, wind angle=0 and Ice loading proportion=100/50

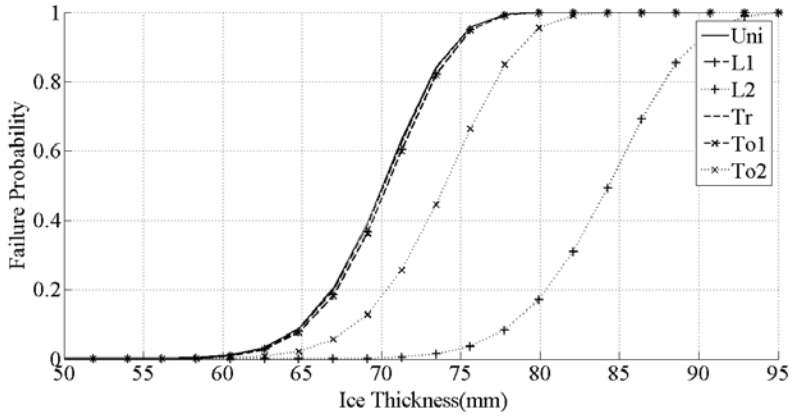


Figure 6-13. Fragility curves for tension tower 3 for different ice shedding scenarios given that Wind speed=20 m/s, wind angle=90 and Ice loading proportion=100/50

6.5.2. The effect of wind speed

Figure 6-14 shows the fragility curves for suspension tower 5 when wind speed is 0 m/s, 20 m/s and 30 m/s, respectively. It is inferred from the results that wind speed has a significant effect on the developed fragility curves, and line components become more vulnerable under higher wind intensities. It is also noted that for different wind speeds, the relative importance of different ice

loading scenarios may change based on the order of most critical components that contribute to the failure of the tower. Finally, figure 6-14 indicates that, as the wind speed increases, the fragility curves become insensitive to the ice shedding scenario.

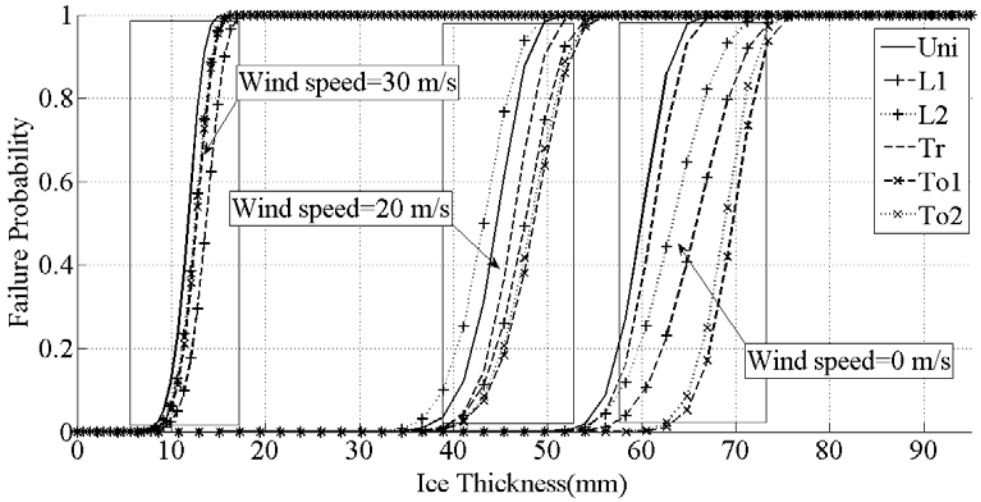


Figure 6-14. Fragility curves for suspension tower 5 for different ice shedding scenarios given that Wind angle=90 and Ice loading proportion=100/50

6.5.3. The effect of ice loading proportion

Current design guidelines propose different ice loading proportions when considering unbalanced ice loading scenarios. The ice loading proportion suggested by CAN/CSA-C22.3 (2006) is 70% and 28% of the reference design ice load on two adjacent spans for all unbalanced ice loading scenarios. However, EN 50341-1 (2001) suggests to use an ice loading proportion of 70% and 30% of the reference design ice load on two adjacent spans for longitudinal bending and torsion ice loading scenarios and an ice loading proportion of 100% and 50% of the reference design ice load for transversal bending scenario. In this study, fragility curves are developed for different ice loading proportion of 100/10, 100/30, 100/50, 100/70, 100/90 and 70/28 on adjacent spans. Figures 6-15 to 6-17 show the effect of ice loading proportion on fragility curves of suspension tower 5 for L2, Tr and To2 loading scenarios, respectively, given

that wind speed is 20 m/s and wind angle is 90 degrees with respect to the main direction of the line.

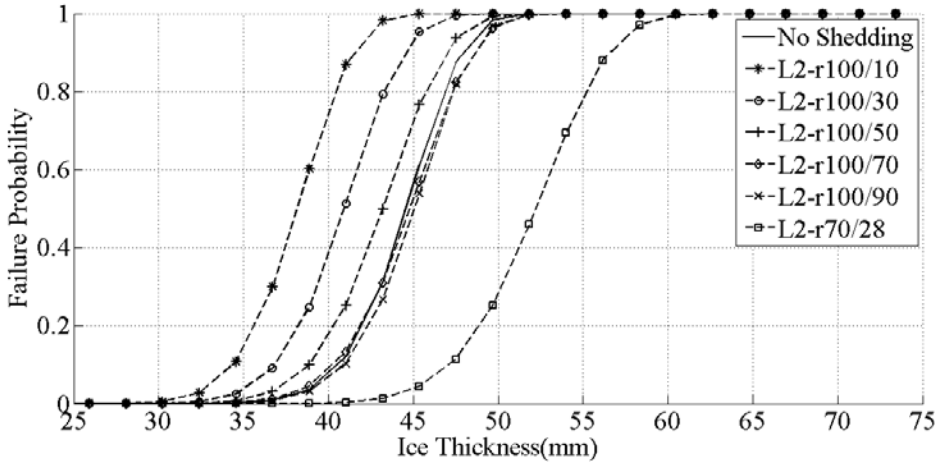


Figure 6-15. The effect of icing rate on fragility curves of suspension tower 5 for longitudinal bending loading scenario (L2)

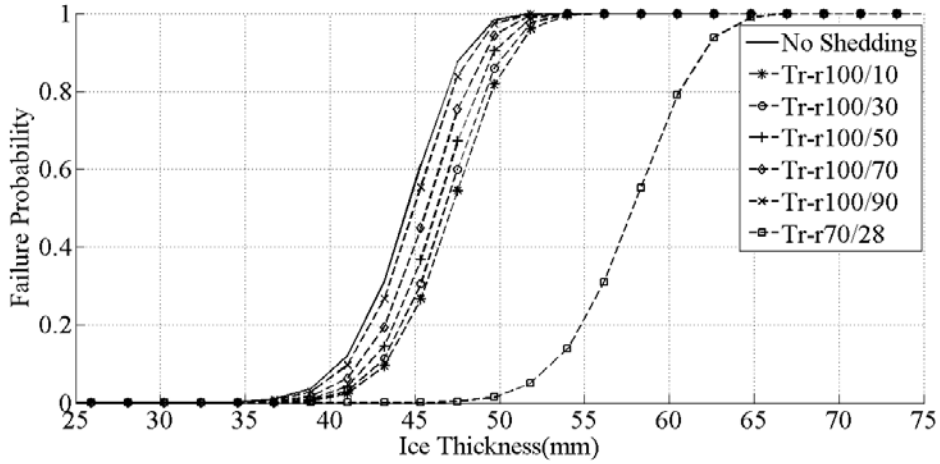


Figure 6-16. The effect of icing rate on fragility curves of suspension tower 5 for transversal bending loading scenario (Tr)

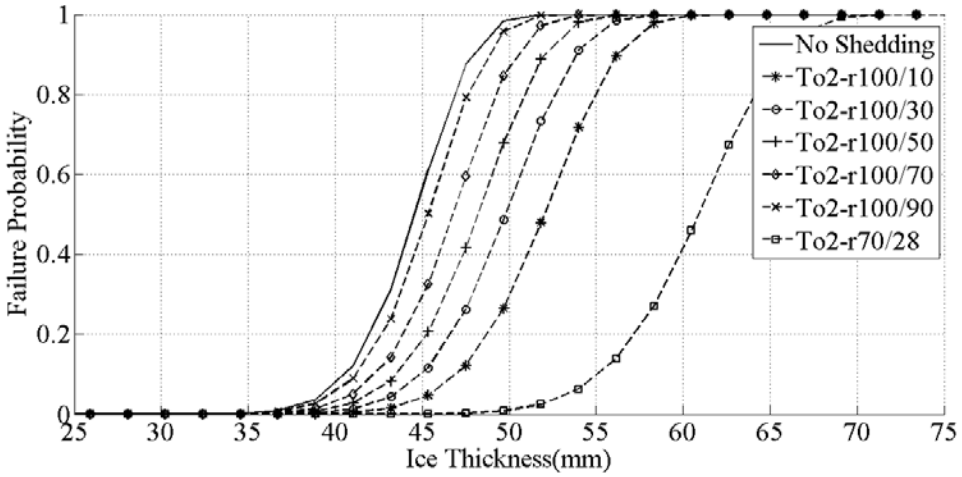


Figure 6-17. The effect of icing rate on fragility curves of suspension tower 5 for torsion loading scenario (To2)

The results indicate that under L2 loading scenario, tower 5 becomes more vulnerable as the difference between the applied ice loads on adjacent spans increases and the tower is subjected to higher longitudinal loads. While under Tr and To2 loading scenarios, the studied tower is more vulnerable when the difference between the applied ice loads on adjacent spans decreases and the tower is subjected to higher vertical ice loads. It is also inferred that the ice loading proportion proposed by current standards are not necessarily the most critical ice loading proportion and caution should be exercised when designing transmission towers for unbalanced ice loads. It is noted that, in determining the design loads, both criticality and likelihood of a loading scenario should be taken into account to achieve a safe and feasible structural design. Therefore, further research is needed to determine the likelihood of different ice shedding scenarios.

6.6 Fragility curves for the transmission line system

Figures 6-18 and 6-19 present the fragility curves for the studied transmission line, all towers and the wire system given that wind angle is 90 and 0 degrees with respect to the main direction of the line, respectively. Non-uniform ice loads are applied to both line sections considering their exposure to the wind and wind speed is 20 m/s. It is noted that no ice shedding is considered in

this section. In addition, It should be mentioned that the fragility of foundations are not addressed in this study and may be considered in future studies when experimental data becomes available.

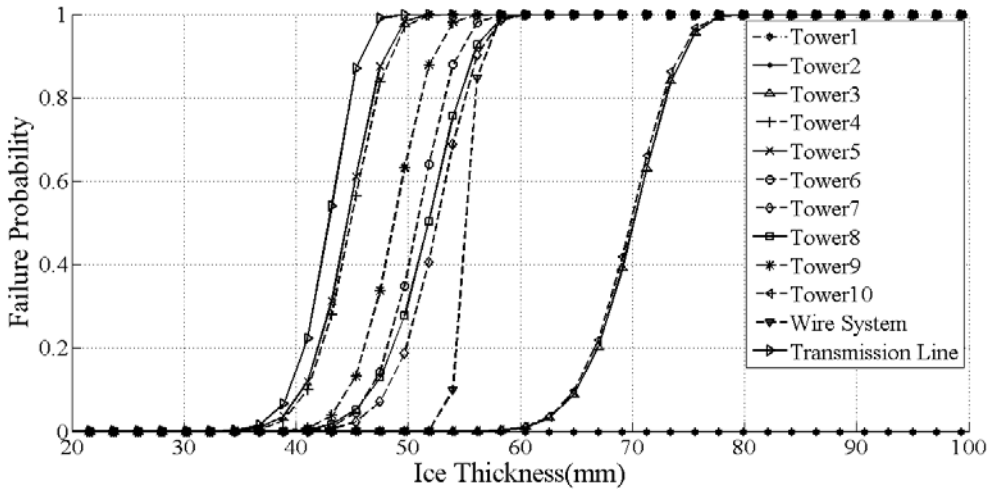


Figure 6-18. Component and system fragility curves for wind speed=20 m/s and wind angle=90

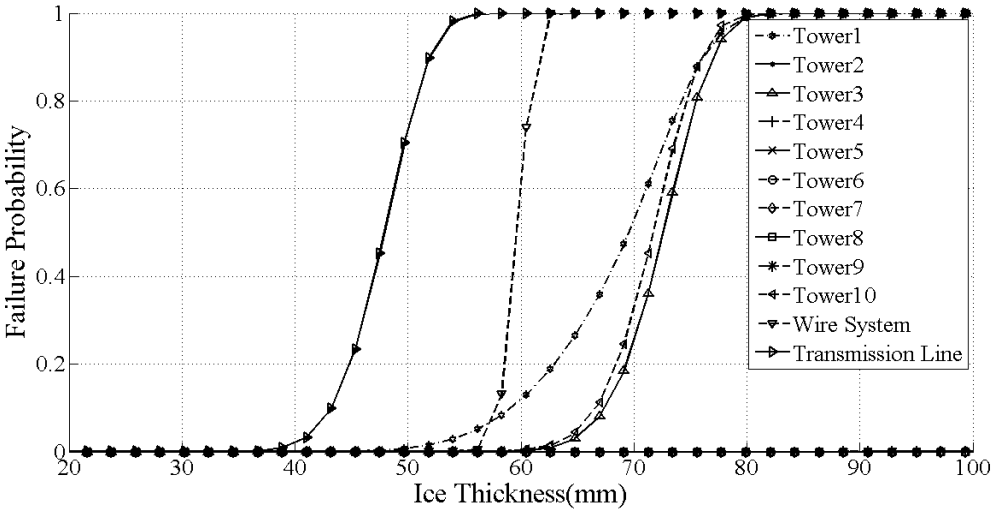


Figure 6-19. Component and system fragility curves for wind speed=20 m/s and wind angle=0

It is inferred from the results that transmission line components contribute differently to the failure of the transmission line for different climatic conditions. For example, as indicated in figure 6-18, the failure of the transmission line is mainly due to the failure of suspension towers in section 2 when wind angle is 90 degrees while figure 6-19 shows that when wind angle is zero

with respect to the main direction of the line, the failure of the transmission line is mainly due to the failure of the only suspension tower in section 1. It is noted that the most critical components and load combinations for a specific transmission line can be determined by incorporating the information about the climatic hazard at the location of the line with these fragility curves (Rezaei et al, "Component Based Structural Reliability analysis of transmission line systems for climatic hazards Using Statistical Learning Theory", working paper).

6.7 Conclusion

This paper describes the application of Statistical Learning Theory (SLT) in component and system fragility analysis of transmission lines subjected to unbalanced ice loads resulting from non-uniform ice formation or ice shedding. Considering the implicit nature of component limit state functions, the optimal demand function for each component of the line is selected from the class of polynomial functions using VC-generalization bounds and Stepwise regression. The unbiased prediction errors determined using CV technique indicate good prediction properties for the selected surrogate demand model of the finite element solver. The results are presented in the form of fragility curves for 3 different unbalanced loading scenarios of longitudinal, transverse and torsional loadings. The effect of various design parameters such as wind speed and direction and ice loading proportion on the developed fragility curves are also investigated. The results show that the proposed methodology can serve as a practical tool to study and compare the vulnerability of transmission line components and systems under various climatic loading scenarios. In addition, Fragility curves such as the ones developed in this study, provide a basis for the reliability analysis by presenting the conditional failure probability of transmission lines and their components given specific climatic conditions which are independent of the

location of the line. For the case study used in this paper, the proposed methodology allowed to show that

- The configuration and type of towers affect the tower fragility curves. For example, in this study, suspension towers are more vulnerable than tension towers for all loading scenarios.
- Wind speed and direction significantly affect both wind and ice loads applied to transmission lines and consequently fragility curves can change considerably based on these parameters.
- For different climatic conditions, the relative importance of different ice loading scenarios may change based on the order of the most critical components that contribute to the system failure.
- Ice loading proportion notably influences the fragility curves of towers. It is shown that the ice shedding scenario affects the most critical ice loading proportion for each tower. In addition, the results indicate that, the ice loading proportion proposed by current standards are not necessarily the most critical ice loading proportion, as a safe and feasible design is only achieved by considering both criticality and likeliness of a loading scenario. This signifies the necessity of further studies on determining the appropriate design point considering the probability of occurrence of each ice shedding scenario for the transmission line location.
- The most critical ice shedding scenario is the one that results in higher vertical and/or longitudinal loads on towers and conductors. For example, ice shedding from the spans far from a tension tower will result in higher longitudinal loads and higher failure probabilities for the tower under unbalanced loads due to lower available movement

freedom of conductor attachment points with tension towers. The length of spans before and after each tower is another factor affecting the resulting vertical and longitudinal loads. For instance, in adjacent spans, ice shedding from the larger span is less critical than ice shedding from the smaller one. In summary, for actual lines where various span lengths and tower types are used, determining the most critical ice shedding scenario is complicated and highlights the need for a thorough fragility analysis.

- Transmission line components contribute differently to the failure of the transmission line system under different climatic conditions, and a comprehensive reliability analysis is required to determine the most critical components and load combinations for a specific transmission line.

CHAPTER 7 MANUSCRIPT 4

Analysis of the effect of climate change on the reliability of overhead transmission lines

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Abstract

Climate change is anticipated to influence the reliability of overhead transmission and distribution lines through impacts on extreme weather events. Changes in the frequency and intensity of wind and ice storms may have a considerable effect on applied loads and can consequently affect the probability of structural failure of different components of the line. This study examines the reliability of transmission lines under a range of assumed changes in the mean and standard deviation of climatic variables affecting transmission lines such as annual extreme wind speed and ice thickness. The methodology used for the reliability analysis of transmission lines under current and future climatic conditions is based on the concepts of Statistical Learning Theory. The sensitivity study provides the information required to improve the capacity of transmission lines and mitigate long-term risks from the effects of a changing climate. The results indicate that climate change as predicted by many researchers can significantly affect the reliability of existing transmission line systems. Hence, relying on the

historic climatic data may not be sufficient to ensure an adequate reliability of transmission line systems in the future. The specification of design loads for the evaluation of existing lines or the design of new lines should consider both future climate models and historical climate data.

Keywords: Structural reliability analysis, Transmission lines, Climate change, Statistical learning theory

7.1 Introduction

Decisions relating to the design, maintenance, replacement or updating of transmission line systems come with a long-term commitment. Although these decisions and the associated investments need to consider future climate conditions, current transmission line guidelines and standards are developed on the basis of historical climate data and do not take into account a probable change in the frequency and intensity of climatic hazards (ASCE 2010, CAN/CSA-C22.3, 2006, CEI/IEC 60826, 2003, EN 50341, 2001). In order to respond to climate change, it is required to determine the impacts of various climate change scenarios on the reliability of these complex structural systems. The results can be used to improve the robustness of transmission lines and mitigate the long-term risks from the effects of a changing climate. This is particularly important considering that many researchers predict more intense and more frequent climatic hazards in the future (Cheng et al 2007, 2011, 2012, 2014, Eichelberger et al. 2008, Zamuda et al. 2013) while historical events have shown to cause severe damages to transmission line networks (Bigras et al. 1998, Le et al. 2002, Sundell et al. 2006).

Although many studies evaluate the impacts of climate change on different infrastructures and energy sectors qualitatively (Boyle et al. 2013, Kezunovic et al. 2008, Peters et al. 2006, Schaeffer et al. 2012, Zamuda et al. 2013), very few studies provide quantitative assessments of

the effects of climate change on the reliability of structures, specially transmission or distribution lines (Bjarnadottir et al. 2014, Madsen 2014). These studies mainly focus on effects of climate change on the reliability of single structures or specific components of a transmission or distribution line. However, this paper presents the reliability index of an overall typical transmission line system as a function of future variations in extreme climatic conditions while considering the interaction and correlation of different components of the line. The results are presented in the form of design specifications as a function of a range of plausible climate change scenarios. The information presented in this paper is useful to develop new policies, climate adaptation strategies and risk mitigation measures that are cost-effective for a wide range of likely climate change scenarios.

In order to perform a comprehensive risk analysis and provide a quantitative assessment of climate change impacts on transmission lines, the following steps need to be completed:

(1) Identify the primary adverse climate impacts that are most critical for transmission lines;

(2) Considering the results of step (1), determine the corresponding random climatic variables that need to be downscaled locally from available Global Climate Models (GCMs);

(3) Identify the consequences of climatic hazards on the structural system.

The main objective of this study is to quantify the impact of future changes in the frequency and intensity of climatic hazards such as wind and ice storms on the structural reliability of transmission line systems using the concepts of statistical learning theory (SLT). It should be noted that the focus of this study is not the local prediction of changes in the probability distribution functions of wind speed and ice built-up at specific locations from global climate change scenarios but rather a parametric study on the effect of changes in the mean and variance

of the maximum annual wind and ice accumulation on the reliability of transmission lines. In other words, this study does not address step (2), and the changes in reliability levels can be linked to specific or multiple climate change scenarios at a later stage. More information about step (1) is presented in sections 7.2 and 7.3 of this study. And, Step (3) is elaborated in section 7.6.

7.2 Effect of climate change on climatic variables that affect transmission lines

Previous experience with electric transmission lines shows that one of the primary causes of structural failure of transmission lines is extreme wind and ice storms. Currently available data suggest that the intensity and frequency of extreme climatic events such as hurricanes and ice storms will increase in the future with increasing global temperatures.

A recent review by the Intergovernmental Panel on Climate Change (IPCC) (Field 2012), presents many studies in which decreasing or increasing historical and future trends in mean or extreme wind speed are reported for various locations around the globe. This report also states that future trends in mean and extreme wind speed are difficult to predict with the currently available data and models. Cheng et al. (2012, 2014) develop a wind gust simulation model to analyze changes in the frequency of future wind gusts for different regions in Canada. The analyses are performed by downscaling data from eight GCM simulations and predict that the frequency of wind gust events could increase in the latter part of this century across Canada. The results also show a greater increase in the frequency of more severe wind gust events. For example, the authors report an increase of less than 10% in the frequency of future daily wind gust events greater than 28 km/hr over the periods 2046-2065 and 2081-2100, while the increase

in the frequency of daily wind gust events greater than 70 km/hr and 90 km/hr are projected to be 10%-20% and 20%-40%, respectively. Knutson and Tuleya (1999) demonstrate that a 2.2°C increase in sea surface temperature results in a surface wind speed increase of about 5%-11% based on the analyses performed on 51 northwest Pacific storm cases.

Cheng et al. (2007, 2011) investigate impacts of climate change on future freezing rain events for eastern and southern-central parts of Canada. The results are expressed in terms of the frequency of future daily freezing rain events. The results indicate that eastern Canada could experience more frequent freezing rain events towards the end of the century (2081-2100) especially during the coldest months (December-February) in comparison to the average historical frequency (1958-2007), while fewer events are expected during the warmer months (November, March, April). General circulating models also predict that for locations above 30°N, extreme winter storms will become more frequent and severe if the amount of environmental CO_2 doubles (Lambert 1995).

7.3 Possible effects of climate change on transmission lines

The anticipated increase in the frequency and severity of extreme weather events are anticipated to have an impact on the reliability of transmission lines since they increase the risk of structural damage to electric transmission and distribution lines. Heavy snowfalls or freezing rain events can cause failure in different components of a line and consequently trigger a cascading failure incident. In addition, concurrent winds during or after winter storms can increase the failure probability of ice-covered structural components.

It is noted that this study does not consider failure scenarios caused by dynamic effects of wind such as galloping which is the high-amplitude, low-frequency wind induced vibration of

overhead conductors in the reliability analysis. Determining the contribution of these failure scenarios to the overall failure probability of components and their sensitivity to climate change can be the subject of future studies.

7.4 Description of the studied transmission line section

The analysis is performed for a line segment that is representative of typical transmission lines with components, layout and spatial scale that are of interest for realistic reliability assessments. As indicated in Figure 7-1, the nine-span segment of the studied 120 kV transmission line consists of two sections separated by tension towers and is modelled with Sap2000®. The diameter and weight per unit length of conductors are 27.78 mm and 15.26 N/m, respectively, and the diameter and weight per unit length of ground wires are 12.57 mm and 7.44 N/m, respectively. Figure 7-2 presents the configurations of tower types for the line while Table 7-1 provides detailed information on the towers. It is noted that three families of towers are used in this line segment which include light suspension towers (LS), heavy suspension towers (HS) and angled towers (STR). In this study, ten towers are included in the system reliability analysis due to the limited spatial extension of climatic hazards. According to CEI/IEC 60826 (2003) and CAN/CSA-C22.3 (2006), in a flat to rolling terrain and based on a span of 400 m, 1 to 5 towers are usually subjected to maximum wind and maximum ice and wind loads while 10 to 20 towers are usually subjected to maximum ice loads.

The modelled transmission line is analyzed for various combinations of wind speed, wind direction and ice accumulation. The results obtained from these analyses are used to train a surrogate demand function, $D(\mathbf{x_d})$, for each component of the line which relates component

demands to climatic variables and can substitute the time consuming finite element solver of the line as described in section 7.5.

The model accounts for the effects of structural flexibilities, component interactions, geometric nonlinearities and uncertainty on the capacity of transmission line components for calculating component and system failure probabilities. Parametric uncertainties of wind and ice loads are also considered in the probabilistic model. However, it is noted that model uncertainties such as uncertainties in determining wind and ice loads applied on towers and conductors given a specific wind speed and ice thickness and structural model uncertainties resulting from variations in the dimension and stiffness of structural elements, are not taken into account in this paper and further study is needed to quantify model uncertainties and incorporate them into the proposed probabilistic framework. The model also considers the effects of wind direction and exposure on ice accumulation. It is noted that the amount of ice built up on a wire is significantly dependent on wind speed and wind direction with maximum ice thickness forming on wires perpendicular to the wind direction. This effect can result in unbalanced ice formation of as much as 70% on adjacent spans in some cases for example where there is a change in the direction of the line (ASCE 2010). This study applies the Simple Model proposed by Jones (1998) to determine the amount of ice accumulation from available meteorological data as indicated in equation 7-1.

$$R_{eq} = \sum_{j=1}^{N} \frac{1}{\rho_i \pi} \left\{ (P_j \rho_0)^2 + (3.6 V_j \omega_j \sin[\varphi_1 - \varphi_2])^2 \right\}^{1/2} \quad (7\text{-}1)$$

where P_j, ρ_0 , ρ_i , φ_1 and φ_2 are the precipitation rate (mm in the jth hour), the density of water (1 gr/cm^3), the density of glaze ice (0.9 gr/cm^3), the wire direction and the wind direction, respectively. V_j is the wind speed (m/s) and ω_j=0.067$P_j^{0.846}$ (Best 1950) is the liquid water content (gr/m^3) of the rain-filled air in the jth hour.

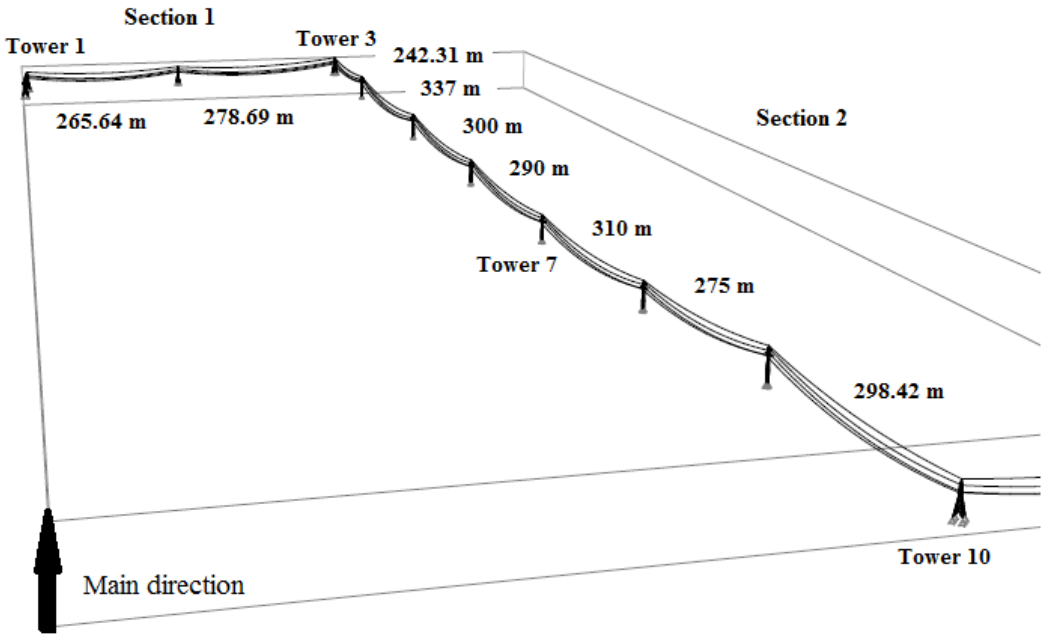


Figure 7-1. Nine span segment of the transmission line under study.

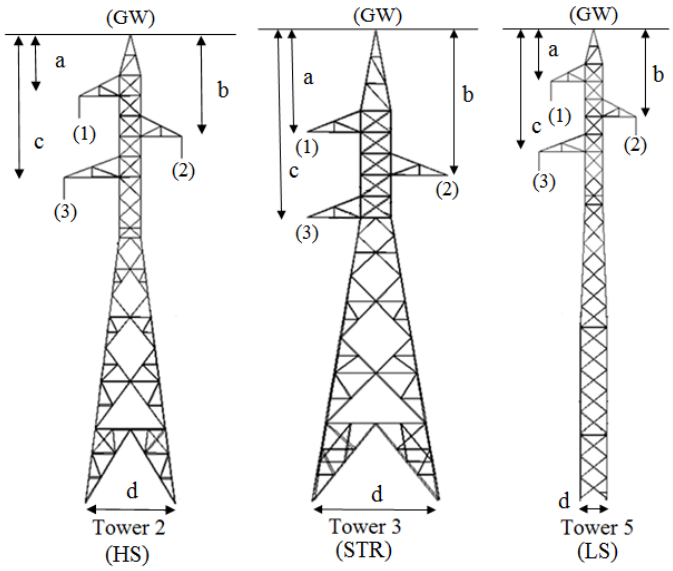


Figure 7-2. The configuration of tower types within the line.

Table 7-1. Description of towers of the line segment

Tower Number	Tower Height(m)	a(m)	b(m)	c(m)	d(m)	Tower Type
1	30.5	6	8.5	11	8.06	STR
2	28.75	3.75	6.25	8.75	5.26	HS
3	27.5	6	8.5	11	7.08	STR
4	33.25	3.75	6.25	8.75	1.9	LS
5	33.25	3.75	6.25	8.75	1.9	LS
6	30.25	3.75	6.25	8.75	1.9	LS
7	30.25	3.75	6.25	8.75	1.9	LS
8	28.75	3.75	6.25	8.75	1.9	LS
9	33.25	3.75	6.25	8.75	1.9	LS
10	27.5	6	8.5	11	7.08	STR

7.5 Methodology used for reliability assessment of transmission lines

Figure 7-3 presents the flowchart of the steps involved in the reliability assessment of the studied line under current and future climatic conditions. Each step is further explained in sections 7.4, 7.5 and 7.6.

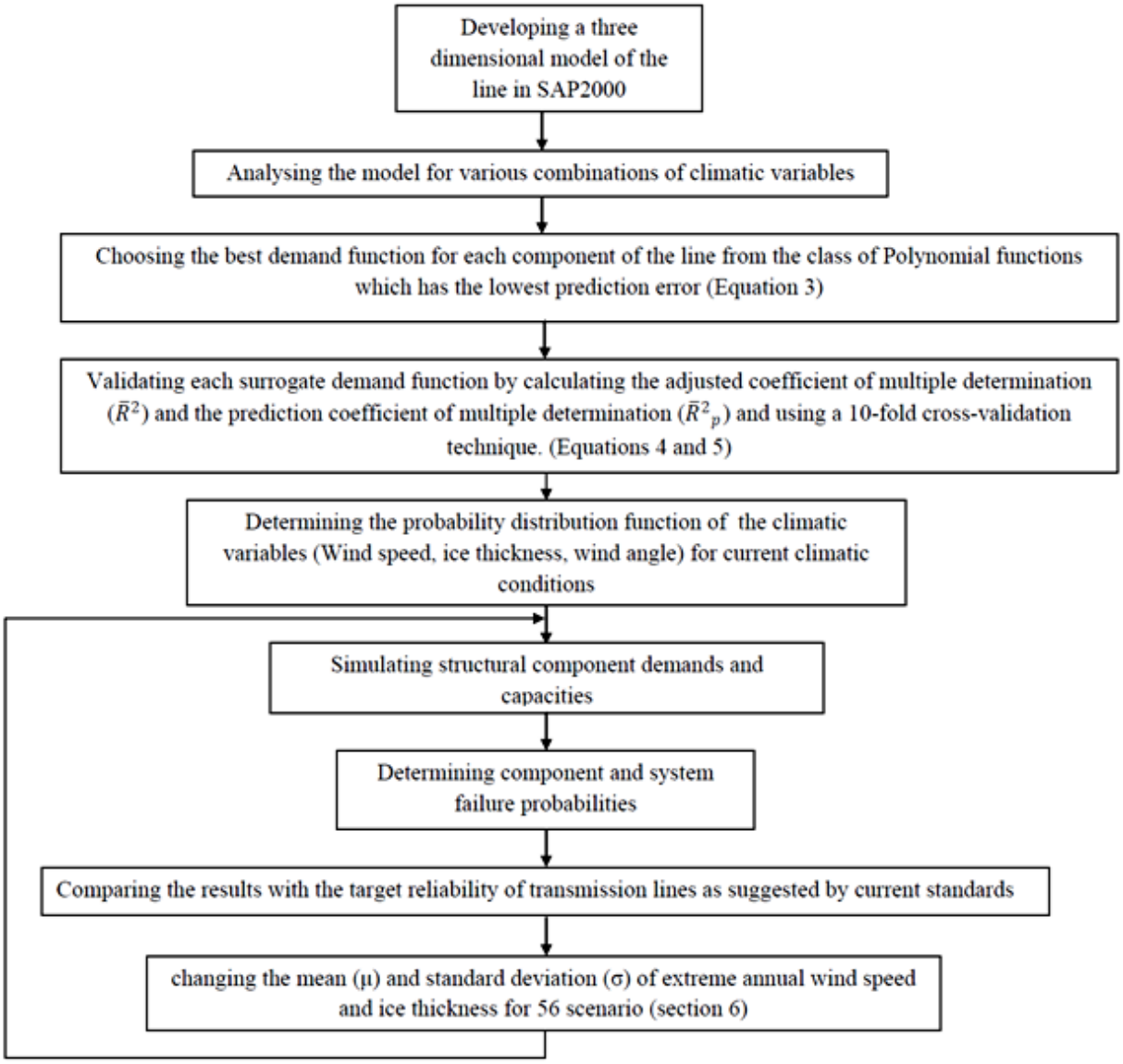


Figure 7-3. Flowchart of the steps involved in the reliability assessment of the studied line under current and future climatic conditions

7.5.1. Defining and validating the limit state function for each component based on the concepts of Statistical Learning Theory (SLT)

One challenge in determining the reliability of complex structural systems such as transmission lines is the implicit nature of the limit state function for each structural component or for the whole system. The general form of a limit state function is indicated in equation 7-2.

$$G(x) = C(xc) - D(xd) \quad (7\text{-}2)$$

where D ($\mathbf{x_d}$) is the demand function, C ($\mathbf{x_c}$) is the capacity function, $\mathbf{x_c}$ is the random vector of variables related to the capacity and $\mathbf{x_d}$ is the random vector of variables related to the structural demand.

It is noted that the implicit nature of the limit state function limits the application of the gradient based reliability analysis methods (Mahadevan and Haldar 2000, Zhao and Ono 1999) which require to estimate the gradient of the performance function, and of the simulation based methods (Karamchandani 1987, Melchers 1999) due to the vast computational effort needed for repeated calls on the finite element solver of the structure. Many researchers have traditionally used the Response Surface Method (RSM) to explicitly define the unknown limit state function (Bucher and Bourgund 1990, Rajashekhar and Ellingwood 1993). Since RSM is based on the Empirical Risk Minimization (ERM) principle and can result in over-fitting due to the rigid and non-adaptive nature of the selected model (Guan and Melchers 2001, Hurtado 2004), this paper adopts the concepts of SLT to substitute D($\mathbf{x_d}$) with a surrogate model that has good generalization (prediction) properties. This surrogate model replaces the finite element solver of the model and reduces the time and computational effort required to perform a Monte Carlo simulation. SLT is based on the Structural Risk Minimization (SRM) inductive principle and unlike ERM based methods; it does not impose strict assumptions over the class of approximating functions. Therefore, it can prevent the high bias produced by the discrepancy between the assumed function and the actual governing function.

In a study presented by Cherkassky and Mulier (Cherkassky and Mulier 2007), it is stated that SLT or Vapnik–Chervonenkis (VC) theory is the best currently available theory for flexible

statistical estimation from finite samples. The theory presents an analytical generalization bound for model selection as shown in equation 7-3 (Cherkassky et al. 1999, Vapnik 2000).

$$E(\omega) = E_{emp}(\omega) / \left(1 - \sqrt{p - p\ln(p) + \frac{\ln(n)}{2n}}\right)_{+} \quad (7\text{-}3)$$

where $E(\omega)$ is the unknown prediction error, $E_{emp}(\omega)$ is the known empirical error, n is the number of training samples and p is the ratio of VC dimension (h) to the sample size. VC dimension is a characteristic of a set of functions which equals the maximum number of samples for which all possible binary labellings can be induced without error. It is noted that in the case of linear real-valued functions, h is the number of free parameters.

In this study, for each structural element, the best model which has the lowest prediction error is selected from the class of polynomial functions. To achieve this goal, the class of polynomial functions are divided into nested subsets (S_k) according to their degree of complexity. Then from the functions in subset S_k, SRM finds the function with lowest empirical risk over the training sample. Using equation 3, the prediction error can be calculated for each subset S_k. The best model with the optimal complexity is the one with the lowest prediction error.

In order to estimate the final unbiased prediction error of the selected model, a 10-fold cross-validation technique is applied. The selected model is validated by calculating the adjusted coefficient of multiple determination ($\bar{R}^2$) and the prediction coefficient of multiple determination ($\bar{R}^2{}_p$) using equations 7-4 and 7-5. A good model will have $\bar{R}^2$ and $\bar{R}^2{}_p$ values near 1.

$$\bar{R}^2 = 1 - \frac{MSE}{MST} = 1 - \left(\frac{n-1}{n-p}\right)(1 - R^2) \quad (7\text{-}4)$$

$$R_p^2 = 1 - \frac{\sum_{i=1}^{n_{te}} (y_{ite} - \hat{y}_{itr})^2}{\sum_{i=1}^{n_{te}} (y_{ite} - \bar{y}_{te})^2} \quad (7\text{-}5)$$

In equation 7-4, *MSE*, *MST*, *n* and *p* are the mean square error, the total mean square of variation in observations, the total number of samples and the number of model variables, respectively. In equation 7-5, n_{te} and $\bar{y}_{te}$ represent the number and mean of the observed responses (y_{ite}) in the test fold, respectively, and $\hat{y}_{itr}$ is the predicted response of the ith observation in the test fold using the fitted model.

Figure 7-4 presents $\bar{R}^2$ and $\bar{R}^2{}_p$ respectively for all members of suspension tower 2. The adjusted coefficient of multiple determination expresses the quality of the fit between the regression model and the training samples while preventing overfitting by penalizing the analyst for adding terms to the model. The prediction coefficient of multiple determination shows the ability of the selected model to predict future samples. The results shown demonstrate the adequacy of the selected model and validate its application for determining the failure probability of components.

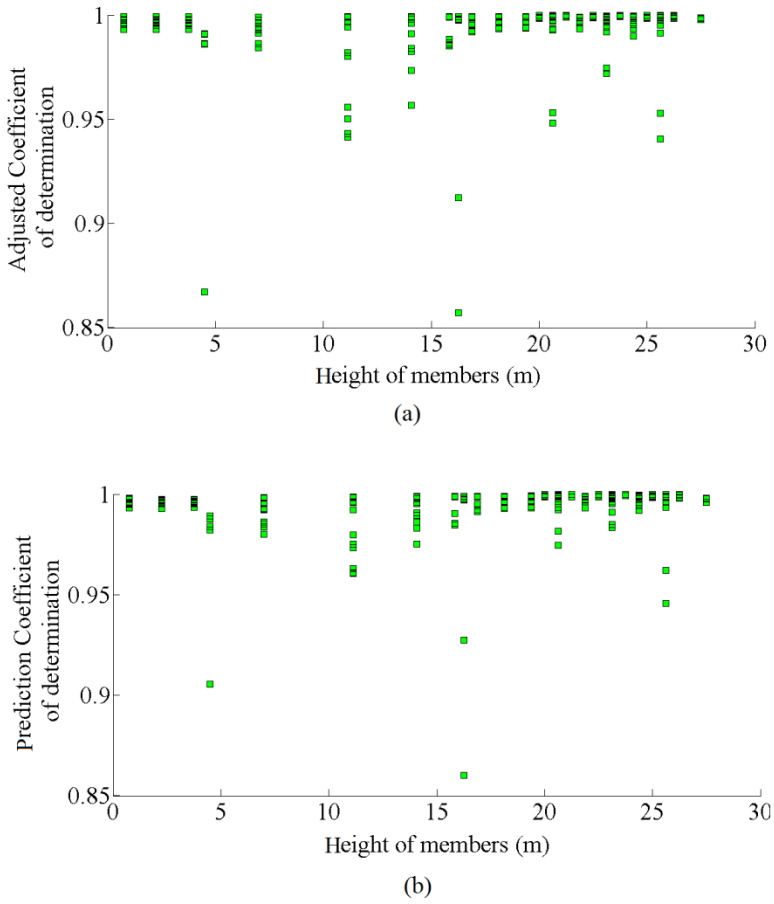


Figure 7-4. (a) Adjusted coefficient of determination and (b) Prediction coefficient of determination for all members of tower 2

7.5.2. Determining the component and system failure probabilities for current climatic conditions

After determining the performance function for each structural component of the transmission line, structural demands are simulated by randomly sampling from the distributions for the climatic variables and substituting into the selected response function for each component. The analysis is performed for an hypothetical line and for the purposes of the analyses, distributions for wind speed and ice accumulation are derived to be consistent with CAN/CSA-C22.3 (2006). Note that for an actual line, site specific data on wind speed and ice accumulation could be used to obtain more accurate and less conservative estimates of atmospheric hazards in comparison to

code derived distributions (Chouinard et al. 1998).The annual extreme ice thickness is assumed to have a Gumbel distribution with a mean and standard deviation of 32.2 (mm) and 4.1 (mm), respectively (CAN/CSA-C22.3, 2006). It is noted that, following CAN/CSA-C22.3 (2006), the 50 year return period extreme concurrent wind speed is assumed as 70% of the 50 year return period extreme wind speed which is 31.17 m/s for the location of the line. Therefore, in this study, the annual extreme concurrent wind speed is assumed to have a Gumbel distribution with a mean and standard deviation of 15.64 (m/sec) and 2 (m/sec), respectively (CAN/CSA-C22.3, 2006). The statistical parameters of the probability distribution of extreme annual ice thickness and extreme annual concurrent wind speed are estimated from the climatic coefficients proposed by CAN/CSA-C22.3 (2006) to derive the 25, 100, 150, 200, 400, and 500 year ice thickness and wind speed variables based on their 50 year design values using the regression technique. The likelihood of wind blowing from different directions is assumed to be uniform. Since the effect of temperature on the failure probability of tower elements is small and failure due to excessive sag of conductors is not considered in this study, temperature is assumed to be deterministic and equal to -5 °C. Figure 7-5 indicates the histograms of the simulated climatic variables of wind speeds, ice thicknesses and wind angles using Monte Carlo simulation for 10^8 events.

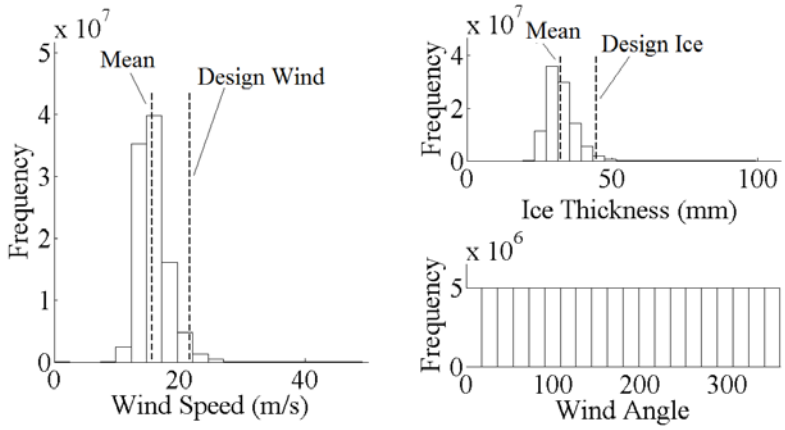


Figure 7-5. Histograms of simulated climatic variables

The capacity for each member of the overhead towers is calculated based on ASCE10-97 (2003). It is assumed that the capacity of overhead tower components have a lognormal distribution with a coefficient of variation of 10% (CAN/CSA-C22.3, 2006).The nominal rated tensile strength of conductors and ground wires are 141.7 kN and 139 kN, respectively. They are assumed to have a lognormal distribution with a coefficient of variation of 3% (CAN/CSA-C22.3, 2006). Three groups of insulators with nominal capacities of 70, 140 and 240 kN are used in the studied line section. It is assumed that the capacity of insulators has a lognormal distribution with a coefficient of variation of 5% (CAN/CSA-C22.3, 2006).In addition, following CEI/IEC 60826 (2003), it is assumed that the probability of each element of the line not having its nominal strength is 10%.

The failure probability of each component of the line is estimated as the ratio of the number of events for which the structural demand exceeds the structural capacity to the total number of simulations. In order to calculate the failure probability of each tower and the whole line, they are both represented as series systems. It should be noted that this assumption is conservative and implies that the tower/transmission line fails once any one of the modelled structural components fails. The other assumption used in this study is that the correlation between failure events of different components is only due to the same climatic conditions and the capacity of members are independent. Also, it should be mentioned that the failure probability of foundations are not considered when estimating the system failure probability and may be considered in future studies when experimental data becomes available. Table 7-2 shows the estimates of the failure probabilities and the corresponding reliability indices for different components of the line and the whole system under current climatic conditions. The results indicate that suspension towers have higher failure probabilities/lower reliability indices compared to other components of the

line. This is due to the fact that current design procedures are based on a specific strength coordination in which the suspension towers are considered as the weakest link in the transmission line system to minimize the impact of a potential failure (CEI/IEC 60826, 2003).

Table 7-2. Failure probabilities and reliability indices of components and the system under current climate.

	Failure Probability	Reliability index
Tower 1	5.70×10^{-5}	3.86
Tower 2	1.80×10^{-3}	2.91
Tower 3	1.40×10^{-5}	4.19
Tower 4	7.80×10^{-3}	2.42
Tower 5	8.60×10^{-3}	2.38
Tower 6	3.80×10^{-3}	2.67
Tower 7	3.50×10^{-3}	2.70
Tower 8	4.50×10^{-3}	2.61
Tower 9	4.80×10^{-3}	2.59
Tower 10	4.60×10^{-5}	3.91
Wire system	2.30×10^{-5}	4.08
Transmission line	1.45×10^{-2}	2.18

7.5.3. Target reliability of transmission lines

The target reliability level of a structure is the reliability level that minimizes the total working-life cost of the structure and maintains a balance between performance and life-cycle cost of the structure. Therefore, in order to determine the acceptable level of reliability for a given structure, an optimization problem between the consequences of failure and the costs of safety measures needs to be solved. The parameters that should be considered in solving this optimization problem include 1) the consequence and nature of failure, 2) the economic losses, 3) the social inconvenience and 4) the amount of expense and effort required to reduce the probability of failure (Cherkassky and Mulier 2007).

For transmission lines, based on their strategic function within the supply network, three reliability levels are proposed (CAN/CSA-C22.3, 2006, CEI/IEC 60826, 2003).

All lines should at least have a reliability level characterized by setting the 50 year return period loads equal to the strengths with an exclusion limit of 10% (level 1).

Lines below 230 kV which constitute the principal source of electric supply to a region or lines above 230 kV should be designed for a reliability level characterized by setting the 150 year return period loads equal to the strengths with an exclusion limit of 10% (level 2).

Lines above 230 kV which constitute the principal source of electric supply to a region or lines should be designed for a reliability level characterized by setting a 500 year return period loads equal to the strengths with an exclusion limit of 10% (level 3).

These reliability levels result in an almost constant yearly failure probability between 1/2T and 1/T, independent of the shape of the system load and strength curves and the coefficients of variation of each variable (CAN/CSA-C22.3, 2006, CEI/IEC 60826, 2003). Where, T is the return period of climatic events. It is noted that, following CAN/CSA-C22.3 (2006) and CEI/IEC 60826 (2003), the studied 120 kV transmission line should have a level 1 reliability with an expected annual system failure probability between 0.01 and 0.02. As indicated in table 7-2, the estimated annual failure probability of the line is 0.0145 for current climatic conditions. This shows that the model provides failure probabilities which are in line with the design standards for current climatic conditions.

7.6 Impact of climate change on the reliability of transmission lines

Estimating and quantifying changes in climate conditions and extreme events is a challenging task. For this purpose, various methodologies based on Extreme Value Theory or General Circulation Models have been used by scientist (García-Cueto and Santillán-Soto 2012). The

World Meteorological Organization (WMO) suggests extreme value analysis in which the same statistical distributions are used over time while the parameters in the statistical models vary over time to describe the temporal evolution of the extremes (Data 2009). The same assumption can be found in the works of Huntingford et al (2003), Beniston et al (2007) and Pryor and Barthelmie (2010). In this study, climatic variables for current and future climate conditions are assumed to follow Gumbel distribution in accordance with CAN/CSA-C22.3 (2006). In order to investigate the effect of climate change on the reliability of transmission lines and their components, a sensitivity analysis is performed by changing the mean (μ) and standard deviation (σ) of extreme annual wind speed and ice thickness for 56 scenarios. Tables 7-3 and 7-4 present the percentage change in the reliability index of the studied transmission line system as the result of changing the distribution parameters (μ, σ) of extreme wind speed and ice thickness individually. A change in the mean value is related to a change in the severity of an event while an increase in the standard deviation corresponds to an increase in the variability of climate related events.

It is inferred from Table 7-3 that both mean value and standard deviation of the wind speed have significant influence on the reliability index of transmission line systems. However, comparing the effect of these two variables, the mean value has more effect on the reliability of the line since an increase of 20% in the mean value of extreme wind speed can reduce the reliability index of the system by more than 30%, while an increase of 20% in the standard deviation of extreme wind speed can reduce the system reliability index by less than 8%. Therefore, as indicated in Table 7-3, a mild reduction of 5% in the mean value of the wind speed will increase the structural reliability even if there is a 10% increase in the standard deviation. Similarly, it is inferred from table 4 that the mean value of extreme ice thickness has more effect on system

reliability index of the transmission line compared to the standard deviation of extreme ice thickness.

Comparing the results from Tables 7-3 and 7-4 demonstrates that changes in the distribution for ice thickness has less effect on the reliability of transmission line systems compared to the wind speed. However, this influence is still significant and a change of 20% in the mean value of extreme ice thickness can reduce the reliability index of the system by more than 17%. It is noted that an increase of 20% in the mean value of extreme annual wind speed and extreme annual ice thickness approximately represents current climatic conditions at the most exposed hazardous regions near the location of the line.

Table 7-3. Percentage change in the transmission line system reliability index as a function of mean and standard deviation of the extreme wind speed distribution

		% of change in μ_{wind}			
		-5	0	10	20
		μ_{wind} (m/s)			
% of change in σ_{wind}	σ_{wind}	21.23	22.35	24.58	26.82
-20	2.28	17.04	9.71	-6.50	-24.67
-10	2.57	12.07	4.89	-10.70	-27.58
0	2.86	6.54	0.00	-14.40	-30.37
10	3.14	2.51	-3.86	-18.01	-32.95
20	3.43	-1.56	-7.58	-20.81	-35.34

Table 7-4. Percentage change in the transmission line system reliability index as a function of mean and standard deviation of the ice thickness distribution

		% of change in μ_{Ice}				
		-20	-10	0	10	20
		μ_{Ice} (mm)				
% of change inσ_{Ice}	σ_{Ice}	25.76	28.98	32.20	35.42	38.64
-20	3.30	17.97	10.96	2.51	-5.56	-14.72
-10	3.70	17.04	9.51	1.42	-6.96	-15.99
0	4.10	15.32	8.53	0.00	-8.45	-17.19
10	4.50	14.01	6.72	-1.56	-9.68	-18.68
20	4.90	12.07	5.05	-2.80	-11.23	-20.50

It is noted that in tables 7-3 and 7-4, the values in bold font show the percentage change in the reliability index of the transmission line system and a negative value represents a reduction in the reliability index while a positive value represents an increase in the system reliability index.

In order to get a better understanding of Tables 7-3 and 7-4, the results are also presented in figures 7-6 and 7-7, respectively.

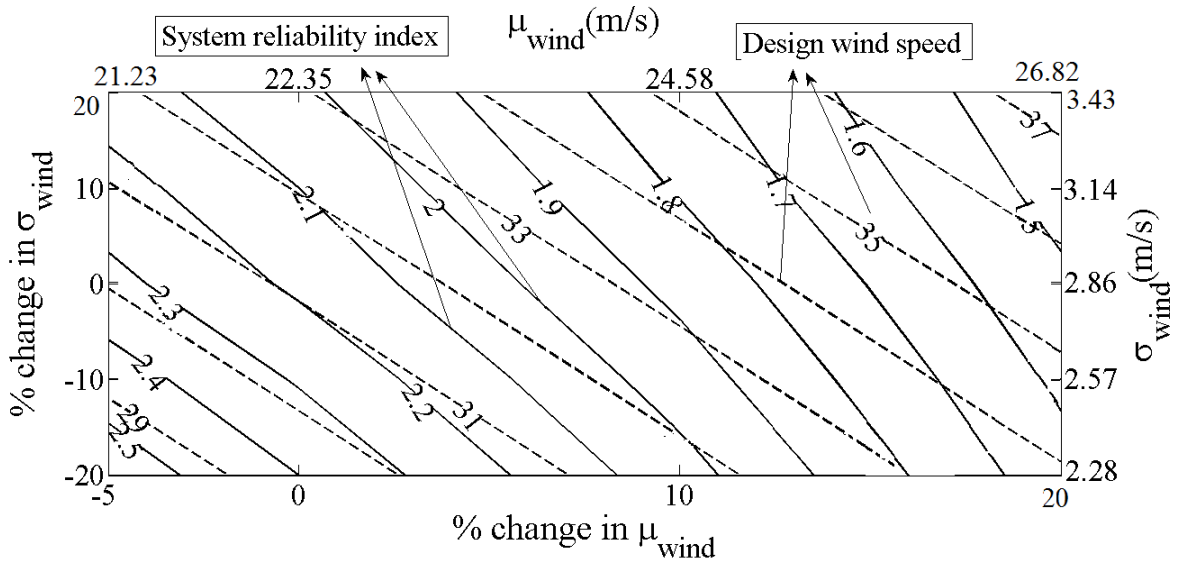


Figure 7-6. The effect of changing the mean and standard deviation of extreme wind speed distribution on the system reliability index and design wind speed

In figure 7-6, the lower horizontal axis is the percentage change in the mean value of extreme wind speed and the upper horizontal axis is the corresponding mean value of extreme wind speed (m/s). The left and right vertical axes are the percentage change in the standard deviation of extreme wind speed and the corresponding value of the standard deviation of extreme wind speed (m/s), respectively. The solid contour lines show the system reliability index, while the dashed contour lines show the 50 year return period wind speed(m/s) which is the design wind speed for this line. It is inferred from this figure that if the mean value of extreme wind speed increases by 10% from 22.35 m/s to 24.58 m/s while the standard deviation is unchanged, the

reliability of the line is reduced from 2.18 to 1.87. In order to maintain the same reliability index, the transmission line should be upgraded for a design wind speed of 33.4 m/s instead of its current design value of 31 m/s. It is noted that, in order to propose an upgrading strategy, the following questions should be answered: 1) which components contribute more to the system failure 2) which components have higher improvement potential for the entire system 3) which upgrading strategy results in a desirable strength coordination between line components and consequently minimizes the impacts of a potential failure. Determining the best upgrading strategy and estimating the reliability of the improved line using the statistical learning theory is the subject of a future study.

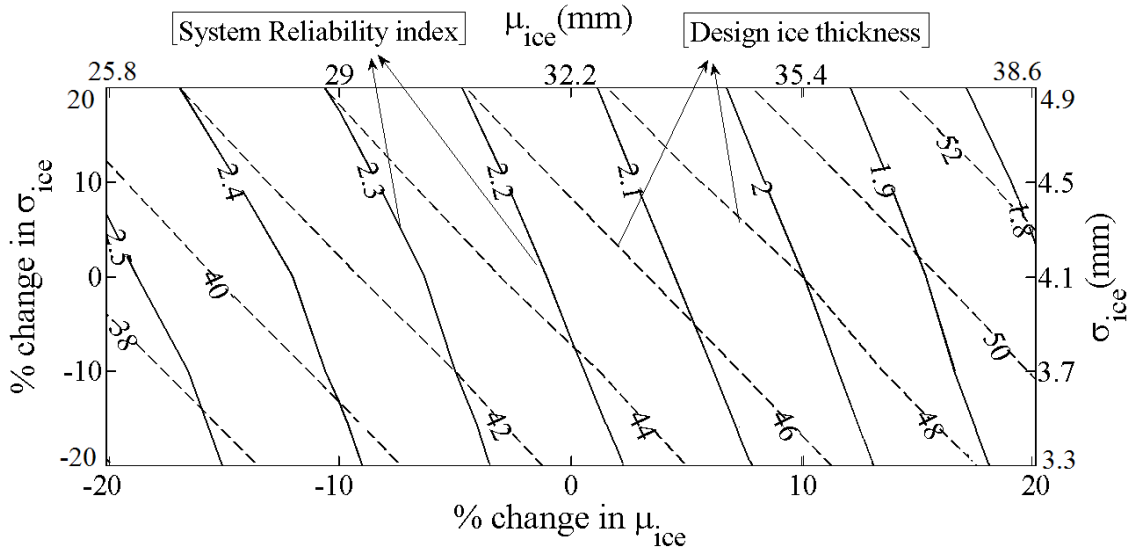


Figure 7-7. The effect of changing the mean and standard deviation of extreme ice thickness distribution on the system reliability index and design ice thickness

In figure 7-7, the lower horizontal axis is the percentage change in the mean value of extreme ice thickness and the upper horizontal axis is the corresponding mean value of extreme ice thickness (mm). The left and right vertical axes are the percentage change in the standard deviation of extreme ice thickness and the corresponding value of standard deviation of extreme ice thickness (mm) , respectively. The solid contour lines show the system reliability index, while the dashed

contour lines show the 50 year return period ice thickness (mm) which is the design ice thickness for this line. It is inferred from this figure that if the mean value of extreme ice thickness increases by 10% from 32.2 mm to 35.42 mm, while the standard deviation remains constant, the reliability index is reduced from 2.18 to 2.0. In order to maintain the same reliability index, the transmission line should be upgraded for a design ice thickness of 48 mm instead of its current design value of 45 mm.

Table 7-5 presents the percentage change in the reliability index of the studied transmission line system as the result of changing the mean value of extreme wind speed and ice thickness distributions simultaneously. Figure 7-8 shows the effect of changing the mean value of the extreme wind speed and ice thickness distributions on the system reliability index and design wind and ice loads.

Table 7-5. Percentage change in the transmission line system reliability index as a function of the mean value of extreme wind speed and ice thickness distributions

			% of change in μ_{wind}			
			-5	0	10	20
			μ_{wind} (m/s)			
			21.23	22.35	24.58	26.82
			$Wind_{50}$ (m/s)			
% of change inμ_{Ice}	μ_{Ice} (mm)	Ice_{50} (mm)	30.05	31.17	33.40	35.64
-20	25.76	38.50	21.46	15.32	2.23	-12.48
-10	28.98	41.70	15.32	8.53	-6.22	-21.47
0	32.2	45.00	6.54	0.00	-14.40	-30.37
10	35.42	48.00	-1.44	-8.45	-23.40	-39.76
20	38.64	51.40	-10.39	-17.19	-32.55	-49.25

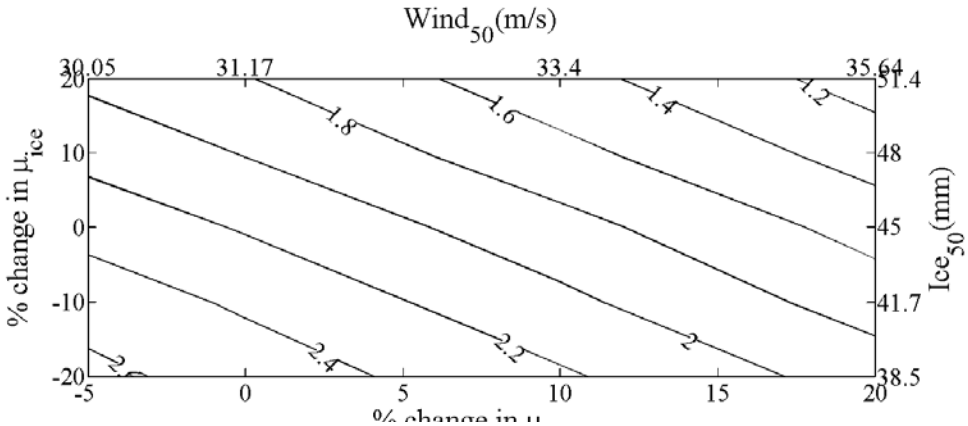


Figure 7-8. The effect of changing the mean value of extreme wind speed and ice thickness distributions on the system reliability index and design wind and ice loads

It is inferred from table 7-5 and figure 7-8 that a simultaneous change in the mean value of concurrent wind speed and ice thickness intensifies their effect on the reliability of transmission lines. This can be explained by the nonlinear nature of the applied wind loads on ice covered components. In other words, ice thickness amplifies the applied wind loads by increasing the projected area of the structural elements exposed to the wind. In addition, it imposes additional vertical loads on the components. The analysis of wind statistical characteristics for various sites across Canada indicates that the coefficient of variation is relatively constant which shows a strong correlation between the mean and standard deviation (Ye 2013). If a similar relation is observed for future climates, an increase in mean severity will also be accompanied by a corresponding increase in the standard deviation which may further reduce the reliability index of existing lines.

7.7 Conclusion

In this study, the sensitivity of the reliability of a typical transmission line system to climate change is investigated using the Statistical Learning Theory and it is shown that a change in the mean value or standard deviation of extreme wind and ice events can significantly alter the

reliability index of existing transmission line systems. The reliability of transmission lines is shown to be more sensitive to an increase in the intensity of extreme wind speed than to an increase in the intensity of extreme ice thickness. For example, it is demonstrated that a 20% change in the mean value of extreme wind speed and extreme ice thickness can reduce the reliability index of the studied transmission line system by more than 30% and 17%, respectively. This effect is even more significant when the increase in wind speed and ice thickness is concurrent due to the applied wind loads on ice covered components. In other words, a climate change as predicted by many researchers can significantly reduce the reliability of existing transmission line systems. Hence, considering the increasing environmental CO_2, relying on the historic climatic data may not be sufficient to ensure an adequate reliability of transmission line systems in the future. The specification of design loads for the evaluation of existing lines or the design of new lines should consider both future climate models and historical climate data.

Future research remains to be done on projecting the potential changes in the probability distribution functions of wind speed and ice built-up from available climate change scenarios and integrating these results into the proposed probabilistic framework.

CHAPTER 8 SUMMARY, CONCLUSIONS, ORIGINAL CONTRIBUTIONS AND RECOMMENDATIONS

8.1 Summary and conclusions

This study presents two probabilistic approaches 1) component-based approach 2) system-based approach for reliability analysis and fragility assessment of complex structural systems with implicit limit state functions such as electric transmission lines. Both approaches are advanced meta-modeling techniques based on machine learning concepts that can overcome the limitations of traditional structural reliability methods for the problem at hand. The results of this study show good prediction abilities for both approaches and validate their application for reliability assessment of transmission lines. Section 8.1.1 elaborates on the summary, results and conclusions of the component-based approach, while section 8.1.2 presents the summary, results and conclusions of the system-based approach. This dissertation also includes two practical applications of the proposed methodology which include 1) developing component and system fragility curves for various non-uniform ice loading scenarios due to ice accumulation or ice shedding and 2) investigating the robustness of the line to climate change and identifying the most effective manner to increase the line robustness. The summary, results and conclusions of each practical application are presented in sections 8.1.3 and 8.1.4, respectively. The results from this study are useful for optimizing the design of new transmission lines, evaluating the vulnerability of existing lines, making well-founded decisions with respect to upgrading strategies and improving current policies and design standards to consider the effects of a changing climate.

8.1.1 The component-based approach for structural reliability analysis and fragility assessment of transmission line components and systems

The component-based approach applies SLT and VC-generalization bounds to select a surrogate demand model of the finite element solver for each tower element or line component from several possible models that has good generalization (prediction) properties. Cross validation technique is used to validate the selected model and determine its unbiased prediction error. The adjusted coefficient of multiple determination ($\bar{R}^2$) and the prediction coefficient of multiple determination ($\bar{R}_p^2$) are calculated for all members of all towers and line components. In addition, SLT is combined with Monte Carlo simulation to calculate the yearly failure probability of each element and component of the line as well as the overall yearly system failure probability. In order to show the effect of failure correlation between different tower elements or different components of the line, the calculated failure probabilities are compared with first and second order Ditlevsen bounds. To provide the engineers with a practical reliability-based decision tool for the purpose of fault diagnosis after an extreme climatic event or for assessing upgrading strategies with limited resources, the components of the studied transmission line are ranked based on two importance measures: 1) the Bayesian reliability importance measure and 2) the improvement potential importance measure. Finally, component and system fragility curves as well as the system design point are provided for the studied 120 kV transmission line located in North America.

A summary of the results is presented in the following:

- The results show that the proposed approach provides a good fit of the demand model to SAP2000 results for most line members as the values of $\bar{R}^2$ and $\bar{R}_p^2$ for 99 percent of members are greater than 0.95.
- Using the code derived climatic hazards, the overall annual failure probability of the studied line is 0.0154 which corresponds to a reliability index of 2.16. The estimated failure probability lies within the expected failure probability range of [0.01-0.02] as suggested by current design guidelines
- A comparison of the failure probabilities of tower members indicate that, as expected, critical tower elements are mainly leg members located at the lower part of towers, where aggregate demands concentrate.
- In addition, a comparison of the failure probabilities of line components show that, suspension towers (Type HS and LS) have higher failure probabilities compared to other components of the line which is consistent with the fact that current design procedures are based on a specific strength coordination in which the suspension towers are designed as weakest links in the transmission line system.
- A comparison of line component rankings based on the Bayesian reliability and the improvement potential importance measures indicate that, for the studied transmission line, tower 5 has the highest contribution to system failure and is the most critical component from a fault diagnosis perspective while tower 4 offers the highest potential system reliability improvement and is the most important component for fund allocation purposes when upgrading the line with limited resources.
- The most probable failure point of the studied line is similar to the design combination of low probability wind during icing and high probability ice suggested by current standards

which means that the studied line most likely fails under low amounts of ice accompanied by high winds blowing perpendicular to the main direction of the line.

8.1.2 The system-based approach for structural reliability analysis and fragility assessment of transmission line systems

The system-based approach, which is represented by a binary classification problem based on SVMs initially aims to determine the system limit state surface and predict the state of the transmission line (survival or failure) for given climatic conditions. A double re-sampling technique (10-fold cross validation) is used to properly tune the SVM parameters such as the kernel function, the kernel parameter (γ) and the penalty parameter (C) and evaluate the prediction ability of the selected model through estimating the prediction error, sensitivity, specificity, precision and F-measure. Both polynomial and RBF kernels are used for this study. The results show superior performance for SVMs with polynomial kernels compared to SVMs with RBF kernels. To further enable the model to predict the state of the line for given climatic conditions with the probability that this prediction is correct, an SVM-based Venn Predictor is used. The VP model provides lower and upper failure probabilities for given climatic conditions, which are useful for developing system fragility curves for the studied transmission line and providing estimates of its failure probability.

A summary of the results is presented in the following:

- The results show good prediction abilities for the polynomial SVM-based VP and validate its application for the fragility assessment and reliability analysis of transmission lines.

- The developed fragility curves indicate that, the line is most vulnerable when wind blows perpendicular to the longest section of the line since this direction is the most detrimental direction for most of the suspension towers which are the weakest components of the line.
- Using the SVM-based VP with polynomial kernels, the lower and upper bounds for the failure probability of the studied line are 0.0133 (reliability index of 2.22) and 0.0317 (reliability index of 1.86), respectively.
- Unlike the component-based approach, the time and effort required for the proposed system-based approach is independent of the number of components/elements within the line, because it directly determines the failure limit state surface of the whole system which makes it an efficient solution for overall reliability assessment of systems comprising numerous components/elements. In condition assessment of large systems such as power networks, this methodology can efficiently be applied to each subsection of the system. Once the vulnerable subsections are identified within the power network, the most efficient retrofit strategy can be determined using the component-based approach for each weak subsection.

8.1.3 Fragility assessment of transmission line components and systems subjected to unbalanced ice loads

In this section, the proposed component-based approach based on SLT and VC-generalization bounds is used to develop component and system fragility curves of the studied transmission line subjected to unbalanced ice loads resulting from non-uniform ice formation or ice shedding. The results are presented in the form of fragility curves for 3 different unbalanced loading scenarios

of longitudinal, transverse and torsional loadings. The effect of various design parameters such as wind speed and direction and ice loading proportion on the developed fragility curves are also investigated. For the studied transmission line the following conclusions are drawn from the developed fragility curves.

- The configuration and type of towers affect the tower fragility curves. For example, in this study, suspension towers are more vulnerable than tension towers for all loading scenarios.

- Wind speed and direction significantly affect both wind and ice loads applied to transmission lines and consequently fragility curves can change considerably based on these parameters.

- For different climatic conditions, the relative importance of different ice loading scenarios may change based on the order of the most critical components (elements) that contribute to the system (tower) failure.

- Ice loading proportion notably influences the fragility curves of towers. It is shown that the ice shedding scenario changes the most critical ice loading proportion for each tower.
- The most critical ice shedding scenario is the one that results in higher vertical and/or longitudinal loads on towers and conductors. This depends on various parameters such as the location of ice shedding and the length of spans before and after each tower. Therefore, for actual lines where various span lengths and tower types are used, determining the most critical ice shedding scenario is complicated and highlights the need for a thorough fragility analysis.

- Transmission line components contribute differently to the failure of the transmission line system under different climatic conditions, and a comprehensive reliability analysis is required to determine the most critical components and load combinations for a specific transmission line.

8.1.4 Analysis of the effect of climate change on the reliability of overhead transmission lines

In this section, the sensitivity of the reliability of the studied transmission line system to climate change is investigated using the statistical learning theory. The reliability index of the line is presented as a function of future variations in extreme climatic conditions while considering the interaction and correlation of different components of the line. The results are presented in the form of design specifications as a function of a range of plausible climate change scenarios.

A summary of the results and conclusions is presented in the following:

- Both mean value and standard deviation of the wind speed have significant influence on the reliability index of transmission line systems. However, comparing the effect of these two variables, the mean value has more effect on the reliability of the line.
- Similarly, the mean value of extreme ice thickness has more effect on system reliability index of the transmission line compared to the standard deviation of extreme ice thickness.
- The reliability of transmission lines is shown to be more sensitive to an increase in the intensity of extreme wind speed than to an increase in the intensity of extreme ice thickness. This effect is even more significant when the increase in wind speed and ice thickness is concurrent due to the applied wind loads on ice covered components.

- Considering the increasing environmental CO2, relying on the historic climatic data may not be sufficient to ensure an adequate reliability of transmission line systems in the future. The specification of design loads for the evaluation of existing lines or the design of new lines should consider both future climate models and historical climate data.

8.2 Original contributions

The present dissertation is the first study that comprehensively considers the effects of structural flexibilities, component interactions, geometric nonlinearities, parametric uncertainties of wind and ice loads, uncertainty of the capacity of transmission line components/elements, as well as wind direction and line exposure effects on ice accumulation for reliability analysis and fragility assessment of transmission line components and systems. The proposed model can be used to validate the reliability of the line and the coordination of the levels of reliability of components to insure that a desired sequence of failure is achieved. The model can also be used to study the robustness of the line to climate change and to identify the most effective manner to increase the line robustness. The original contributions of this study are summarized in the following.

- The novel application of Statistical Learning Theory and VC-generalization bounds to address the problem of component-based reliability analysis and fragility assessment of complex structural systems with implicit limit state functions with an emphasis on electric transmission lines subjected to extreme wind and ice storms.
- The novel application of Probabilistic Support Vector Machine and Venn Predictor for determining the system limit state surface and performing system-based reliability analysis and fragility assessment of transmission lines subjected to extreme wind and ice storms.

- Applying the results from the component-based approach to quantify the relative importance of transmission line components and ranking them based on two importance measures which are useful for determining upgrading strategies with limited resources and for fault diagnosis purposes after an extreme climatic event.
- Developing component and system fragility curves for transmission lines under unbalanced ice loads due non-uniform ice accumulation and/or ice shedding.
- Considering wind angle as a random variable and taking into account the effect of wind direction and line exposure to wind, on ice accumulation in fragility assessment of the studied line.
- Investigating the effect of various design parameters such as wind speed and direction and ice loading proportion on the component and system fragility curves.
- Investigating the effect of climate change on the reliability of transmission lines and proposing design considerations for mitigating the risk of future line failures due to climate change.

8.3 Future recommendations

This study aims to develop a probabilistic framework for reliability analysis and fragility assessment of transmission lines subjected to extreme wind and ice storms. However, as previously mentioned in section 1.4, it has certain limitations. Further research will help to refine and improve future applications of the proposed methodologies. The following fundamental and applied research is recommended:

- Developing fragility curves and determining the failure probability of different types of overhead tower foundations, and determining their contribution to the line failure.

- Applying the system-based approach proposed in this study (PSVMs) to quantify model uncertainties in reliability analysis of overhead towers including the uncertainties in tower member dimensions, materials and stiffness.
- Applying the system-based approach to develop the system limit state function and fragility curves for overhead towers and estimate their failure probabilities under dynamic loads due to galloping, ice shedding or conductor rupture.
- Projecting the potential changes in the probability distribution functions of extreme wind speed and ice built-up from available climate change scenarios and integrating these results into the proposed probabilistic framework to provide realistic predictions of the future reliability of existing transmission lines.

APPENDIX A TECHNICAL INFORMATION AND EQUATIONS USED TO DETERMINE CLIMATIC LOADS ON LINE COMPONENTS

A.1 Overview

This appendix presents some technical information which together with chapter 3 helps the reader better understand the mathematical procedure required to the determine wind and ice loads on transmission line components.

A.2 Definition of terrain categories and the corresponding roughness factors, K_R (CAN/CSA-C22.3 2006)

Table A.1 Classification of terrain categories

Terrain category	Roughness characteristics	K_R
A	flat coastal areas	1.08
B	Open country with very few obstacles, for example airports or cultivated fields with few trees or buildings	1.00
C	Terrain with numerous small obstacles of low height (hedges, trees and buildings)	0.85
D	Suburban areas or terrain with many tall trees	0.67

A.3 Equations for the combined wind factor for conductors, G_c (CAN/CSA-C22.3 2006)

$$G_c = 0.2914Ln(h) + 1.0468 \text{ (terrain type A)} \quad \text{(A-1)}$$

$$G_c = 0.3733Ln(h) + 0.9762 \text{ (terrain type B)} \quad \text{(A-2)}$$

$G_c = 0.4936Ln(h) + 0.9124$ (terrain type C) (A-3)

$G_c = 0.6153Ln(h) + 0.8144$ (terrain type D) (A-4)

A.4 Equations for the combined wind factor for towers and insulators, G_t (CAN/CSA-C22.3 2006)

$G_t = -0.0002h^2 + 0.0232h + 1.4661$ (terrain type A) (A-5)

$G_t = -0.0002h^2 + 0.0274h + 1.6820$ (terrain type B) (A-6)

$G_t = -0.0002h^2 + 0.0298h + 2.2744$ (terrain type C) (A-7)

$G_t = -0.0002h^2 + 0.0384h + 2.9284$ (terrain type D) (A-8)

A.5 Equation for the span factor for conductors, G_L (CAN/CSA-C22.3 2006)

$G_L = 4 \times 10^{-10} \times L^3 - 5 \times 10^{-7} \times L^2 - 10^{-4} \times L + 1.0403$ (A-9)

A.6 Equation for the drag coefficient for flat sided structures, C_{xt1}, C_{xt2} (CAN/CSA-C22.3 2006)

$C_{xt} = 4.1727\chi^2 - 6.1681\chi + 4.0088$ (A-10)

A.7 Equations for diameter factor, k_d, and height factor, k_h (CAN/CSA-C22.3 2006)

The equations presented in this section is derived for precipitation icing. The governing equations to estimate modification factors for ice thickness due to in-cloud icing can be found in CAN/CSA-C22.3 (2006).

$$k_d = 0.35 \times \frac{d}{30} + 0.65 \quad \text{(A-11)}$$

$$k_h = 0.075 \times \frac{z}{10} + 0.925 \quad \text{(A-12)}$$

APPENDIX B TECHNICAL INFORMATION ABOUT THE STATISTICAL DISTRIBUTION FUNCTIONS AND THEIR APPLICATION TO CLIMATIC LOADS AND STRENGTH OF OVERHEAD LINES

B.1 Overview

This appendix presents the distribution functions that are of interest to implement the reliability concepts presented in this study.

In all the functions described hereafter, *m* and *σ* represent the mean value of observations and the standard deviation of observation, respectively.

B.2 Normal distribution (Gaussian distribution)

Although the Normal distribution is not directly used in this study, knowing its characteristics helps the reader to better understand the characteristics of the Lognormal distribution. Therefore, some information regarding the format of its probability density function and cumulative distribution function is presented in this section.

B.2.1 Probability density function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{x-m}{\sigma}\right)^2\right],\ \sigma > 0 \qquad \text{(B-1)}$$

B.2.2 Cumulative distribution function:

$$F(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{x} exp\left[-\frac{1}{2}\left(\frac{x-m}{\sigma}\right)^2\right] dx \qquad \text{(B-2)}$$

B.3 Lognormal distribution

The log-normal distribution is defined as a distribution where $\ln(x - p_1)$ follows a normal distribution.

B.3.1 Probability density function:

$$f(x) = \frac{1}{p_2(x-p_1)\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\frac{\ln(x-p_1)-p_3}{p_2}\right)^2\right],\ x > p_1,\ p_2 > 0,\ p_3 > 0 \quad \text{(B-3)}$$

B.3.2 Cumulative distribution function:

$$F(x) = \frac{1}{p_2\sqrt{2\pi}} \int_{p_1}^{x} \frac{1}{x-p_1} exp\left[-\frac{1}{2}\left(\frac{ln(x-p_1)-p_3}{p_2}\right)^2\right] dx \quad \text{(B-4)}$$

where p_1 is the lower threshold of the distribution, p_2 is the standard deviation of the variable $\ln(x - p_1)$, and p_3 is the mean value of the variable $\ln(x - p_1)$.

$$p_2 = \sqrt{\ln\left[1 + \frac{\sigma^2}{(m-p_1)^2}\right]} \quad \text{(B-5)}$$

$$p_3 = \ln\left[\frac{(m-p_1)^2}{\sqrt{(m-p_1)^2+\sigma^2}}\right] \quad \text{(B-6)}$$

As mentioned earlier in this study, according to CAN/CSA-C22.3 (2006), CEI/IEC 60826 (2003), the lognormal distribution can adequately present the statistical distribution of the strength of tower elements or line components, and is used in this study.

B.4 Gumbel distribution

B.4.1 Probability density function:

$$f(x) = \frac{1}{p_2} exp\left[-\frac{x-p_1}{p_2} - \exp(-\frac{x-p_1}{p_2})\right],\ p_2 > 0 \quad \text{(B-7)}$$

B.4.2 Cumulative distribution function:

$$F(x) = \exp[-\exp\left(-\frac{x-p_1}{p_2}\right)] \quad \text{(B-8)}$$

where the parameters p_1 and p_2 are defined as:

$$p_2 = \frac{\sigma}{C_1} \quad \text{(B-9)}$$

$$p_1 = m - \frac{C_2}{C_1}\sigma \quad \text{(B-10)}$$

The parameters C_1, and C_2 depend on the number (n) of values considered in a measurement series. If an infinite number of observations is available $C_1 = \frac{\pi}{\sqrt{6}} = 1.28255$ and $C_2 = 0.577216$. more information regarding the governing equations and other values of C_1, and C_2 can be found in CAN/CSA-C22.3 (2006).

As mentioned in section 3.2, according to current design guidelines including CAN/CSA-C22.3 (2006), CEI/IEC 60826 (2003) and CENELEC EN 50341 (2001), for the loading of transmission lines, the distribution of annual maximum wind velocities and ice thicknesses can be represented, with good approximation, by the Gumbel distribution which is used in this study.

APPENDIX C FLOWCHARTS FOR THE COMPONENT-BASED AND THE SYSTEM-BASED APPROACHES

C.1 Overview

This appendix presents the flowcharts of the steps involved in both component-based and system-based approaches proposed in this study.

C.2 The component-based approach

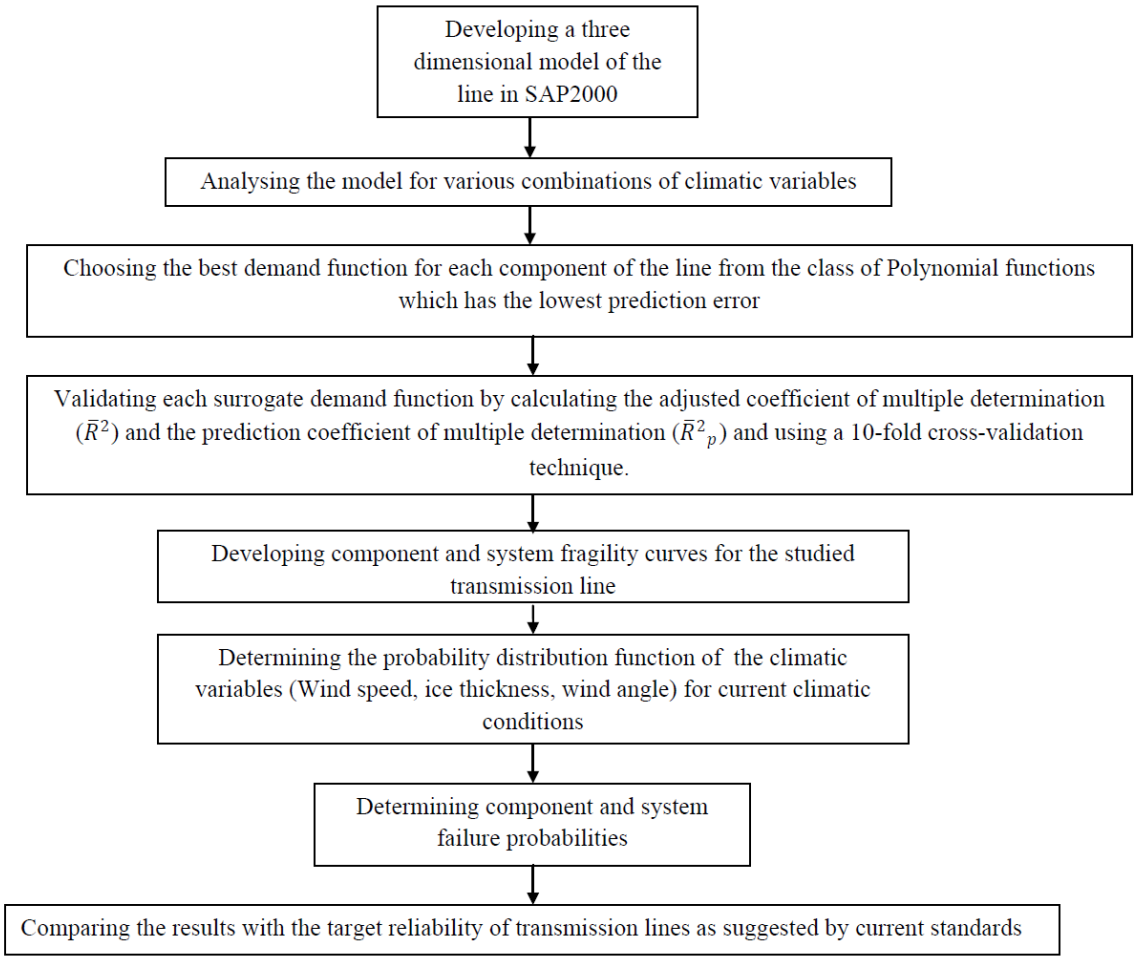


Figure C.1 Flowchart of the steps involved in the component-based approach

C.3 The system-based approach

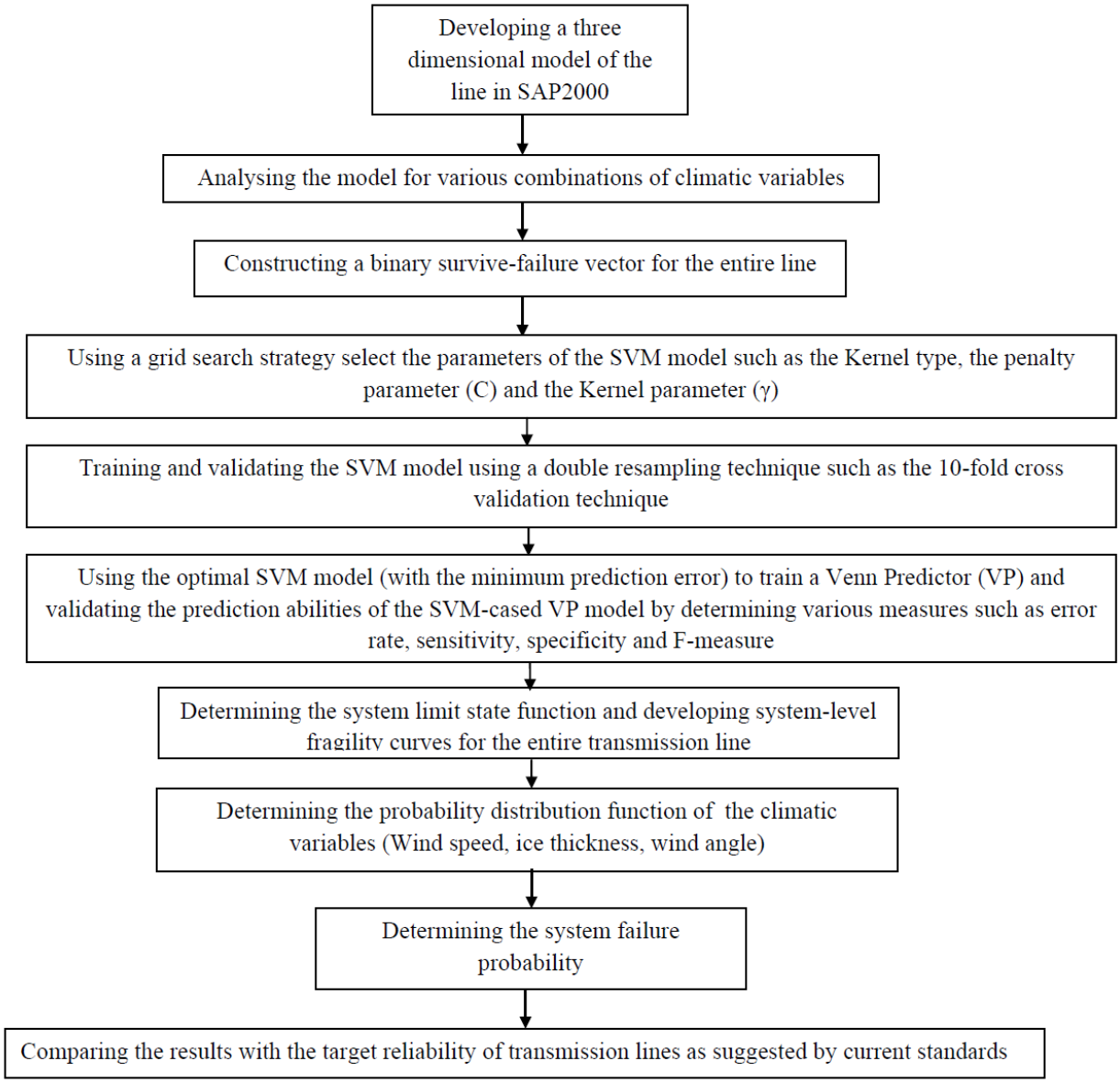


Figure C.2 Flowchart of the steps involved in the system-based approach

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